GEOMETRY · COMMON CORE HIGH SCHOOL MATH

Question Collection

Main • Backward Branches • Forward 1 • Forward 2 (later courses)

105 question sets • 3160 questions

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Precise definition of a circle

MAIN	Geometry main question	**HSG-CO.A.1**
BACKWARD 3	Use the center, radius, and diameter of a circle	**7.G.B.4**
BACKWARD 4	Find the distance from a center to points on a grid	**8.G.B.8**
FORWARD 1	Test definitions of circle vocabulary with counterexamples	**HSG-CO.A.1**
FORWARD 2	Write precise definitions of circle parts from the definition of a circ..	**HSG-CO.A.1**

SET 1 • HSG-CO.A.1

MAIN

Geometry main question • High School — Geometry • Geometry: Congruence • HSG-CO.A.1

Write a precise definition of a circle. Base it on the notions of point, plane, and distance (you may use ordinary words such as "set", "all", and "given"). Name the center and the radius in your definition.

Definition: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-M-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 1 • MAIN • Q1/5

Kai writes: "A circle is a round curve with a center."

Explain what makes this imprecise, and rewrite it as a precise definition.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-M-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 1 • MAIN • Q2/5

Lee writes: "The circle with center O and radius 4 is all the points of the plane that are within 4 units of O."

What figure does Lee's sentence describe? Rewrite it so that it defines the circle.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-M-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 1 • MAIN • Q3/5

Mia writes: "A circle is all the points that are 5 cm from point O."

What must she add to make the definition precise? Explain why.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-M-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 1 • MAIN • Q4/5

Ana writes: "A circle is the set of points in a plane at a distance from a point O in that plane."

Which condition is missing? Rewrite the definition so that it is precise.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-M-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 1 • MAIN • Q5/5

SET 1 • HSG-CO.A.1

BACKWARD 3

Use the center, radius, and diameter of a circle • Grade 7 • Geometry • 7.G.B.4

A circle has center O and radius 7 cm. Point P is in the plane of the circle, 7 cm from O. Is P on the circle? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-B3-Q1

Grade 7 • Geometry • 7.G.B.4 • Set 1 • BACKWARD 3 • Q1/5

A circle has a diameter of 18 in. What is its radius?

Radius: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-B3-Q2

Grade 7 • Geometry • 7.G.B.4 • Set 1 • BACKWARD 3 • Q2/5

A circle has center C and radius 4. Point D is in the same plane, 3 units from C. Where is D?

A. Inside the circle

B. Outside the circle

C. On the circle

D. There is not enough information

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-B3-Q3

Grade 7 • Geometry • 7.G.B.4 • Set 1 • BACKWARD 3 • Q3/5

True or false?

Every radius of a circle has the same length.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-B3-Q4

Grade 7 • Geometry • 7.G.B.4 • Set 1 • BACKWARD 3 • Q4/5

True or false?

A point 10 units from the center of a circle of radius 5 lies on the circle.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-B3-Q5

Grade 7 • Geometry • 7.G.B.4 • Set 1 • BACKWARD 3 • Q5/5

SET 1 • HSG-CO.A.1

BACKWARD 4

Find the distance from a center to points on a grid • Grade 8 • Geometry • 8.G.B.8

Find the distance from O(0, 0) to P(3, 4).

Distance: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-B4-Q1

Grade 8 • Geometry • 8.G.B.8 • Set 1 • BACKWARD 4 • Q1/5

Find the distance from C(1, 1) to Q(1, 6).

Distance: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-B4-Q2

Grade 8 • Geometry • 8.G.B.8 • Set 1 • BACKWARD 4 • Q2/5

Which point is 5 units from O(0, 0)?

A. (5, 5)

B. (2, 3)

C. (4, 4)

D. (−4, 3)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-B4-Q3

Grade 8 • Geometry • 8.G.B.8 • Set 1 • BACKWARD 4 • Q3/5

True or false?

The points (5, 0) and (0, −5) are the same distance from the origin.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-B4-Q4

Grade 8 • Geometry • 8.G.B.8 • Set 1 • BACKWARD 4 • Q4/5

True or false?

The point (3, 3) is 3 units from the origin.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-B4-Q5

Grade 8 • Geometry • 8.G.B.8 • Set 1 • BACKWARD 4 • Q5/5

SET 1 • HSG-CO.A.1

FORWARD 1

Test definitions of circle vocabulary with counterexamples • High School — Geometry • Geometry: Congruence • HSG-CO.A.1

Proposed definition: "A circle is a set of points that are all the same distance from a point O."

Give a set that fits this wording but is NOT a circle.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-F1-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 1 • FORWARD 1 • Q1/5

Proposed definition: "A radius of a circle is a segment from the center."

Give a counterexample, and repair the definition.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-F1-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 1 • FORWARD 1 • Q2/5

Which set fits the description "all points 2 units from point O" but is NOT a circle?

A. A circle of radius 2 centered at O in a plane

B. A sphere of radius 2 centered at O

C. A disk of radius 2

D. A segment of length 2 from O

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-F1-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 1 • FORWARD 1 • Q3/5

Proposed definition: "A chord is a segment inside a circle."

Give a counterexample, and repair the definition.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-F1-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 1 • FORWARD 1 • Q4/5

Proposed definition: "A diameter is a segment that passes through the center of a circle."

Give a counterexample, and repair the definition.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-F1-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 1 • FORWARD 1 • Q5/5

SET 1 • HSG-CO.A.1

FORWARD 2

Write precise definitions of circle parts from the definition of a circle • High School — Geometry • Geometry: Congruence • HSG-CO.A.1

Write a precise definition of a chord of a circle.

Definition: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-F2-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 1 • FORWARD 2 • Q1/5

Write a precise definition of a diameter of a circle with center O.

Definition: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-F2-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 1 • FORWARD 2 • Q2/5

Segment PQ is a diameter of a circle with center O. Write a precise definition of a semicircle with endpoints P and Q.

Definition: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-F2-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 1 • FORWARD 2 • Q3/5

Which is a precise definition of the interior of a circle with center O and radius r?

A. All points of the plane whose distance from O is at most r

B. All points of the circle

C. All points whose distance from O is less than r

D. All points of the plane of the circle whose distance from O is less than r

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-F2-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 1 • FORWARD 2 • Q4/5

Two circles in the same plane have the same center. Write a precise definition of this pair ("concentric circles") and explain why two different concentric circles never meet.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-F2-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 1 • FORWARD 2 • Q5/5

Precise definition of an angle

MAIN	Geometry main question	**HSG-CO.A.1**
BACKWARD 1	See an angle as two rays with a common endpoint, measured by…	**4.MD.C.5.a**
BACKWARD 2	Identify rays and their endpoints	**4.G.A.1**
FORWARD 1	Define angle relationships from the definition of an angle	**HSG-CO.A.1**
FORWARD 2	Prove angle facts from the angle definitions	**HSG-CO.C.9**

SET 97 • HSG-CO.A.1

MAIN

Geometry main question • High School — Geometry • Geometry: Congruence • HSG-CO.A.1

Write a precise definition of an angle. Name its vertex and its sides in your definition.

Definition: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S97-M-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 97 • MAIN • Q1/5

Sam writes: "An angle is the space between two lines."

Explain two things that are wrong, and rewrite it as a precise definition.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S97-M-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 97 • MAIN • Q2/5

Jo writes: "An angle is two rays."

Give a pair of rays that fits Jo's wording but is not an angle, and repair the definition.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S97-M-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 97 • MAIN • Q3/5

A 1° angle is defined using distance around a circular arc. Complete the definition, and explain why the measure of an angle does not depend on the radius of the circle used.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S97-M-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 97 • MAIN • Q4/5

Rays QP and QR have the common endpoint Q, and ray RS starts at R. Using the definition, explain which pair of rays forms an angle and name the angle, and which pair does not.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S97-M-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 97 • MAIN • Q5/5

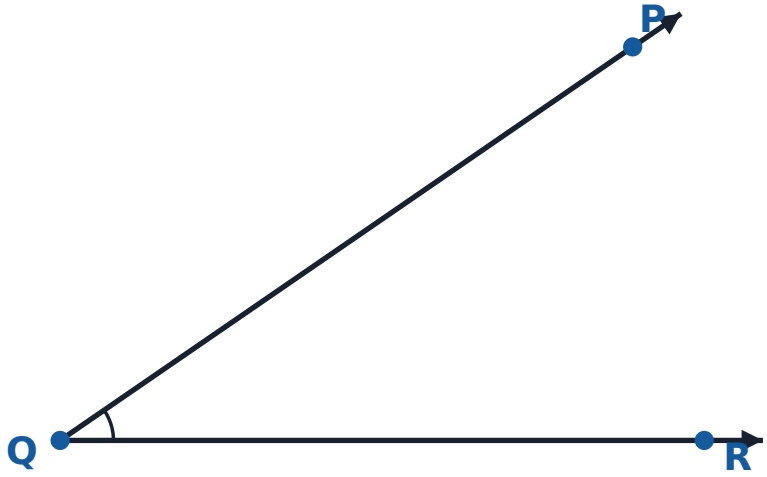
SET 97 • HSG-CO.A.1

BACKWARD 1

See an angle as two rays with a common endpoint, measured by an arc • Grade 4 • Measurement & Data • 4.MD.C.5.a

Name the vertex and the two sides of ∠PQR.

Answer: _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S97-B1-Q1

Grade 4 • Measurement & Data • 4.MD.C.5.a • Set 97 • BACKWARD 1 • Q1/5

An angle has its vertex at the center of a circle and cuts off $\frac{1}{4}$ of the circle. What is its measure?

Measure: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S97-B1-Q2

Grade 4 • Measurement & Data • 4.MD.C.5.a • Set 97 • BACKWARD 1 • Q2/5

An angle measures 60°. What fraction of a circle centered at its vertex does it cut off?

Fraction: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S97-B1-Q3

Grade 4 • Measurement & Data • 4.MD.C.5.a • Set 97 • BACKWARD 1 • Q3/5

True or false?

The measure of an angle depends on how long its sides are drawn.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S97-B1-Q4

Grade 4 • Measurement & Data • 4.MD.C.5.a • Set 97 • BACKWARD 1 • Q4/5

True or false?

A one-degree angle turns through $\frac{1}{360}$ of a full circle.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S97-B1-Q5

Grade 4 • Measurement & Data • 4.MD.C.5.a • Set 97 • BACKWARD 1 • Q5/5

SET 97 • HSG-CO.A.1

BACKWARD 2

Identify rays and their endpoints • Grade 4 • Geometry • 4.G.A.1

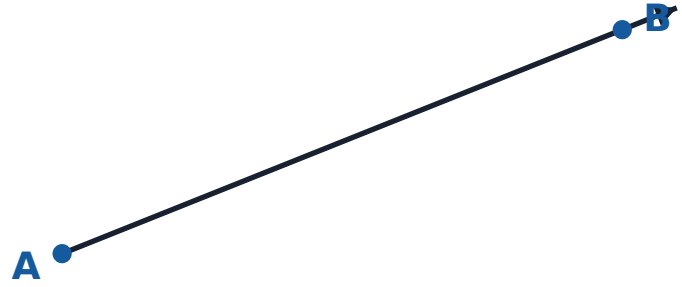
What is shown in the figure?

A. Segment AB

B. Ray BA

C. Line AB

D. Ray AB



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S97-B2-Q1

Grade 4 • Geometry • 4.G.A.1 • Set 97 • BACKWARD 2 • Q1/5

How many endpoints does a ray have?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S97-B2-Q2

Grade 4 • Geometry • 4.G.A.1 • Set 97 • BACKWARD 2 • Q2/5

Which figure goes on without end in exactly one direction?

A. A line segment

B. A point

C. A ray

D. A line

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S97-B2-Q3

Grade 4 • Geometry • 4.G.A.1 • Set 97 • BACKWARD 2 • Q3/5

True or false?

Ray AB and ray BA are the same ray.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S97-B2-Q4

Grade 4 • Geometry • 4.G.A.1 • Set 97 • BACKWARD 2 • Q4/5

True or false?

Rays QA and QB share the endpoint Q.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S97-B2-Q5

Grade 4 • Geometry • 4.G.A.1 • Set 97 • BACKWARD 2 • Q5/5

SET 97 • HSG-CO.A.1

FORWARD 1

Define angle relationships from the definition of an angle • High School — Geometry • Geometry: Congruence • HSG-CO.A.1

Write a precise definition of adjacent angles.

Definition: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S97-F1-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 97 • FORWARD 1 • Q1/5

Write a precise definition of vertical angles.

Definition: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S97-F1-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 97 • FORWARD 1 • Q2/5

Write a precise definition of a linear pair of angles.

Definition: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S97-F1-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 97 • FORWARD 1 • Q3/5

Proposed definition: "Two angles are vertical angles if they have the same vertex and are not adjacent."

Give a counterexample and repair the definition.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S97-F1-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 97 • FORWARD 1 • Q4/5

Which is a precise definition of the bisector of ∠ABC?

A. Any line through B

B. A ray from B that lies outside ∠ABC

C. Ray BD, where D is in the interior of ∠ABC and m∠ABD = m∠DBC

D. The segment that joins A and C

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S97-F1-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 97 • FORWARD 1 • Q5/5

SET 97 • HSG-CO.A.1

FORWARD 2

Prove angle facts from the angle definitions • High School — Geometry • Geometry: Congruence • HSG-CO.C.9

∠1 and ∠2 form a linear pair, and ∠2 and ∠3 form a linear pair, where ∠1 and ∠3 are vertical angles. Prove that m∠1 = m∠3.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S97-F2-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 97 • FORWARD 2 • Q1/5

Ray BD bisects ∠ABC and m∠ABD = (2x + 10)°, m∠DBC = (4x − 20)°. Find m∠ABC.

m∠ABC = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S97-F2-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 97 • FORWARD 2 • Q2/5

∠PQR and ∠RQS form a linear pair, and m∠PQR is 4 times m∠RQS. Find both angles.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S97-F2-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 97 • FORWARD 2 • Q3/5

Rays OA and OB are opposite rays, and ray OC is not on line AB. Ray OD bisects ∠AOC and ray OE bisects ∠COB. Prove that m∠DOE = 90°.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S97-F2-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 97 • FORWARD 2 • Q4/5

True or false?

If two angles are adjacent and supplementary, their noncommon sides are opposite rays.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S97-F2-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 97 • FORWARD 2 • Q5/5

SET 98 • HSG-CO.A.1 • CONGRUENCE: DEFINITIONS AND TRANSFORMATIONS

Precise definition of a line segment

MAIN	Geometry main question	**HSG-CO.A.1**
BACKWARD 1	Identify points, lines, rays, and segments	**4.G.A.1**
BACKWARD 2	Find distances between points on a number line	**6.NS.C.8**
FORWARD 1	Define figures built from segments	**HSG-CO.A.1**
FORWARD 2	Test segment-based definitions with counterexamples	**HSG-CO.A.1**

SET 98 • HSG-CO.A.1

MAIN

Geometry main question • High School — Geometry • Geometry: Congruence • HSG-CO.A.1

Points A and B lie on line ℓ. Which is a precise definition of segment AB?

A. Point A and every point of line ℓ on the same side of A as B

B. Every point of the plane that is closer to A than to B

C. Every point of line ℓ

D. Points A and B together with every point C of line ℓ for which AC + CB = AB

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S98-M-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 98 • MAIN • Q1/5

Write a precise definition of segment AB using the word "between".

Definition: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S98-M-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 98 • MAIN • Q2/5

Ravi writes: "A segment is part of a line."

Give a part of a line that is not a segment, and rewrite the definition precisely.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S98-M-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 98 • MAIN • Q3/5

Points A, B, and C are on one line with AC = 3, CB = 5, and AB = 8. Is C on segment AB? Use the definition to explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S98-M-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 98 • MAIN • Q4/5

Points A, B, and D are on one line with AD = 3, DB = 11, and AB = 8. Is D on segment AB? Use the definition to explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S98-M-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 98 • MAIN • Q5/5

SET 98 • HSG-CO.A.1

BACKWARD 1

Identify points, lines, rays, and segments • Grade 4 • Geometry • 4.G.A.1

How many endpoints does a line segment have? How many does a ray have?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S98-B1-Q1

Grade 4 • Geometry • 4.G.A.1 • Set 98 • BACKWARD 1 • Q1/5

True or false?

A line segment has exactly two endpoints.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S98-B1-Q2

Grade 4 • Geometry • 4.G.A.1 • Set 98 • BACKWARD 1 • Q2/5

Which figure has two endpoints?

A. A ray

B. A line

C. An angle

D. A segment

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S98-B1-Q3

Grade 4 • Geometry • 4.G.A.1 • Set 98 • BACKWARD 1 • Q3/5

True or false?

A line has two endpoints.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S98-B1-Q4

Grade 4 • Geometry • 4.G.A.1 • Set 98 • BACKWARD 1 • Q4/5

Which name describes the part of line AB from A to B, including A and B?

A. Ray BA

B. Line AB

C. Segment AB

D. Ray AB

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S98-B1-Q5

Grade 4 • Geometry • 4.G.A.1 • Set 98 • BACKWARD 1 • Q5/5

SET 98 • HSG-CO.A.1

BACKWARD 2

Find distances between points on a number line • Grade 6 • The Number System • 6.NS.C.8

On a number line, A is at 2 and B is at 9. Find AB.

AB = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S98-B2-Q1

Grade 6 • The Number System • 6.NS.C.8 • Set 98 • BACKWARD 2 • Q1/5

On a number line, A is at −3 and C is at 4. Find AC.

AC = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S98-B2-Q2

Grade 6 • The Number System • 6.NS.C.8 • Set 98 • BACKWARD 2 • Q2/5

On a number line, A is at 1, C is at 4, B is at 10. Which is true?

A. AB < AC

B. AC + CB > AB

C. AC + CB = AB

D. AC = CB

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S98-B2-Q3

Grade 6 • The Number System • 6.NS.C.8 • Set 98 • BACKWARD 2 • Q3/5

True or false?

On a number line, the point at 5 is between the points at 2 and 8.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S98-B2-Q4

Grade 6 • The Number System • 6.NS.C.8 • Set 98 • BACKWARD 2 • Q4/5

True or false?

On a number line, the point at 11 is between the points at 2 and 8.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S98-B2-Q5

Grade 6 • The Number System • 6.NS.C.8 • Set 98 • BACKWARD 2 • Q5/5

SET 98 • HSG-CO.A.1

FORWARD 1

Define figures built from segments • High School — Geometry • Geometry: Congruence • HSG-CO.A.1

Write a precise definition of ray AB. Use the word "between".

Definition: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S98-F1-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 98 • FORWARD 1 • Q1/5

Write a precise definition of the midpoint of segment AB.

Definition: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S98-F1-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 98 • FORWARD 1 • Q2/5

Write a precise definition of a triangle using segments.

Definition: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S98-F1-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 98 • FORWARD 1 • Q3/5

Write a precise definition of congruent segments.

Definition: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S98-F1-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 98 • FORWARD 1 • Q4/5

Write a precise definition of a segment bisector.

Definition: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S98-F1-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 98 • FORWARD 1 • Q5/5

SET 98 • HSG-CO.A.1

FORWARD 2

Test segment-based definitions with counterexamples • High School — Geometry • Geometry: Congruence • HSG-CO.A.1

Proposed definition: "Point M is the midpoint of segment AB if AM = MB."

Give a counterexample, and repair the definition.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S98-F2-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 98 • FORWARD 2 • Q1/5

Proposed definition: "A triangle is three segments."

Give a counterexample, and repair the definition.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S98-F2-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 98 • FORWARD 2 • Q2/5

Proposed definition: "C is between A and B if $AC + CB = AB$."

Is this enough when A, B, and C are any three points of a plane? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S98-F2-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 98 • FORWARD 2 • Q3/5

True or false?

"The midpoint of AB is a point M with AM = MB" is precise enough, because every such point lies on segment AB.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S98-F2-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 98 • FORWARD 2 • Q4/5

Proposed definition: "A segment bisector is a line that cuts a segment into two parts."

Give a counterexample and repair it.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S98-F2-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 98 • FORWARD 2 • Q5/5

Precise definition of perpendicular lines

MAIN	Geometry main question	**HSG-CO.A.1**
BACKWARD 1	Recognize perpendicular lines and right angles	**4.G.A.1**
BACKWARD 2	Find the other angles when two lines cross	**7.G.B.5**
FORWARD 1	Prove consequences of the definition of perpendicular lines	**HSG-CO.C.9**
FORWARD 2	Reason about perpendicular lines with a third line, in a plane and…	**HSG-CO.C.9**

SET 2 • HSG-CO.A.1

MAIN

Geometry main question • High School — Geometry • Geometry: Congruence • HSG-CO.A.1

Ben writes: "Perpendicular lines are lines that make a T."

Explain what is imprecise, and write a precise definition.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-M-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 2 • MAIN • Q1/5

Write a precise definition of perpendicular lines.

Definition: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-M-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 2 • MAIN • Q2/5

Lines a and b intersect at P. Two angles at P form a linear pair and have equal measures.

Use the definition to decide whether a and b are perpendicular. Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-M-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 2 • MAIN • Q3/5

Line m passes through (0, 0) and (2, 1). Line n passes through (0, 0) and (−1, 2). Use the definition (intersect and form a right angle) to decide whether m and n are perpendicular. Show how you know the angle at (0, 0) is a right angle.

Answer: _____

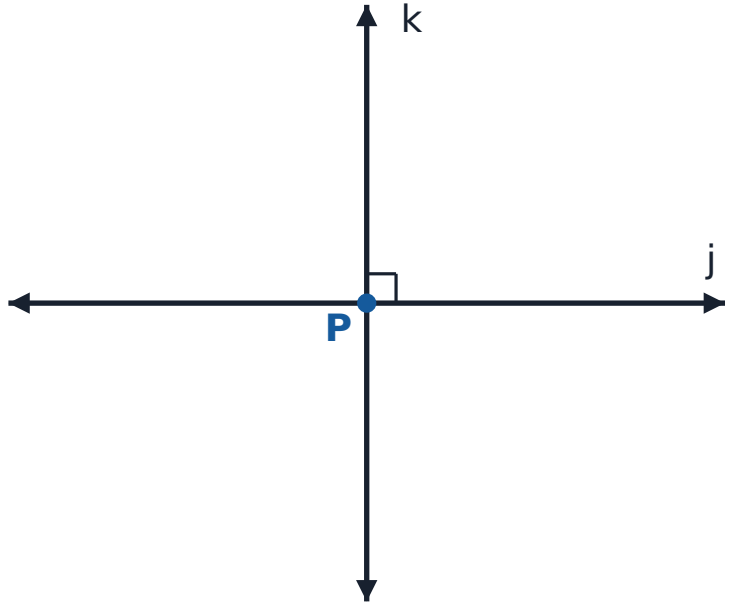
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-M-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 2 • MAIN • Q4/5

Lines j and k meet at point P, and one of the angles they form is marked as a right angle.

Are j and k perpendicular? Use the definition to explain.



Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-M-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 2 • MAIN • Q5/5

SET 2 • HSG-CO.A.1

BACKWARD 1

Recognize perpendicular lines and right angles • Grade 4 • Geometry • 4.G.A.1

True or false?

Lines that meet at a 90° angle are perpendicular.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-B1-Q1

Grade 4 • Geometry • 4.G.A.1 • Set 2 • BACKWARD 1 • Q1/5

True or false?

Any two lines that cross are perpendicular.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-B1-Q2

Grade 4 • Geometry • 4.G.A.1 • Set 2 • BACKWARD 1 • Q2/5

Two perpendicular lines cross. How many right angles do they form?

Number: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-B1-Q3

Grade 4 • Geometry • 4.G.A.1 • Set 2 • BACKWARD 1 • Q3/5

What is true of two sides of a rectangle that share a vertex?

A. They are perpendicular

B. They are parallel

C. They never meet

D. They form a 60° angle

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-B1-Q4

Grade 4 • Geometry • 4.G.A.1 • Set 2 • BACKWARD 1 • Q4/5

Which pair of streets is most like perpendicular lines?

A. Two straight streets that never meet

B. Two streets that curve around a park

C. Two streets that meet at a sharp point

D. Two streets that cross at a square corner

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-B1-Q5

Grade 4 • Geometry • 4.G.A.1 • Set 2 • BACKWARD 1 • Q5/5

SET 2 • HSG-CO.A.1

BACKWARD 2

Find the other angles when two lines cross • Grade 7 • Geometry • 7.G.B.5

Two lines cross. One of the angles measures 90°. Find the other three angles.

Angles: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-B2-Q1

Grade 7 • Geometry • 7.G.B.5 • Set 2 • BACKWARD 2 • Q1/5

Two lines cross. One of the angles measures 70°. Find the other three angles.

Angles: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-B2-Q2

Grade 7 • Geometry • 7.G.B.5 • Set 2 • BACKWARD 2 • Q2/5

True or false?

Vertical angles formed by two crossing lines are equal.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-B2-Q3

Grade 7 • Geometry • 7.G.B.5 • Set 2 • BACKWARD 2 • Q3/5

Two angles form a linear pair. One measures 90°. What does the other measure?

A. 180°

B. 45°

C. 90°

D. 270°

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-B2-Q4

Grade 7 • Geometry • 7.G.B.5 • Set 2 • BACKWARD 2 • Q4/5

True or false?

Two lines cross and one angle measures 100°. The angle next to it also measures 100°.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-B2-Q5

Grade 7 • Geometry • 7.G.B.5 • Set 2 • BACKWARD 2 • Q5/5

SET 2 • HSG-CO.A.1

FORWARD 1

Prove consequences of the definition of perpendicular lines • High School — Geometry • Geometry: Congruence • HSG-CO.C.9

Lines ℓ and m are perpendicular and meet at point P. Prove that all four angles at P are right angles.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-F1-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 2 • FORWARD 1 • Q1/5

Two lines meet at point P and form a linear pair of congruent angles. Prove that the lines are perpendicular.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-F1-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 2 • FORWARD 1 • Q2/5

Lines a and b intersect, and one of the angles they form measures (3x + 15)°. For what value of x are a and b perpendicular? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-F1-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 2 • FORWARD 1 • Q3/5

Two intersecting lines form a pair of vertical angles that are supplementary. Prove that the lines are perpendicular.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-F1-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 2 • FORWARD 1 • Q4/5

Lines p and q meet at O. Ana says: "Two adjacent angles at O add to 180°, so p ⊥ q." Explain why her reason does not prove perpendicularity, and state a fact that does.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-F1-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 2 • FORWARD 1 • Q5/5

SET 2 • HSG-CO.A.1

FORWARD 2

Reason about perpendicular lines with a third line, in a plane and in space • High School — Geometry • Geometry: Congruence • HSG-CO.C.9

In a plane, lines a and b are both perpendicular to line t. Prove that a and b are parallel.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-F2-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 2 • FORWARD 2 • Q1/5

True or false?

In three-dimensional space, two lines that are each perpendicular to the same line must be parallel.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-F2-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 2 • FORWARD 2 • Q2/5

In a plane, line a is parallel to line b, and line t is perpendicular to line a. What is true about t and b?

A. t and b do not meet

B. t is perpendicular to b

C. t is parallel to b

D. The angle between t and b cannot be known

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-F2-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 2 • FORWARD 2 • Q3/5

In a coordinate plane, line t is the x-axis. Line a is perpendicular to t at (0, 0), and line b is perpendicular to t and passes through (5, 2). Write an equation of line b.

Equation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-F2-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 2 • FORWARD 2 • Q4/5

True or false?

In a plane, through a point on a line there is exactly one line perpendicular to that line.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-F2-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 2 • FORWARD 2 • Q5/5

Precise definition of parallel lines

MAIN	Geometry main question	**HSG-CO.A.1**
BACKWARD 1	Recognize parallel lines	**4.G.A.1**
BACKWARD 2	Decide whether two lines in the coordinate plane intersect	**8.EE.C.8**
BACKWARD 3	Visualize lines and planes in a box	**7.G.A.3**
FORWARD 1	Prove that lines are parallel from angle facts	**HSG-CO.C.9**
FORWARD 2	Use the parallel postulate	**HSG-CO.C.9**

SET 99 • HSG-CO.A.1

MAIN

Geometry main question • High School — Geometry • Geometry: Congruence • HSG-CO.A.1

Write a precise definition of parallel lines.

Definition: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S99-M-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 99 • MAIN • Q1/5

Lines ℓ and m do not intersect. Which additional fact makes them parallel?

A. They lie in the same plane

B. They both look horizontal in a drawing

C. They are drawn the same length

D. They are both perpendicular to one segment in space

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S99-M-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 99 • MAIN • Q2/5

Eli writes: "Parallel lines are lines that never meet."

Give an example of two lines that fit Eli's sentence but are not parallel, and repair the definition.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S99-M-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 99 • MAIN • Q3/5

Box ABCD-EFGH has top face ABCD directly above bottom face EFGH (A above E, B above F, C above G, D above H). Using the definition, decide whether line AB is parallel to line HG, and whether line AB is parallel to line FG. Explain each.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S99-M-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 99 • MAIN • Q4/5

Line p is $y = 2x + 1$ and line q is $y = 2x - 3$. Use the definition to decide whether p and q are parallel. Show that they do not intersect.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S99-M-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 99 • MAIN • Q5/5

SET 99 • HSG-CO.A.1

BACKWARD 1

Recognize parallel lines • Grade 4 • Geometry • 4.G.A.1

True or false?

Two lines in a plane that never meet are parallel.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S99-B1-Q1

Grade 4 • Geometry • 4.G.A.1 • Set 99 • BACKWARD 1 • Q1/5

What is true of the opposite sides of a rectangle?

A. They are parallel

B. They are perpendicular

C. They meet at one point

D. They are neither parallel nor perpendicular

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S99-B1-Q2

Grade 4 • Geometry • 4.G.A.1 • Set 99 • BACKWARD 1 • Q2/5

True or false?

Any two lines that look slanted the same amount in a drawing are parallel.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S99-B1-Q3

Grade 4 • Geometry • 4.G.A.1 • Set 99 • BACKWARD 1 • Q3/5

Which pair of streets is most like parallel lines?

A. Two streets that meet at a corner

B. Two streets that cross

C. A street and a bridge above it going another way

D. Two straight streets on flat ground that never meet

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S99-B1-Q4

Grade 4 • Geometry • 4.G.A.1 • Set 99 • BACKWARD 1 • Q4/5

How many times do two parallel lines intersect?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S99-B1-Q5

Grade 4 • Geometry • 4.G.A.1 • Set 99 • BACKWARD 1 • Q5/5

SET 99 • HSG-CO.A.1

BACKWARD 2

Decide whether two lines in the coordinate plane intersect • Grade 8 • Expressions & Equations • 8.EE.C.8

Do the lines $y = x + 2$ and $y = x - 1$ intersect? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S99-B2-Q1

Grade 8 • Expressions & Equations • 8.EE.C.8 • Set 99 • BACKWARD 2 • Q1/5

Do the lines $y = 2x$ and $y = -x + 6$ intersect? If so, where?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S99-B2-Q2

Grade 8 • Expressions & Equations • 8.EE.C.8 • Set 99 • BACKWARD 2 • Q2/5

Which line never meets $y = 3x + 1$?

A. $y = x + 1$

B. $y = 3$

C. $y = -3x + 1$

D. $y = 3x - 4$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S99-B2-Q3

Grade 8 • Expressions & Equations • 8.EE.C.8 • Set 99 • BACKWARD 2 • Q3/5

True or false?

Two different lines with the same slope never intersect.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S99-B2-Q4

Grade 8 • Expressions & Equations • 8.EE.C.8 • Set 99 • BACKWARD 2 • Q4/5

True or false?

The lines $y = 2x + 5$ and $y = 5$ never meet.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S99-B2-Q5

Grade 8 • Expressions & Equations • 8.EE.C.8 • Set 99 • BACKWARD 2 • Q5/5

SET 99 • HSG-CO.A.1

BACKWARD 3

Visualize lines and planes in a box • Grade 7 • Geometry • 7.G.A.3

True or false?

Any two edges of a box that never meet are parallel.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S99-B3-Q1

Grade 7 • Geometry • 7.G.A.3 • Set 99 • BACKWARD 3 • Q1/5

In a box, the top front edge and the bottom back edge (running the same direction) are

A. perpendicular

B. skew

C. parallel

D. the same line

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S99-B3-Q2

Grade 7 • Geometry • 7.G.A.3 • Set 99 • BACKWARD 3 • Q2/5

How many faces of a box are there?

Faces: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S99-B3-Q3

Grade 7 • Geometry • 7.G.A.3 • Set 99 • BACKWARD 3 • Q3/5

Which pair of box edges are skew?

A. A top front edge and the bottom front edge below it

B. A top front edge and a bottom side edge that runs front to back

C. Two top edges that meet at a corner

D. Two vertical edges

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S99-B3-Q4

Grade 7 • Geometry • 7.G.A.3 • Set 99 • BACKWARD 3 • Q4/5

True or false?

Two vertical edges of a box are parallel.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S99-B3-Q5

Grade 7 • Geometry • 7.G.A.3 • Set 99 • BACKWARD 3 • Q5/5

SET 99 • HSG-CO.A.1

FORWARD 1

Prove that lines are parallel from angle facts • High School — Geometry • Geometry: Congruence • HSG-CO.C.9

Lines ℓ and m are cut by transversal t, and a pair of alternate interior angles are congruent. Explain why ℓ and m cannot meet (so they are parallel).

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S99-F1-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 99 • FORWARD 1 • Q1/5

Lines ℓ and m are cut by a transversal. A pair of corresponding angles measure $(5x - 10)°$ and $(3x + 20)°$. For what x are ℓ and m parallel?

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S99-F1-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 99 • FORWARD 1 • Q2/5

Same-side interior angles formed by lines a, b, and a transversal measure 115° and 65°. Are a and b parallel? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S99-F1-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 99 • FORWARD 1 • Q3/5

In a plane, line a ∥ line b and line b ∥ line c, where a and c are different lines. Prove that a ∥ c.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S99-F1-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 99 • FORWARD 1 • Q4/5

Which fact proves that two coplanar lines cut by a transversal are parallel?

A. A linear pair adds to 180°

B. Two angles are acute

C. A pair of alternate interior angles are congruent

D. A pair of vertical angles are congruent

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S99-F1-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 99 • FORWARD 1 • Q5/5

SET 99 • HSG-CO.A.1

FORWARD 2

Use the parallel postulate • High School — Geometry • Geometry: Congruence • HSG-CO.C.9

True or false?

In a plane, through a point not on a line there is exactly one line parallel to the given line.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S99-F2-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 99 • FORWARD 2 • Q1/5

Point P is not on line ℓ. Line m passes through P and is parallel to ℓ. Line n also passes through P and is parallel to ℓ. What can you conclude about m and n? Why?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S99-F2-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 99 • FORWARD 2 • Q2/5

In the coordinate plane, find an equation of the line through (2, 5) parallel to $y = -3x + 1$, and explain why it is the only one.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S99-F2-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 99 • FORWARD 2 • Q3/5

Which statement is the parallel postulate?

A. Vertical angles are congruent

B. All right angles are congruent

C. Through any two points there is exactly one line

D. Through a point not on a line there is exactly one parallel to the line

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S99-F2-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 99 • FORWARD 2 • Q4/5

Why does the parallel postulate say "a point not on the line"?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S99-F2-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 99 • FORWARD 2 • Q5/5

Transformations as functions: inputs, outputs, and preimages

MAIN	Geometry main question	**HSG-CO.A.2**
BACKWARD 1	Evaluate a rule for a given input	**8.F.A.1**
BACKWARD 2	Apply coordinate rules for translations, reflections, and rotations	**8.G.A.3**
BACKWARD 3	Read and locate ordered pairs in all four quadrants	**6.NS.C.6.c**
FORWARD 1	Compose two transformations and write one rule	**HSG-CO.A.2**
FORWARD 2	Find the inverse of a transformation	**HSG-CO.A.2**

SET 3 • HSG-CO.A.2

MAIN

Geometry main question • High School — Geometry • Geometry: Congruence • HSG-CO.A.2

The transformation T is defined by T(x, y) = (x + 4, y − 3). Find T(2, 5).

T(2, 5) = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-M-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.A.2 • Set 3 • MAIN • Q1/5

The transformation R is defined by $R(x, y) = (-y, x)$. Find $R(3, -1)$.

$R(3, -1) =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-M-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.A.2 • Set 3 • MAIN • Q2/5

The transformation F takes each point (x, y) to (x, −y). The output point is F(P) = (4, 7). Find the input point P.

P = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-M-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.A.2 • Set 3 • MAIN • Q3/5

Every point of the plane is moved 5 units right and 2 units up. Which function describes this transformation?

A. $T(x, y) = (x - 5, y - 2)$

B. $T(x, y) = (5x, 2y)$

C. $T(x, y) = (x + 2, y + 5)$

D. $T(x, y) = (x + 5, y + 2)$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-M-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.A.2 • Set 3 • MAIN • Q4/5

The transformation D is defined by D(x, y) = (2x, 2y). Find every input point P whose output is itself (D(P) = P). Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-M-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.A.2 • Set 3 • MAIN • Q5/5

SET 3 • HSG-CO.A.2

BACKWARD 1

Evaluate a rule for a given input • Grade 8 • Functions • 8.F.A.1

f(x) = 3x − 2. Find f(4).

f(4) = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-B1-Q1

Grade 8 • Functions • 8.F.A.1 • Set 3 • BACKWARD 1 • Q1/5

A rule takes a number, doubles it, and adds 1. What is the output for the input 6?

Output: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-B1-Q2

Grade 8 • Functions • 8.F.A.1 • Set 3 • BACKWARD 1 • Q2/5

g(x) = x + 5. Which input gives the output 12?

Input: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-B1-Q3

Grade 8 • Functions • 8.F.A.1 • Set 3 • BACKWARD 1 • Q3/5

True or false?

A function assigns exactly one output to each input.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-B1-Q4

Grade 8 • Functions • 8.F.A.1 • Set 3 • BACKWARD 1 • Q4/5

True or false?

The rule "the input 3 gives the outputs 4 and 5" describes a function.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-B1-Q5

Grade 8 • Functions • 8.F.A.1 • Set 3 • BACKWARD 1 • Q5/5

SET 3 • HSG-CO.A.2

BACKWARD 2

Apply coordinate rules for translations, reflections, and rotations • Grade 8 • Geometry • 8.G.A.3

Translate the point (2, −1) 3 units left and 4 units up. Where does it go?

Image: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-B2-Q1

Grade 8 • Geometry • 8.G.A.3 • Set 3 • BACKWARD 2 • Q1/5

Reflect the point (5, 2) across the y-axis.

Image: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-B2-Q2

Grade 8 • Geometry • 8.G.A.3 • Set 3 • BACKWARD 2 • Q2/5

Rotate the point (4, 1) by 180° about the origin.

A. (1, 4)

B. (−1, 4)

C. (4, −1)

D. (−4, −1)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-B2-Q3

Grade 8 • Geometry • 8.G.A.3 • Set 3 • BACKWARD 2 • Q3/5

True or false?

Reflecting the point (3, 6) across the x-axis gives (3, −6).

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-B2-Q4

Grade 8 • Geometry • 8.G.A.3 • Set 3 • BACKWARD 2 • Q4/5

True or false?

Rotating the point (0, 5) by 90° counterclockwise about the origin gives (5, 0).

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-B2-Q5

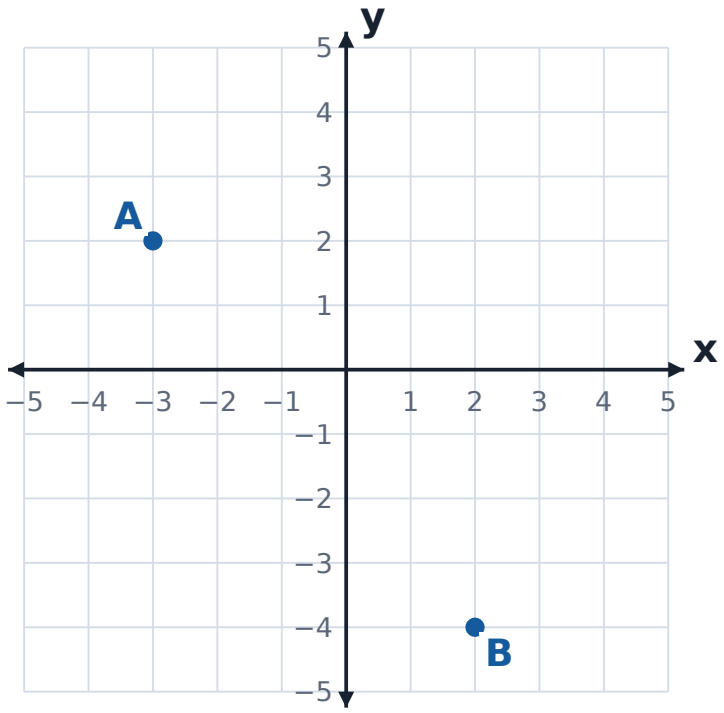
Grade 8 • Geometry • 8.G.A.3 • Set 3 • BACKWARD 2 • Q5/5

SET 3 • HSG-CO.A.2

BACKWARD 3

Read and locate ordered pairs in all four quadrants • Grade 6 • The Number System • 6.NS.C.6.c

What are the coordinates of point A?



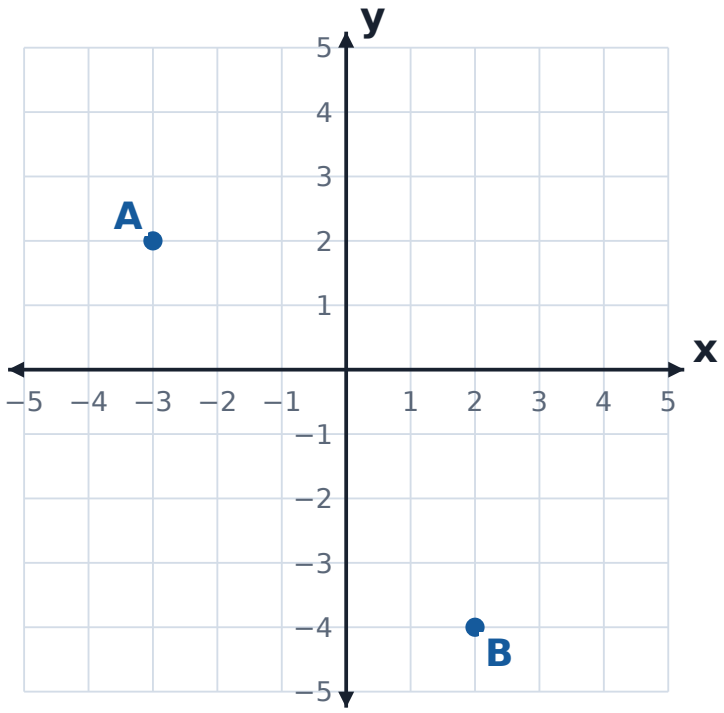
A = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-B3-Q1

Grade 6 • The Number System • 6.NS.C.6.c • Set 3 • BACKWARD 3 • Q1/5

What are the coordinates of point B?



B = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-B3-Q2

Grade 6 • The Number System • 6.NS.C.6.c • Set 3 • BACKWARD 3 • Q2/5

Which point is in Quadrant III?

A. (−2, −5)

B. (−2, 5)

C. (2, 5)

D. (2, −5)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-B3-Q3

Grade 6 • The Number System • 6.NS.C.6.c • Set 3 • BACKWARD 3 • Q3/5

True or false?

The point (0, −3) lies on the y-axis.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-B3-Q4

Grade 6 • The Number System • 6.NS.C.6.c • Set 3 • BACKWARD 3 • Q4/5

True or false?

The point $(-2, 5)$ is 2 units to the right of the y-axis.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-B3-Q5

Grade 6 • The Number System • 6.NS.C.6.c • Set 3 • BACKWARD 3 • Q5/5

SET 3 • HSG-CO.A.2

FORWARD 1

Compose two transformations and write one rule • High School — Geometry • Geometry: Congruence • HSG-CO.A.2

$T(x, y) = (x + 2, y)$ and $R(x, y) = (-x, y)$. Find $R(T(1, 4))$.

$R(T(1, 4)) =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-F1-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.A.2 • Set 3 • FORWARD 1 • Q1/5

$T(x, y) = (x + 2, y)$ and $R(x, y) = (-x, y)$. Write a single rule for $R(T(x, y))$.

$R(T(x, y)) =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-F1-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.A.2 • Set 3 • FORWARD 1 • Q2/5

$T(x, y) = (x + 2, y)$ and $R(x, y) = (-x, y)$. Find $T(R(1, 4))$.

$T(R(1, 4)) =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-F1-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.A.2 • Set 3 • FORWARD 1 • Q3/5

True or false?

For any two transformations, doing them in either order gives the same result.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-F1-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.A.2 • Set 3 • FORWARD 1 • Q4/5

$T(x, y) = (x + 3, y - 1)$ and $S(x, y) = (x - 1, y + 5)$. Which rule is $S(T(x, y))$?

A. $(x + 2, y + 4)$

B. $(x - 3, y - 5)$

C. $(x + 2, y - 4)$

D. $(x + 4, y - 6)$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-F1-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.A.2 • Set 3 • FORWARD 1 • Q5/5

SET 3 • HSG-CO.A.2

FORWARD 2

Find the inverse of a transformation • High School — Geometry • Geometry: Congruence • HSG-CO.A.2

Find a rule for the inverse of $T(x, y) = (x - 6, y + 2)$.

Inverse: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-F2-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.A.2 • Set 3 • FORWARD 2 • Q1/5

$R(x, y) = (-y, x)$ is a 90° counterclockwise rotation about the origin. Find a rule for its inverse.

Inverse: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-F2-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.A.2 • Set 3 • FORWARD 2 • Q2/5

True or false?

Every rotation is its own inverse.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-F2-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.A.2 • Set 3 • FORWARD 2 • Q3/5

D(x, y) = (3x, 3y). Which rule undoes D?

A. $(x - 3, y - 3)$

B. $(3y, 3x)$

C. $(-3x, -3y)$

D. $(\frac{x}{3}, \frac{y}{3})$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-F2-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.A.2 • Set 3 • FORWARD 2 • Q4/5

True or false?

The transformation P(x, y) = (x, 0), which moves every point straight up or down to the x-axis, has an inverse transformation.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-F2-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.A.2 • Set 3 • FORWARD 2 • Q5/5

Compare transformations that preserve distance and angle with those that do not

MAIN	Geometry main question	**HSG-CO.A.2**
BACKWARD 1	Find the distance between two points	**8.G.B.8**
BACKWARD 2	Know that rigid motions keep lengths and angle measures	**8.G.A.1**
FORWARD 1	Classify transformations as rigid motions, similarity transformatio...	**HSG-SRT.A.2**
FORWARD 2	Prove with variables that a transformation preserves or scales ev...	**HSG-SRT.A.2**

SET 4 • HSG-CO.A.2

MAIN

Geometry main question • High School — Geometry • Geometry: Congruence • HSG-CO.A.2

Which transformation does NOT preserve distance?

A. $R(x, y) = (-x, y)$

B. $T(x, y) = (x + 4, y - 1)$

C. $H(x, y) = (2x, y)$

D. $Q(x, y) = (-y, x)$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-M-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.A.2 • Set 4 • MAIN • Q1/5

H(x, y) = (2x, y). Points A(0, 0) and B(3, 4) are 5 units apart.

Find H(A), H(B), and the distance between them. Does H preserve distance?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-M-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.A.2 • Set 4 • MAIN • Q2/5

The translation $T(x, y) = (x - 2, y + 5)$ maps $A(1, 1)$ and $B(4, 5)$ to A' and B'. Show that $A'B' = AB$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-M-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.A.2 • Set 4 • MAIN • Q3/5

$V(x, y) = (x, 3y)$. Does V preserve the measure of every angle? Use the right angle at $O(0, 0)$ between the rays through $(1, 1)$ and $(1, -1)$ to decide.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-M-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.A.2 • Set 4 • MAIN • Q4/5

S(x, y) = (3x, 3y). Does S preserve distance? Does S preserve angle measure? Explain each.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-M-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.A.2 • Set 4 • MAIN • Q5/5

SET 4 • HSG-CO.A.2

BACKWARD 1

Find the distance between two points • Grade 8 • Geometry • 8.G.B.8

Find the distance between (1, 2) and (4, 6).

Distance: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-B1-Q1

Grade 8 • Geometry • 8.G.B.8 • Set 4 • BACKWARD 1 • Q1/5

Find the distance between $(-2, 3)$ and $(4, 3)$.

Distance: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-B1-Q2

Grade 8 • Geometry • 8.G.B.8 • Set 4 • BACKWARD 1 • Q2/5

Find the distance between (0, 0) and (5, 12).

Distance: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-B1-Q3

Grade 8 • Geometry • 8.G.B.8 • Set 4 • BACKWARD 1 • Q3/5

True or false?

The distance from (0, 0) to (1, 1) is 2.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-B1-Q4

Grade 8 • Geometry • 8.G.B.8 • Set 4 • BACKWARD 1 • Q4/5

Find the distance between $(2, -1)$ and $(5, 3)$.

A. 25

B. $\sqrt{7}$

C. 5

D. 7

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-B1-Q5

Grade 8 • Geometry • 8.G.B.8 • Set 4 • BACKWARD 1 • Q5/5

SET 4 • HSG-CO.A.2

BACKWARD 2

Know that rigid motions keep lengths and angle measures • Grade 8 • Geometry • 8.G.A.1

True or false?

A rotation keeps every segment the same length.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-B2-Q1

Grade 8 • Geometry • 8.G.A.1 • Set 4 • BACKWARD 2 • Q1/5

True or false?

A reflection can change the measure of an angle.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-B2-Q2

Grade 8 • Geometry • 8.G.A.1 • Set 4 • BACKWARD 2 • Q2/5

A triangle is translated. What changes?

A. Its angle measures

B. Its side lengths

C. The positions of its vertices

D. Its area

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-B2-Q3

Grade 8 • Geometry • 8.G.A.1 • Set 4 • BACKWARD 2 • Q3/5

A segment 7 cm long is reflected across a line. How long is its image?

Length: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-B2-Q4

Grade 8 • Geometry • 8.G.A.1 • Set 4 • BACKWARD 2 • Q4/5

A 35° angle is rotated 90° about a point. What is the measure of its image?

Measure: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-B2-Q5

Grade 8 • Geometry • 8.G.A.1 • Set 4 • BACKWARD 2 • Q5/5

SET 4 • HSG-CO.A.2

FORWARD 1

Classify transformations as rigid motions, similarity transformations, or neither • High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2

Which transformation preserves angle measure but not distance?

A. $(x, y) \rightarrow (4x, y)$

B. $(x, y) \rightarrow (x + 4, y)$

C. $(x, y) \rightarrow (x, y + 4)$

D. $(x, y) \rightarrow (4x, 4y)$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-F1-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 4 • FORWARD 1 • Q1/5

Classify $(x, y) \to (-2x, -2y)$ as a rigid motion, a similarity transformation that is not rigid, or neither. Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-F1-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 4 • FORWARD 1 • Q2/5

Classify $(x, y) \to (x + y, y)$ as a rigid motion, a similarity transformation that is not rigid, or neither. Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-F1-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 4 • FORWARD 1 • Q3/5

True or false?

Every similarity transformation is a rigid motion.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-F1-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 4 • FORWARD 1 • Q4/5

True or false?

A transformation that multiplies every distance by the same number k > 0 always changes angle measures.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-F1-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 4 • FORWARD 1 • Q5/5

SET 4 • HSG-CO.A.2

FORWARD 2

Prove with variables that a transformation preserves or scales every distance • High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2

Prove that $T(x, y) = (x + a, y + b)$ preserves the distance between ANY two points (x_1, y_1) and (x_2, y_2).

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-F2-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 4 • FORWARD 2 • Q1/5

Prove that D(x, y) = (3x, 3y) multiplies the distance between ANY two points by 3.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-F2-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 4 • FORWARD 2 • Q2/5

$H(x, y) = (2x, y)$. The distance between the images of (x_1, y_1) and (x_2, y_2) is $\sqrt{(4(x_2 - x_1)^2 + (y_2 - y_1)^2)}$.

For which pairs of points is the distance unchanged?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-F2-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 4 • FORWARD 2 • Q3/5

True or false?

One example in which a distance stays the same is enough to prove that a transformation preserves every distance.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-F2-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 4 • FORWARD 2 • Q4/5

True or false?

If a transformation preserves the distance between every pair of points, it also preserves every angle measure.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-F2-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 4 • FORWARD 2 • Q5/5

Rotations and reflections that carry a rectangle, parallelogram, or trapezoid onto itself

MAIN	Geometry main question	**HSG-CO.A.3**
BACKWARD 1	Identify lines of symmetry of figures	**4.G.A.3**
BACKWARD 2	Know the side and angle properties of parallelograms, rectangles,…	**5.G.B.3**
BACKWARD 3	Rotate points by 90° and 180° about the origin	**8.G.A.3**
FORWARD 1	Use a symmetry to deduce congruent parts of the figure	**HSG-CO.B.6**
FORWARD 2	Combine symmetries and identify a figure from its symmetries	**HSG-CO.A.5**

SET 5 • HSG-CO.A.3

MAIN

Geometry main question • High School — Geometry • Geometry: Congruence • HSG-CO.A.3

Rectangle ABCD is not a square. Describe every rotation (with angle greater than 0° and at most 360°) and every reflection that carries the rectangle onto itself.



Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-M-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.A.3 • Set 5 • MAIN • Q1/5

Parallelogram PQRS has no right angles, and its sides are not all the same length. Describe every rotation (angle greater than 0° and at most 360°, about any point) and every reflection that carries it onto itself.

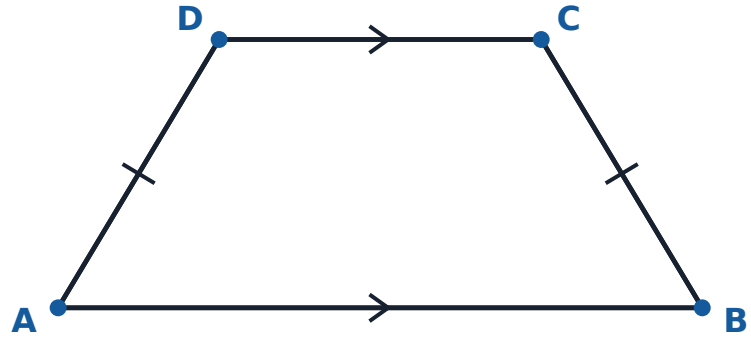
Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-M-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.A.3 • Set 5 • MAIN • Q2/5

Trapezoid ABCD has exactly one pair of parallel sides, the bases AB and DC (AB ≠ DC), and congruent legs AD = BC. Describe every rotation (angle greater than 0° and at most 360°, about any point) and every reflection that carries it onto itself.



Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-M-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.A.3 • Set 5 • MAIN • Q3/5

Rectangle ABCD has A(1, 1), B(7, 1), C(7, 4), D(1, 4). Give the center of every rotation that carries it onto itself (angle greater than 0° and at most 360°), and the equation of every line of reflection that carries it onto itself.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-M-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.A.3 • Set 5 • MAIN • Q4/5

Trapezoid EFGH has exactly one pair of parallel sides, EF ∥ HG, with EF = 4, HG = 8, and legs FG = 5 and EH = 3. Which rotations (angle greater than 0° and at most 360°, about any point) and which reflections carry it onto itself?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-M-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.A.3 • Set 5 • MAIN • Q5/5

SET 5 • HSG-CO.A.3

BACKWARD 1

Identify lines of symmetry of figures • Grade 4 • Geometry • 4.G.A.3

How many lines of symmetry does a square have?

Number: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-B1-Q1

Grade 4 • Geometry • 4.G.A.3 • Set 5 • BACKWARD 1 • Q1/5

How many lines of symmetry does a rectangle that is not a square have?

Number: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-B1-Q2

Grade 4 • Geometry • 4.G.A.3 • Set 5 • BACKWARD 1 • Q2/5

True or false?

Each diagonal of a rectangle that is not a square is a line of symmetry of the rectangle.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-B1-Q3

Grade 4 • Geometry • 4.G.A.3 • Set 5 • BACKWARD 1 • Q3/5

Which figure has exactly one line of symmetry?

A. An isosceles triangle that is not equilateral

B. A square

C. A circle

D. A scalene triangle

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-B1-Q4

Grade 4 • Geometry • 4.G.A.3 • Set 5 • BACKWARD 1 • Q4/5

True or false?

An isosceles trapezoid has a line of symmetry.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-B1-Q5

Grade 4 • Geometry • 4.G.A.3 • Set 5 • BACKWARD 1 • Q5/5

SET 5 • HSG-CO.A.3

BACKWARD 2

Know the side and angle properties of parallelograms, rectangles, and trapezoids • Grade 5 • Geometry • 5.G.B.3

True or false?

In every parallelogram, the diagonals have equal lengths.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-B2-Q1

Grade 5 • Geometry • 5.G.B.3 • Set 5 • BACKWARD 2 • Q1/5

True or false?

Every rectangle is a parallelogram.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-B2-Q2

Grade 5 • Geometry • 5.G.B.3 • Set 5 • BACKWARD 2 • Q2/5

Which property do all rectangles have?

A. Diagonals that are perpendicular

B. Four sides of equal length

C. Four right angles

D. Exactly one pair of parallel sides

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-B2-Q3

Grade 5 • Geometry • 5.G.B.3 • Set 5 • BACKWARD 2 • Q3/5

A parallelogram has one angle of 70°. Find its other three angles.

Angles: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-B2-Q4

Grade 5 • Geometry • 5.G.B.3 • Set 5 • BACKWARD 2 • Q4/5

An isosceles trapezoid has a 65° angle at one end of its longer base. What is the angle at the other end of that base?

Angle: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-B2-Q5

Grade 5 • Geometry • 5.G.B.3 • Set 5 • BACKWARD 2 • Q5/5

SET 5 • HSG-CO.A.3

BACKWARD 3

Rotate points by 90° and 180° about the origin • Grade 8 • Geometry • 8.G.A.3

Rotate the point (3, 1) by 180° about the origin.

Image: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-B3-Q1

Grade 8 • Geometry • 8.G.A.3 • Set 5 • BACKWARD 3 • Q1/5

Rotate the point (2, 0) by 90° counterclockwise about the origin.

Image: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-B3-Q2

Grade 8 • Geometry • 8.G.A.3 • Set 5 • BACKWARD 3 • Q2/5

True or false?

A 180° rotation about the origin sends (x, y) to (−x, −y).

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-B3-Q3

Grade 8 • Geometry • 8.G.A.3 • Set 5 • BACKWARD 3 • Q3/5

Rotate the point $(-4, 2)$ by 90° counterclockwise about the origin.

A. $(4, -2)$

B. $(-4, -2)$

C. $(2, 4)$

D. $(-2, -4)$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-B3-Q4

Grade 8 • Geometry • 8.G.A.3 • Set 5 • BACKWARD 3 • Q4/5

True or false?

A rotation of 360° about a point moves every point to a new position.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-B3-Q5

Grade 8 • Geometry • 8.G.A.3 • Set 5 • BACKWARD 3 • Q5/5

SET 5 • HSG-CO.A.3

FORWARD 1

Use a symmetry to deduce congruent parts of the figure • High School — Geometry • Geometry: Congruence • HSG-CO.B.6

Isosceles trapezoid ABCD (AB ∥ DC) is carried onto itself by the reflection across the line through the midpoints of AB and DC. Which vertex does A go to? Use this to name a pair of congruent angles.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-F1-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.B.6 • Set 5 • FORWARD 1 • Q1/5

A 180° rotation about point O carries parallelogram PQRS onto itself, sending P to R and Q to S. Name two pairs of congruent sides that follow.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-F1-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.B.6 • Set 5 • FORWARD 1 • Q2/5

True or false?

Because a reflection carries an isosceles trapezoid onto itself, its two diagonals are congruent.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-F1-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.B.6 • Set 5 • FORWARD 1 • Q3/5

A reflection carries rectangle ABCD onto itself and swaps vertices A and B (and C and D). Which equality follows?

A. AB = BC

B. AC = AB

C. AD = AB

D. AD = BC

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-F1-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.B.6 • Set 5 • FORWARD 1 • Q4/5

True or false?

If a 180° rotation about a point carries a quadrilateral onto itself, the quadrilateral must be a rectangle.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-F1-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.B.6 • Set 5 • FORWARD 1 • Q5/5

SET 5 • HSG-CO.A.3

FORWARD 2

Combine symmetries and identify a figure from its symmetries • High School — Geometry • Geometry: Congruence • HSG-CO.A.5

Rectangle ABCD (not a square) has center O. Reflect it across the line through the midpoints of AB and CD, then across the line through the midpoints of BC and DA. Which single transformation has the same result?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-F2-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.A.5 • Set 5 • FORWARD 2 • Q1/5

A quadrilateral is carried onto itself by exactly two transformations: the 360° rotation and one reflection across a line through two of its vertices. Which could it be?

A. A parallelogram that is not a rectangle or a rhombus

B. A square

C. A rectangle that is not a square

D. A kite that is not a rhombus

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-F2-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.A.5 • Set 5 • FORWARD 2 • Q2/5

True or false?

A quadrilateral that is carried onto itself by a 180° rotation about a point must be a rectangle.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-F2-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.A.5 • Set 5 • FORWARD 2 • Q3/5

A quadrilateral is carried onto itself by a 180° rotation but by no reflection. What is the most specific name for it?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-F2-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.A.5 • Set 5 • FORWARD 2 • Q4/5

True or false?

Doing the same reflection twice gives the same result as the 360° rotation (the identity).

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-F2-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.A.5 • Set 5 • FORWARD 2 • Q5/5

Rotations and reflections that carry a regular polygon onto itself

MAIN	Geometry main question	**HSG-CO.A.3**
BACKWARD 1	Find the angle of a fraction of a full turn	**4.MD.C.5.b**
BACKWARD 2	Recognize regular polygons	**5.G.B.3**
BACKWARD 3	Find lines of symmetry of regular polygons	**4.G.A.3**
FORWARD 1	Work backward from a symmetry to the regular polygon	**HSG-CO.A.3**
FORWARD 2	Combine symmetries of a regular polygon	**HSG-CO.A.5**

SET 6 • HSG-CO.A.3

MAIN

Geometry main question • High School — Geometry • Geometry: Congruence • HSG-CO.A.3

Describe every rotation about its center, with angle greater than 0° and at most 360°, that carries a regular pentagon onto itself.

Rotations: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-M-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.A.3 • Set 6 • MAIN • Q1/5

How many reflections carry a regular hexagon onto itself? Describe their lines.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-M-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.A.3 • Set 6 • MAIN • Q2/5

Describe every rotation about its center, with angle greater than 0° and at most 360°, that carries a regular octagon onto itself.

Rotations: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-M-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.A.3 • Set 6 • MAIN • Q3/5

How many reflections carry a regular nonagon (9 sides) onto itself? Describe their lines.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-M-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.A.3 • Set 6 • MAIN • Q4/5

Describe every reflection that carries an equilateral triangle onto itself.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-M-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.A.3 • Set 6 • MAIN • Q5/5

SET 6 • HSG-CO.A.3

BACKWARD 1

Find the angle of a fraction of a full turn • Grade 4 • Measurement & Data • 4.MD.C.5.b

How many degrees are in $\frac{1}{5}$ of a full turn?

Degrees: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-B1-Q1

Grade 4 • Measurement & Data • 4.MD.C.5.b • Set 6 • BACKWARD 1 • Q1/5

How many degrees are in $\frac{1}{8}$ of a full turn?

Degrees: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-B1-Q2

Grade 4 • Measurement & Data • 4.MD.C.5.b • Set 6 • BACKWARD 1 • Q2/5

How many degrees are in $\frac{3}{6}$ of a full turn?

Degrees: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-B1-Q3

Grade 4 • Measurement & Data • 4.MD.C.5.b • Set 6 • BACKWARD 1 • Q3/5

True or false?

$\frac{1}{12}$ of a full turn is 30°.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-B1-Q4

Grade 4 • Measurement & Data • 4.MD.C.5.b • Set 6 • BACKWARD 1 • Q4/5

True or false?

$\frac{1}{9}$ of a full turn is 36°.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-B1-Q5

Grade 4 • Measurement & Data • 4.MD.C.5.b • Set 6 • BACKWARD 1 • Q5/5

SET 6 • HSG-CO.A.3

BACKWARD 2

Recognize regular polygons • Grade 5 • Geometry • 5.G.B.3

Which is a regular polygon?

A. A right triangle

B. A rhombus that is not a square

C. A rectangle that is not a square

D. An equilateral triangle

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-B2-Q1

Grade 5 • Geometry • 5.G.B.3 • Set 6 • BACKWARD 2 • Q1/5

True or false?

A polygon with all sides equal must be regular.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-B2-Q2

Grade 5 • Geometry • 5.G.B.3 • Set 6 • BACKWARD 2 • Q2/5

How many sides does a regular decagon have?

Sides: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-B2-Q3

Grade 5 • Geometry • 5.G.B.3 • Set 6 • BACKWARD 2 • Q3/5

What is the name of the regular polygon with 4 sides?

Name: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-B2-Q4

Grade 5 • Geometry • 5.G.B.3 • Set 6 • BACKWARD 2 • Q4/5

True or false?

A regular hexagon has 6 equal angles.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-B2-Q5

Grade 5 • Geometry • 5.G.B.3 • Set 6 • BACKWARD 2 • Q5/5

SET 6 • HSG-CO.A.3

BACKWARD 3

Find lines of symmetry of regular polygons • Grade 4 • Geometry • 4.G.A.3

How many lines of symmetry does an equilateral triangle have?

Number: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-B3-Q1

Grade 4 • Geometry • 4.G.A.3 • Set 6 • BACKWARD 3 • Q1/5

How many lines of symmetry does a regular pentagon have?

Number: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-B3-Q2

Grade 4 • Geometry • 4.G.A.3 • Set 6 • BACKWARD 3 • Q2/5

True or false?

Every line of symmetry of a regular polygon passes through its center.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-B3-Q3

Grade 4 • Geometry • 4.G.A.3 • Set 6 • BACKWARD 3 • Q3/5

A line of symmetry of a square passes through

A. two adjacent vertices

B. exactly one vertex and no midpoint

C. no point of the square

D. two opposite vertices, or the midpoints of two opposite sides

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-B3-Q4

Grade 4 • Geometry • 4.G.A.3 • Set 6 • BACKWARD 3 • Q4/5

True or false?

A regular hexagon has exactly 3 lines of symmetry.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-B3-Q5

Grade 4 • Geometry • 4.G.A.3 • Set 6 • BACKWARD 3 • Q5/5

SET 6 • HSG-CO.A.3

FORWARD 1

Work backward from a symmetry to the regular polygon • High School — Geometry • Geometry: Congruence • HSG-CO.A.3

The smallest positive rotation that carries a regular polygon onto itself is 24°. How many sides does it have?

Sides: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-F1-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.A.3 • Set 6 • FORWARD 1 • Q1/5

A regular polygon has 20 lines of symmetry. What is the smallest positive rotation that carries it onto itself?

Rotation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-F1-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.A.3 • Set 6 • FORWARD 1 • Q2/5

True or false?

A regular polygon can have 50° as its smallest positive rotation onto itself.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-F1-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.A.3 • Set 6 • FORWARD 1 • Q3/5

Which can be the smallest positive rotation that carries a regular polygon onto itself?

A. 35°

B. 70°

C. 80°

D. 40°

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-F1-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.A.3 • Set 6 • FORWARD 1 • Q4/5

Counting the 360° rotation, how many rotations and reflections in all carry a regular 12-gon onto itself?

Number: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-F1-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.A.3 • Set 6 • FORWARD 1 • Q5/5

SET 6 • HSG-CO.A.3

FORWARD 2

Combine symmetries of a regular polygon • High School — Geometry • Geometry: Congruence • HSG-CO.A.5

Regular hexagon ABCDEF is labeled counterclockwise. It is rotated 120° counterclockwise about its center. Which vertex position does A move to?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-F2-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.A.5 • Set 6 • FORWARD 2 • Q1/5

Regular hexagon ABCDEF is labeled counterclockwise. Rotate it 60° counterclockwise about its center, then reflect it across line AD (through the original positions of A and D). Which original vertex position does vertex B end at?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-F2-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.A.5 • Set 6 • FORWARD 2 • Q2/5

True or false?

For a regular pentagon, a reflection followed by a different reflection that carries it onto itself is a rotation that carries it onto itself.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-F2-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.A.5 • Set 6 • FORWARD 2 • Q3/5

Square ABCD is labeled counterclockwise. Reflect it across line AC, then rotate it 90° counterclockwise about its center. Which original vertex position does vertex B end at?

A. The position of C

B. The position of A

C. The position of D

D. The position of B

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-F2-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.A.5 • Set 6 • FORWARD 2 • Q4/5

Two successive 72° rotations of a regular pentagon about its center equal which single transformation?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-F2-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.A.5 • Set 6 • FORWARD 2 • Q5/5

Definition of a reflection

MAIN	Geometry main question	**HSG-CO.A.4**
BACKWARD 1	Find the midpoint of a horizontal or vertical segment	**6.NS.C.8**
BACKWARD 2	Recognize perpendicular lines on a grid	**4.G.A.1**
BACKWARD 3	Reflect points with coordinate rules	**8.G.A.3**
FORWARD 1	Find lines of reflection and images in slanted, non-axis cases	**HSG-CO.A.4**
FORWARD 2	Prove from the definition that reflections keep lengths	**HSG-CO.A.4**

SET 7 • HSG-CO.A.4

MAIN

Geometry main question • High School — Geometry • Geometry: Congruence • HSG-CO.A.4

Complete a precise definition. A reflection across line ℓ maps each point P not on ℓ to the point P′ such that _____. Each point on ℓ maps to _____.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-M-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.A.4 • Set 7 • MAIN • Q1/5

Noa writes: "A reflection across ℓ flips each point over ℓ."

Explain what is imprecise, and write a precise definition.

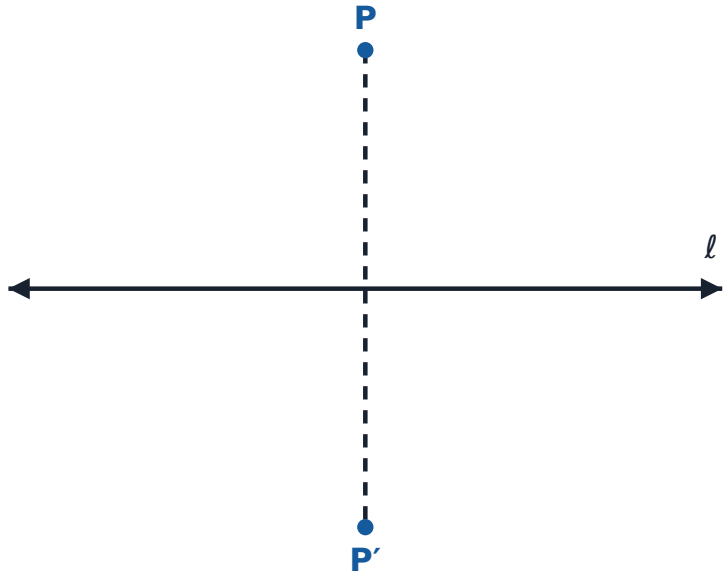
Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-M-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.A.4 • Set 7 • MAIN • Q2/5

Point P′ is the image of point P under a reflection across line ℓ, and P is not on ℓ. Which must be true?



A. ℓ is perpendicular to PP′ and passes through its midpoint

B. ℓ passes through P

C. PP′ is parallel to ℓ

D. P and P′ are on the same side of ℓ

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-M-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.A.4 • Set 7 • MAIN • Q3/5

Line ℓ is y = 1. Use the definition to check that the reflection across ℓ maps P(2, 5) to P′(2, −3): test both conditions.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-M-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.A.4 • Set 7 • MAIN • Q4/5

A reflection maps P(1, 2) to P′(5, 2). Use the definition to find the line of reflection.

Line: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-M-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.A.4 • Set 7 • MAIN • Q5/5

SET 7 • HSG-CO.A.4

BACKWARD 1

Find the midpoint of a horizontal or vertical segment • Grade 6 • The Number System • 6.NS.C.8

Find the midpoint of the segment from (2, 4) to (8, 4).

Midpoint: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-B1-Q1

Grade 6 • The Number System • 6.NS.C.8 • Set 7 • BACKWARD 1 • Q1/5

Find the midpoint of the segment from (1, 1) to (1, 9).

Midpoint: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-B1-Q2

Grade 6 • The Number System • 6.NS.C.8 • Set 7 • BACKWARD 1 • Q2/5

On a number line, what number is halfway between −3 and 7?

Number: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-B1-Q3

Grade 6 • The Number System • 6.NS.C.8 • Set 7 • BACKWARD 1 • Q3/5

True or false?

The midpoint of the segment from (−4, 0) to (6, 0) is (1, 0).

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-B1-Q4

Grade 6 • The Number System • 6.NS.C.8 • Set 7 • BACKWARD 1 • Q4/5

True or false?

The midpoint of a segment is closer to one endpoint than to the other.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-B1-Q5

Grade 6 • The Number System • 6.NS.C.8 • Set 7 • BACKWARD 1 • Q5/5

SET 7 • HSG-CO.A.4

BACKWARD 2

Recognize perpendicular lines on a grid • Grade 4 • Geometry • 4.G.A.1

True or false?

A horizontal line and a vertical line are perpendicular.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-B2-Q1

Grade 4 • Geometry • 4.G.A.1 • Set 7 • BACKWARD 2 • Q1/5

True or false?

Any two segments of the same length lie on parallel lines.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-B2-Q2

Grade 4 • Geometry • 4.G.A.1 • Set 7 • BACKWARD 2 • Q2/5

Which pair of segments lie on perpendicular lines?

A. From (0, 0) to (4, 0) and from (2, −1) to (2, 3)

B. From (1, 1) to (2, 3) and from (1, 1) to (3, 2)

C. From (0, 0) to (3, 1) and from (0, 2) to (3, 3)

D. From (0, 0) to (3, 1) and from (0, 0) to (1, 3)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-B2-Q3

Grade 4 • Geometry • 4.G.A.1 • Set 7 • BACKWARD 2 • Q3/5

A segment goes from (2, 1) to (2, 6). Describe a segment through (5, 3) whose line is perpendicular to the line containing the first segment.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-B2-Q4

Grade 4 • Geometry • 4.G.A.1 • Set 7 • BACKWARD 2 • Q4/5

Which describes two lines that are both perpendicular to the x-axis?

A. They are parallel or the same line

B. They are perpendicular to each other

C. They meet at the origin

D. They always meet

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-B2-Q5

Grade 4 • Geometry • 4.G.A.1 • Set 7 • BACKWARD 2 • Q5/5

SET 7 • HSG-CO.A.4

BACKWARD 3

Reflect points with coordinate rules • Grade 8 • Geometry • 8.G.A.3

Reflect the point (4, −3) across the x-axis.

Image: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-B3-Q1

Grade 8 • Geometry • 8.G.A.3 • Set 7 • BACKWARD 3 • Q1/5

Reflect the point $(-1, 5)$ across the y-axis.

Image: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-B3-Q2

Grade 8 • Geometry • 8.G.A.3 • Set 7 • BACKWARD 3 • Q2/5

Reflect the point (2, 7) across the line y = x.

A. (7, 2)

B. (−7, −2)

C. (−2, 7)

D. (2, −7)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-B3-Q3

Grade 8 • Geometry • 8.G.A.3 • Set 7 • BACKWARD 3 • Q3/5

True or false?

Reflecting the point $(-6, 1)$ across the y-axis gives $(6, 1)$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-B3-Q4

Grade 8 • Geometry • 8.G.A.3 • Set 7 • BACKWARD 3 • Q4/5

True or false?

Reflecting (3, 3) across the x-axis gives (−3, 3).

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-B3-Q5

Grade 8 • Geometry • 8.G.A.3 • Set 7 • BACKWARD 3 • Q5/5

SET 7 • HSG-CO.A.4

FORWARD 1

Find lines of reflection and images in slanted, non-axis cases • High School — Geometry • Geometry: Congruence • HSG-CO.A.4

A reflection maps P(0, 0) to P′(4, 2). Find an equation of the line of reflection.

Equation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-F1-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.A.4 • Set 7 • FORWARD 1 • Q1/5

A reflection maps $A(-2, 3)$ to $A'(4, 3)$. Where does it map $B(0, -1)$?

Image: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-F1-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.A.4 • Set 7 • FORWARD 1 • Q2/5

True or false?

A reflection maps P to P′ and Q to Q′, where P and Q are not on the line of reflection. Then PP′ is parallel to QQ′ (or on the same line).

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-F1-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.A.4 • Set 7 • FORWARD 1 • Q3/5

A reflection maps A(1, 3) to A′(3, 1). Is the line of reflection y = x? Use the definition to decide.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-F1-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.A.4 • Set 7 • FORWARD 1 • Q4/5

A reflection maps P(−1, 4) to P′(5, 4). Which point is on the line of reflection?

A. (−1, 0)

B. (5, 0)

C. (2, −3)

D. (4, 2)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-F1-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.A.4 • Set 7 • FORWARD 1 • Q5/5

SET 7 • HSG-CO.A.4

FORWARD 2

Prove from the definition that reflections keep lengths • High School — Geometry • Geometry: Congruence • HSG-CO.A.4

Points P and Q are on the same side of line ℓ, and line PQ is not perpendicular to ℓ. A reflection across ℓ maps them to P′ and Q′. Let PP′ meet ℓ at M and QQ′ meet ℓ at N. Prove that PQ = P′Q′.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-F2-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.A.4 • Set 7 • FORWARD 2 • Q1/5

Point P is on line ℓ and point Q is not. The reflection across ℓ maps Q to Q′. Prove that PQ = PQ′.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-F2-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.A.4 • Set 7 • FORWARD 2 • Q2/5

Using coordinates with ℓ as the x-axis, the reflection is $(x, y) \to (x, -y)$. Prove that it keeps the distance between any two points (x_1, y_1) and (x_2, y_2).

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-F2-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.A.4 • Set 7 • FORWARD 2 • Q3/5

True or false?

Under a reflection, a point not on the line of reflection and its image are on the same side of the line.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-F2-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.A.4 • Set 7 • FORWARD 2 • Q4/5

In a proof that a reflection across ℓ keeps the length of segment PQ, which fact is used first?

A. ℓ is the perpendicular bisector of PP′ and of QQ′

B. ℓ bisects $\angle PQP'$

C. ℓ is parallel to PQ

D. ℓ passes through P

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-F2-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.A.4 • Set 7 • FORWARD 2 • Q5/5

Definition of a translation

MAIN	Geometry main question	**HSG-CO.A.4**
BACKWARD 1	Recognize parallel segments on a grid	**4.G.A.1**
BACKWARD 2	Translate points with coordinate rules	**8.G.A.3**
FORWARD 1	Use the definition with points and their images	**HSG-CO.A.4**
FORWARD 2	Prove from the definition that translations keep lengths	**HSG-CO.A.4**

SET 100 • HSG-CO.A.4

MAIN

Geometry main question • High School — Geometry • Geometry: Congruence • HSG-CO.A.4

Points A and B are different. Write a precise definition of the translation that maps A to B.

Definition: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S100-M-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.A.4 • Set 100 • MAIN • Q1/5

Tia writes: "A translation slides a figure."

Explain what is imprecise, and write a precise definition of the translation that maps A to B.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S100-M-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.A.4 • Set 100 • MAIN • Q2/5

The translation that maps A(1, 2) to B(4, 6) maps P(−2, 0) to P′. Use the definition (same length, parallel, same direction) to find P′, and check the length.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S100-M-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.A.4 • Set 100 • MAIN • Q3/5

A translation maps A to A′ and B to B′. Explain, using the definition, why AA′ and BB′ are parallel (or on one line) and have the same length.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S100-M-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.A.4 • Set 100 • MAIN • Q4/5

Is $(x, y) \to (x + 3, y - 4)$ a translation? Use the definition to explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S100-M-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.A.4 • Set 100 • MAIN • Q5/5

SET 100 • HSG-CO.A.4

BACKWARD 1

Recognize parallel segments on a grid • Grade 4 • Geometry • 4.G.A.1

True or false?

Two segments that each go 3 right and 1 up lie on parallel lines (or on one line).

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S100-B1-Q1

Grade 4 • Geometry • 4.G.A.1 • Set 100 • BACKWARD 1 • Q1/5

True or false?

Any two segments of the same length are parallel.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S100-B1-Q2

Grade 4 • Geometry • 4.G.A.1 • Set 100 • BACKWARD 1 • Q2/5

Which pair of segments is parallel?

A. From (0, 0) to (3, 1) and from (0, 2) to (3, 3)

B. From (0, 0) to (4, 0) and from (0, 0) to (0, 4)

C. From (1, 1) to (2, 3) and from (1, 1) to (3, 2)

D. From (0, 0) to (3, 1) and from (0, 0) to (1, 3)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S100-B1-Q3

Grade 4 • Geometry • 4.G.A.1 • Set 100 • BACKWARD 1 • Q3/5

A segment goes from (1, 1) to (4, 3). Give the other endpoint of a parallel segment of the same length that starts at (0, 5).

Endpoint: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S100-B1-Q4

Grade 4 • Geometry • 4.G.A.1 • Set 100 • BACKWARD 1 • Q4/5

Which describes two lines that are both perpendicular to the x-axis?

A. They are parallel or the same line

B. They are perpendicular to each other

C. They meet at the origin

D. They always meet

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S100-B1-Q5

Grade 4 • Geometry • 4.G.A.1 • Set 100 • BACKWARD 1 • Q5/5

SET 100 • HSG-CO.A.4

BACKWARD 2

Translate points with coordinate rules • Grade 8 • Geometry • 8.G.A.3

Translate the point $(-1, 5)$ 6 units right and 2 units down.

Image: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S100-B2-Q1

Grade 8 • Geometry • 8.G.A.3 • Set 100 • BACKWARD 2 • Q1/5

Translate the point (3, −2) 4 units left.

Image: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S100-B2-Q2

Grade 8 • Geometry • 8.G.A.3 • Set 100 • BACKWARD 2 • Q2/5

A translation moves (0, 0) to (2, −3). Where does it move (1, 1)?

A. (3, 4)

B. (2, −3)

C. (−1, 4)

D. (3, −2)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S100-B2-Q3

Grade 8 • Geometry • 8.G.A.3 • Set 100 • BACKWARD 2 • Q3/5

True or false?

Translating (3, 3) by 2 units left gives (1, 3).

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S100-B2-Q4

Grade 8 • Geometry • 8.G.A.3 • Set 100 • BACKWARD 2 • Q4/5

True or false?

Translating (3, 3) by 2 units left gives (3, 1).

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S100-B2-Q5

Grade 8 • Geometry • 8.G.A.3 • Set 100 • BACKWARD 2 • Q5/5

SET 100 • HSG-CO.A.4

FORWARD 1

Use the definition with points and their images • High School — Geometry • Geometry: Congruence • HSG-CO.A.4

A translation maps $A(2, -1)$ to $A'(-3, 4)$. Where does it map $B(0, 6)$?

Image: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S100-F1-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.A.4 • Set 100 • FORWARD 1 • Q1/5

A translation maps P to P′(4, 1) and moves every point 2 left and 3 up. Find P.

P = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S100-F1-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.A.4 • Set 100 • FORWARD 1 • Q2/5

True or false?

Under a translation, all the segments that join points to their images are parallel (or on one line) and have the same length.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S100-F1-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.A.4 • Set 100 • FORWARD 1 • Q3/5

A translation maps triangle ABC with A(0, 0), B(2, 0), C(0, 3) so that C goes to (5, 7). Find the images of A and B.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S100-F1-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.A.4 • Set 100 • FORWARD 1 • Q4/5

A translation maps (1, 2) to (4, 0). Which point maps to (0, 0)?

A. (−4, 0)

B. (−3, 2)

C. (3, −2)

D. (1, 2)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S100-F1-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.A.4 • Set 100 • FORWARD 1 • Q5/5

SET 100 • HSG-CO.A.4

FORWARD 2

Prove from the definition that translations keep lengths • High School — Geometry • Geometry: Congruence • HSG-CO.A.4

A translation maps P to P′ and Q to Q′, where Q is not on line PP′. Prove that PQ = P′Q′.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S100-F2-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.A.4 • Set 100 • FORWARD 2 • Q1/5

True or false?

A translation maps P to P′ and Q to Q′. Then PP′Q′Q is always a parallelogram.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S100-F2-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.A.4 • Set 100 • FORWARD 2 • Q2/5

Prove that $T(x, y) = (x + a, y + b)$ maps every nonvertical line $y = mx + k$ to a line with the same slope m (so the image is parallel to the line or is the line itself).

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S100-F2-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.A.4 • Set 100 • FORWARD 2 • Q3/5

A translation maps P to P′ and Q to Q′, where Q IS on line PP′. Prove that PQ = P′Q′ in this case too.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S100-F2-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.A.4 • Set 100 • FORWARD 2 • Q4/5

In the proof that a translation keeps the length of PQ (Q not on line PP′), which figure is used?

A. Rectangle PQRS

B. A circle centered at P

C. Parallelogram PP′Q′Q

D. Triangle PQQ′ only

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S100-F2-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.A.4 • Set 100 • FORWARD 2 • Q5/5

Definition of a rotation

MAIN	Geometry main question	**HSG-CO.A.4**
BACKWARD 1	Describe turns in degrees, clockwise and counterclockwise	**4.MD.C.5.b**
BACKWARD 2	Points on a circle are the same distance from its center	**7.G.B.4**
BACKWARD 3	Rotate points 90°, 180°, and 270° about the origin	**8.G.A.3**
FORWARD 1	Rotate about a center that is not the origin	**HSG-CO.A.4**
FORWARD 2	Find the center and angle of a rotation from points and their imag..	**HSG-CO.A.4**

SET 8 • HSG-CO.A.4

MAIN

Geometry main question • High School — Geometry • Geometry: Congruence • HSG-CO.A.4

Complete a precise definition. A counterclockwise rotation about point O through angle θ maps each point P ≠ O to the point P′ such that _____ and _____. Point O maps to _____.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-M-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.A.4 • Set 8 • MAIN • Q1/5

A rotation about point O maps point P to P′ (P ≠ P′). Which must be true?

A. O is the midpoint of PP′

B. O is on the perpendicular bisector of PP′

C. OP′ = 2 · OP

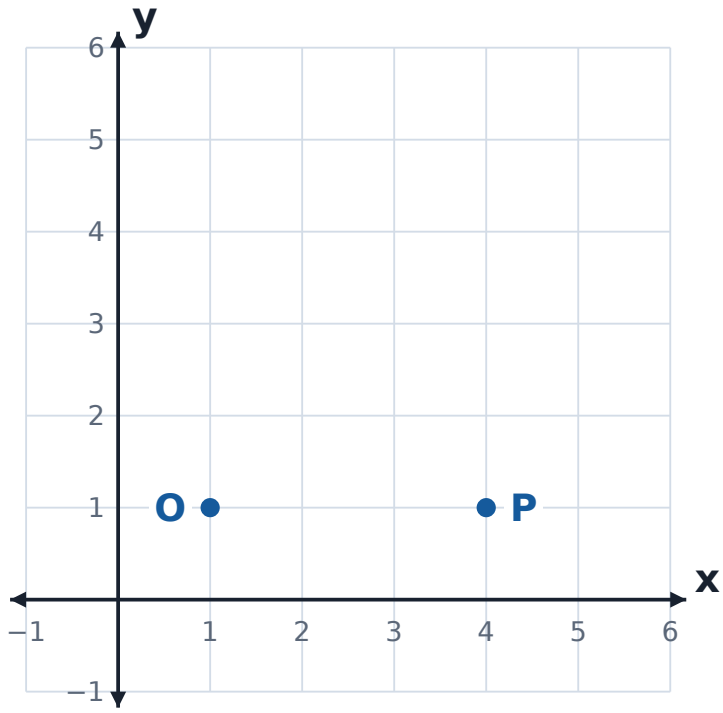
D. PP′ is perpendicular to OP

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-M-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.A.4 • Set 8 • MAIN • Q2/5

A rotation of 90° counterclockwise about O(1, 1) maps P(4, 1) to P′. Use the definition to find P′.



P′ = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-M-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.A.4 • Set 8 • MAIN • Q3/5

A rotation about point O through 70° maps a point P ≠ O to P′. Use the definition to explain why P′ lies on the circle with center O through P, and why O is the only point that the rotation leaves where it is.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-M-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.A.4 • Set 8 • MAIN • Q4/5

A rotation maps P(2, 0) to P′(0, 2). Explain why its center must lie on the line y = x.

Explanation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-M-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.A.4 • Set 8 • MAIN • Q5/5

SET 8 • HSG-CO.A.4

BACKWARD 1

Describe turns in degrees, clockwise and counterclockwise • Grade 4 • Measurement & Data • 4.MD.C.5.b

A ray points right. It turns 90° counterclockwise. Which way does it point now?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-B1-Q1

Grade 4 • Measurement & Data • 4.MD.C.5.b • Set 8 • BACKWARD 1 • Q1/5

A ray points up. It turns 180°. Which way does it point now?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-B1-Q2

Grade 4 • Measurement & Data • 4.MD.C.5.b • Set 8 • BACKWARD 1 • Q2/5

True or false?

A clockwise turn of 270° ends in the same direction as a counterclockwise turn of 90°.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-B1-Q3

Grade 4 • Measurement & Data • 4.MD.C.5.b • Set 8 • BACKWARD 1 • Q3/5

A ray points up. It turns 90° clockwise. Which way does it point now?

A. Down

B. Left

C. Right

D. Up

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-B1-Q4

Grade 4 • Measurement & Data • 4.MD.C.5.b • Set 8 • BACKWARD 1 • Q4/5

How many degrees are in $\frac{3}{4}$ of a full turn?

Degrees: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-B1-Q5

Grade 4 • Measurement & Data • 4.MD.C.5.b • Set 8 • BACKWARD 1 • Q5/5

SET 8 • HSG-CO.A.4

BACKWARD 2

Points on a circle are the same distance from its center • Grade 7 • Geometry • 7.G.B.4

A circle has center O and radius 6. Point P is on the circle. What is OP?

OP = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-B2-Q1

Grade 7 • Geometry • 7.G.B.4 • Set 8 • BACKWARD 2 • Q1/5

True or false?

If OA = OB, then A and B lie on one circle with center O.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-B2-Q2

Grade 7 • Geometry • 7.G.B.4 • Set 8 • BACKWARD 2 • Q2/5

Which point is on the circle with center (0, 0) and radius 5?

A. (4, 4)

B. (3, 4)

C. (2, 3)

D. (5, 5)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-B2-Q3

Grade 7 • Geometry • 7.G.B.4 • Set 8 • BACKWARD 2 • Q3/5

A circle has center O and diameter 10 cm. How far is every point of the circle from O?

Distance: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-B2-Q4

Grade 7 • Geometry • 7.G.B.4 • Set 8 • BACKWARD 2 • Q4/5

True or false?

Two points the same distance from O are always the same point.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-B2-Q5

Grade 7 • Geometry • 7.G.B.4 • Set 8 • BACKWARD 2 • Q5/5

SET 8 • HSG-CO.A.4

BACKWARD 3

Rotate points 90°, 180°, and 270° about the origin • Grade 8 • Geometry • 8.G.A.3

Rotate (5, 2) by 90° counterclockwise about the origin.

Image: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-B3-Q1

Grade 8 • Geometry • 8.G.A.3 • Set 8 • BACKWARD 3 • Q1/5

Rotate $(-3, 4)$ by $180°$ about the origin.

Image: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-B3-Q2

Grade 8 • Geometry • 8.G.A.3 • Set 8 • BACKWARD 3 • Q2/5

Rotate (1, 6) by 90° clockwise about the origin.

A. (−6, 1)

B. (1, −6)

C. (−1, −6)

D. (6, −1)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-B3-Q3

Grade 8 • Geometry • 8.G.A.3 • Set 8 • BACKWARD 3 • Q3/5

True or false?

Rotating (2, 3) by 270° counterclockwise about the origin gives (3, −2).

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-B3-Q4

Grade 8 • Geometry • 8.G.A.3 • Set 8 • BACKWARD 3 • Q4/5

True or false?

Rotating (4, 0) by 90° counterclockwise about the origin gives (0, −4).

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-B3-Q5

Grade 8 • Geometry • 8.G.A.3 • Set 8 • BACKWARD 3 • Q5/5

SET 8 • HSG-CO.A.4

FORWARD 1

Rotate about a center that is not the origin • High School — Geometry • Geometry: Congruence • HSG-CO.A.4

Rotate P(5, 4) by 90° clockwise about C(2, 3).

Image: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-F1-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.A.4 • Set 8 • FORWARD 1 • Q1/5

Rotate P(1, 5) by 180° about C(−1, 2).

Image: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-F1-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.A.4 • Set 8 • FORWARD 1 • Q2/5

Rotate P(3, −2) by 90° counterclockwise about C(0, −2).

Image: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-F1-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.A.4 • Set 8 • FORWARD 1 • Q3/5

Rotate P(1, 4) by 270° counterclockwise about C(1, 1).

A. (4, 4)

B. (4, 1)

C. (1, −2)

D. (−2, 1)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-F1-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.A.4 • Set 8 • FORWARD 1 • Q4/5

True or false?

Rotating P(4, 0) by 90° counterclockwise about C(2, 0) gives the same image as rotating P by 90° counterclockwise about the origin.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-F1-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.A.4 • Set 8 • FORWARD 1 • Q5/5

SET 8 • HSG-CO.A.4

FORWARD 2

Find the center and angle of a rotation from points and their images • High School — Geometry • Geometry: Congruence • HSG-CO.A.4

A rotation maps B(1, 0) to B′(0, 1) and A(3, 1) to A′(−1, 3). The center is where the perpendicular bisectors of BB′ and AA′ meet. Find the center and the angle of rotation.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-F2-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.A.4 • Set 8 • FORWARD 2 • Q1/5

A rotation maps P(4, 1) to P′(2, 3) and Q(4, 3) to Q′(0, 3). Find the center and the angle of rotation.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-F2-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.A.4 • Set 8 • FORWARD 2 • Q2/5

True or false?

The center of a nonzero rotation moves to a new position under the rotation.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-F2-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.A.4 • Set 8 • FORWARD 2 • Q3/5

A rotation about C maps P to P′. $CP = 5$ and $PP' = 5\sqrt{2}$. Which could be the angle of rotation?

A. 60°

B. 90°

C. 45°

D. 180°

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-F2-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.A.4 • Set 8 • FORWARD 2 • Q4/5

A rotation about the origin maps (5, 0) to (3, 4). What is the counterclockwise angle of rotation, to the nearest degree?

Angle: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-F2-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.A.4 • Set 8 • FORWARD 2 • Q5/5

Draw the image of a figure under a rotation, reflection, or translation

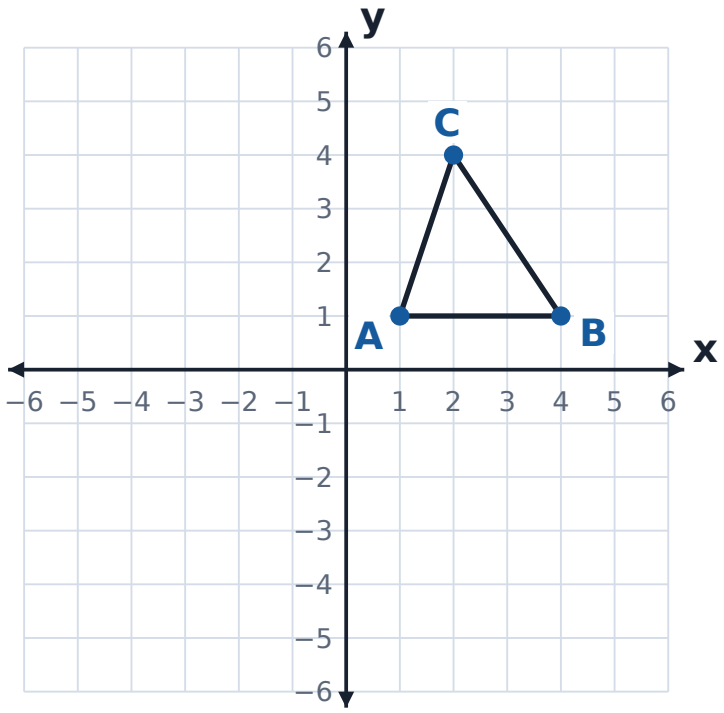
MAIN	Geometry main question	**HSG-CO.A.5**
BACKWARD 1	Plot and read points in all four quadrants	**6.NS.C.6.c**
BACKWARD 2	Find image points with coordinate rules	**8.G.A.3**
BACKWARD 3	Count the distance from a point to a horizontal or vertical line	**6.NS.C.8**
FORWARD 1	Draw the image under a sequence of two transformations	**HSG-CO.A.5**
FORWARD 2	Draw rotations and reflections off the grid with compass, protract…	**HSG-CO.A.5**

SET 9 • HSG-CO.A.5

MAIN

Geometry main question • High School — Geometry • Geometry: Congruence • HSG-CO.A.5

Draw the image of triangle ABC under the translation 5 units left and 2 units down. Label the image A′B′C′.

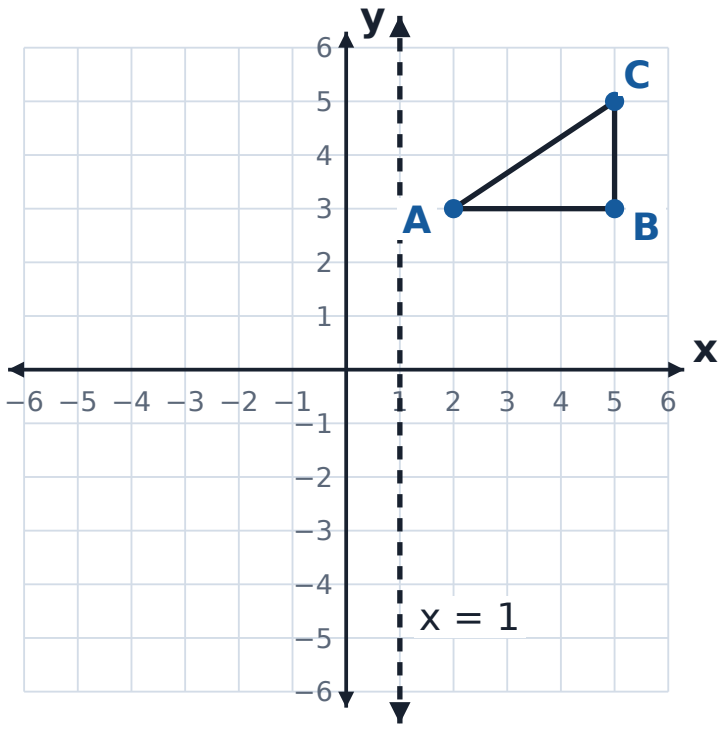


REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-M-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.A.5 • Set 9 • MAIN • Q1/5

Draw the image of triangle ABC under the reflection across the line x = 1. Label the image A′B′C′.

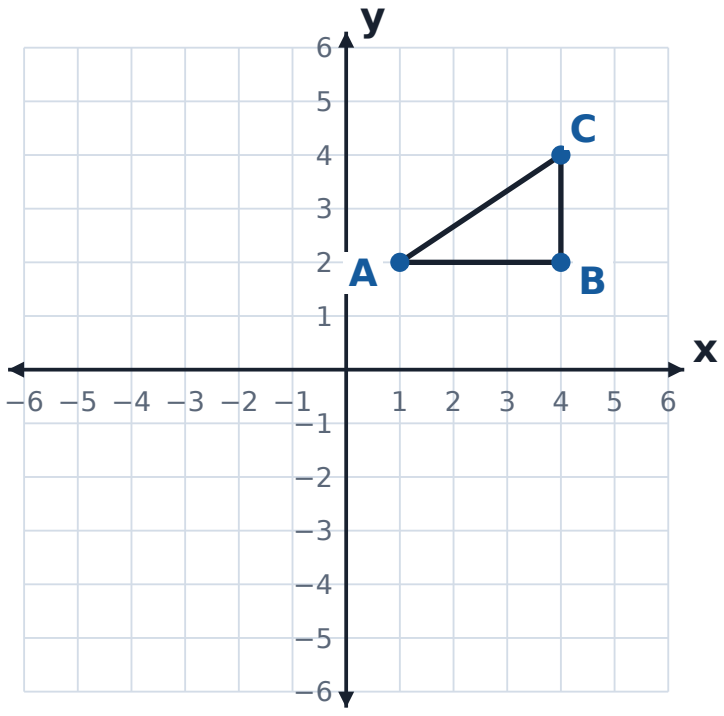


REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-M-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.A.5 • Set 9 • MAIN • Q2/5

Draw the image of triangle ABC under the rotation of 90° counterclockwise about the origin. Label the image A′B′C′.

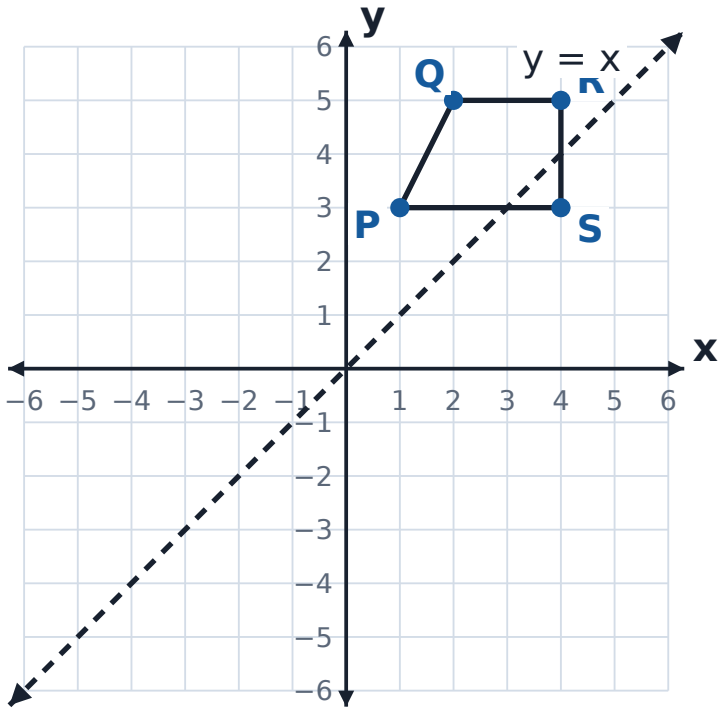


REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-M-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.A.5 • Set 9 • MAIN • Q3/5

Draw the image of quadrilateral PQRS under the reflection across the line $y = x$. Label the image $P'Q'R'S'$.

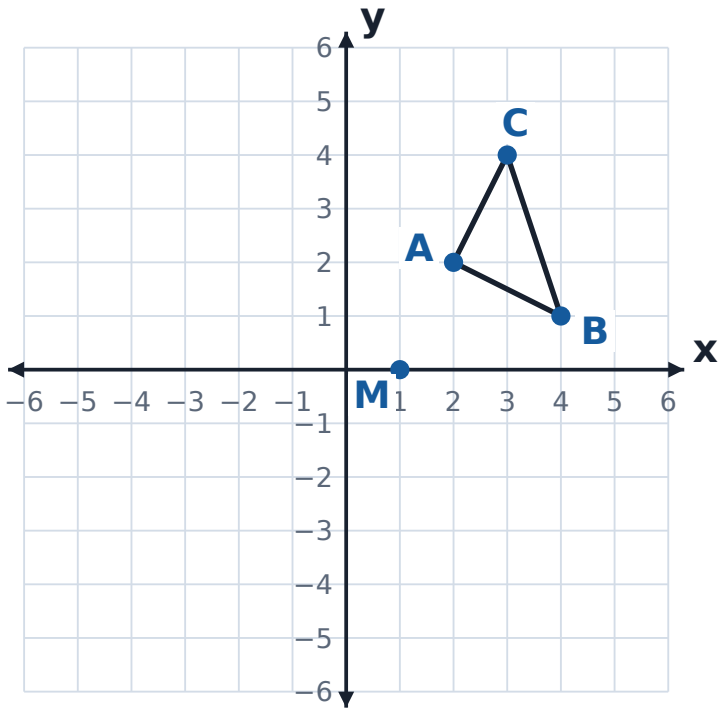


REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-M-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.A.5 • Set 9 • MAIN • Q4/5

Draw the image of triangle ABC under the rotation of 180° about the point M(1, 0). Label the image A′B′C′.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-M-Q5

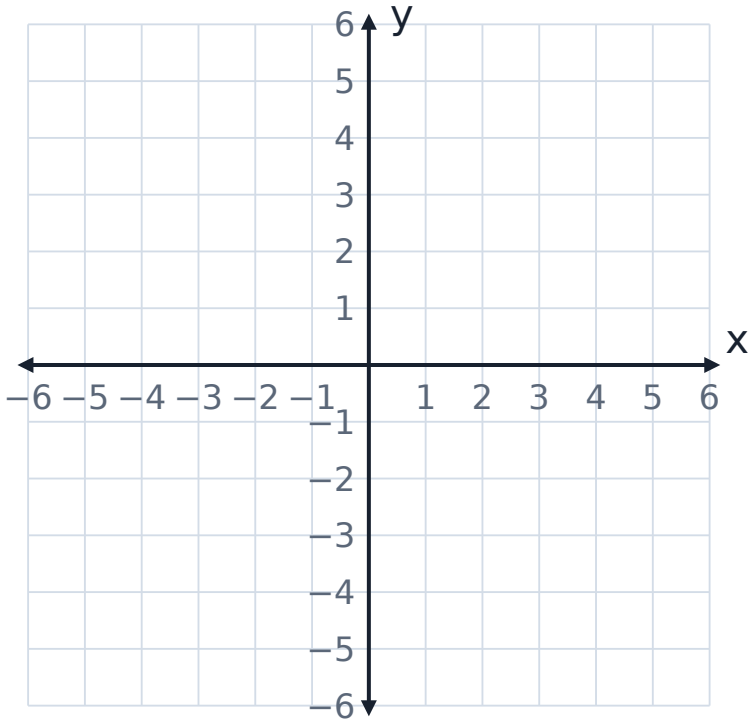
High School — Geometry • Geometry: Congruence • HSG-CO.A.5 • Set 9 • MAIN • Q5/5

SET 9 • HSG-CO.A.5

BACKWARD 1

Plot and read points in all four quadrants • Grade 6 • The Number System • 6.NS.C.6.c

Plot and label the points A(−3, 2), B(0, −4), and C(5, 1).



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-B1-Q1

Grade 6 • The Number System • 6.NS.C.6.c • Set 9 • BACKWARD 1 • Q1/5

A point is 4 units left of the y-axis and 1 unit below the x-axis. What are its coordinates?

Point: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-B1-Q2

Grade 6 • The Number System • 6.NS.C.6.c • Set 9 • BACKWARD 1 • Q2/5

Which point is 2 units above the x-axis and on the y-axis?

A. (2, 2)

B. (0, −2)

C. (0, 2)

D. (2, 0)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-B1-Q3

Grade 6 • The Number System • 6.NS.C.6.c • Set 9 • BACKWARD 1 • Q3/5

True or false?

The point (3, −2) is in Quadrant IV.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-B1-Q4

Grade 6 • The Number System • 6.NS.C.6.c • Set 9 • BACKWARD 1 • Q4/5

True or false?

To plot (2, 5), start at the origin and move 2 units up and 5 units right.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-B1-Q5

Grade 6 • The Number System • 6.NS.C.6.c • Set 9 • BACKWARD 1 • Q5/5

SET 9 • HSG-CO.A.5

BACKWARD 2

Find image points with coordinate rules • Grade 8 • Geometry • 8.G.A.3

Find the image of $(3, -2)$ under a translation 4 units left and 1 unit up.

Image: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-B2-Q1

Grade 8 • Geometry • 8.G.A.3 • Set 9 • BACKWARD 2 • Q1/5

Find the image of $(-2, 5)$ under a reflection across the x-axis.

Image: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-B2-Q2

Grade 8 • Geometry • 8.G.A.3 • Set 9 • BACKWARD 2 • Q2/5

Find the image of (3, 1) under a 90° clockwise rotation about the origin.

A. (−3, −1)

B. (3, −1)

C. (1, −3)

D. (−1, 3)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-B2-Q3

Grade 8 • Geometry • 8.G.A.3 • Set 9 • BACKWARD 2 • Q3/5

True or false?

The image of $(4, -1)$ under a reflection across the y-axis is $(-4, -1)$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-B2-Q4

Grade 8 • Geometry • 8.G.A.3 • Set 9 • BACKWARD 2 • Q4/5

True or false?

The image of (2, 2) under a 180° rotation about the origin is (2, −2).

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-B2-Q5

Grade 8 • Geometry • 8.G.A.3 • Set 9 • BACKWARD 2 • Q5/5

SET 9 • HSG-CO.A.5

BACKWARD 3

Count the distance from a point to a horizontal or vertical line • Grade 6 • The Number System • 6.NS.C.8

How far is the point (5, 3) from the vertical line x = 1?

Distance: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-B3-Q1

Grade 6 • The Number System • 6.NS.C.8 • Set 9 • BACKWARD 3 • Q1/5

How far is the point $(-2, -1)$ from the horizontal line $y = 3$?

Distance: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-B3-Q2

Grade 6 • The Number System • 6.NS.C.8 • Set 9 • BACKWARD 3 • Q2/5

Which point is 3 units from the line x = 2, on its left?

A. (−3, 4)

B. (−1, 4)

C. (2, 1)

D. (5, 4)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-B3-Q3

Grade 6 • The Number System • 6.NS.C.8 • Set 9 • BACKWARD 3 • Q3/5

True or false?

The point (0, 4) is 4 units from the x-axis.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-B3-Q4

Grade 6 • The Number System • 6.NS.C.8 • Set 9 • BACKWARD 3 • Q4/5

True or false?

The point (6, 2) is 6 units from the line $x = 2$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-B3-Q5

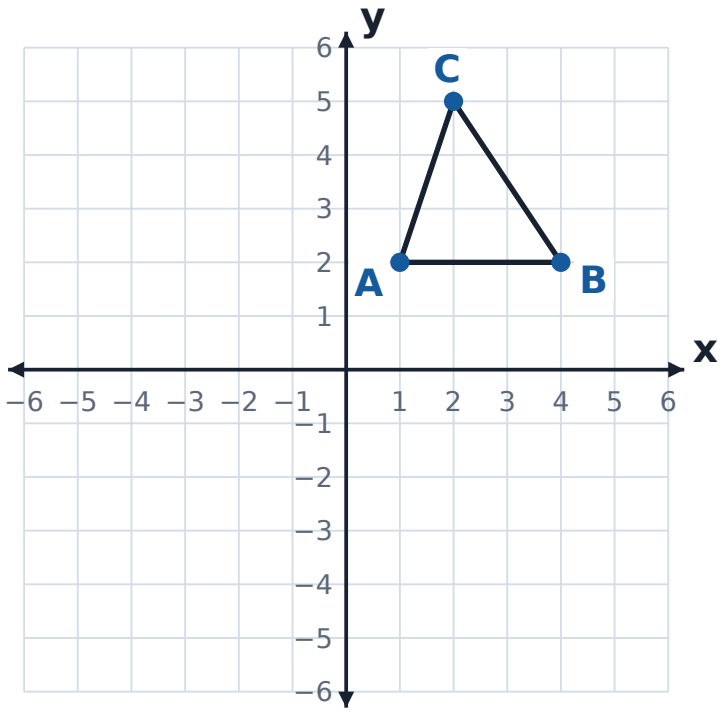
Grade 6 • The Number System • 6.NS.C.8 • Set 9 • BACKWARD 3 • Q5/5

SET 9 • HSG-CO.A.5

FORWARD 1

Draw the image under a sequence of two transformations • High School — Geometry • Geometry: Congruence • HSG-CO.A.5

Reflect triangle ABC across the y-axis, then translate the result 3 units down. Draw and label the final image A″B″C″.

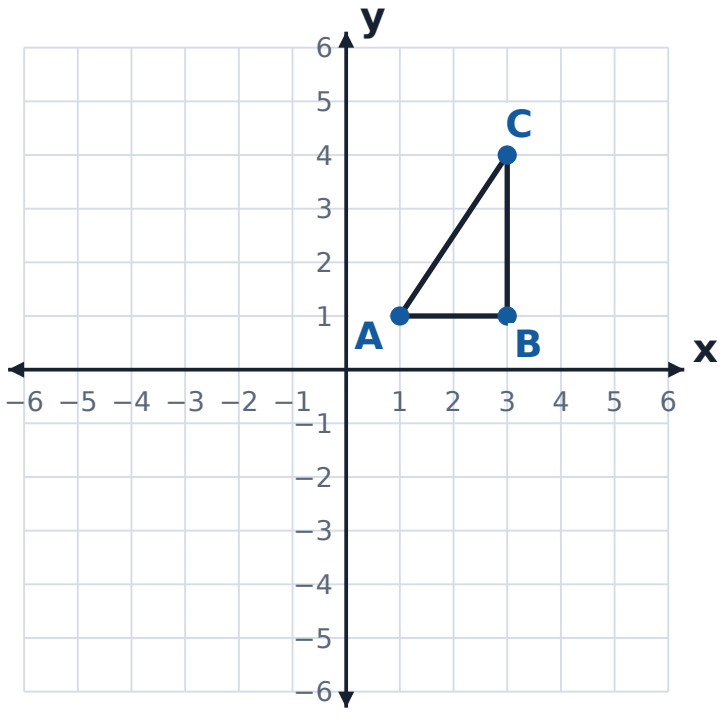


REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-F1-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.A.5 • Set 9 • FORWARD 1 • Q1/5

Rotate triangle ABC by 90° counterclockwise about the origin, then reflect the result across the x-axis. Draw and label the final image A″B″C″.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-F1-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.A.5 • Set 9 • FORWARD 1 • Q2/5

Triangle ABC has A(2, 1). It is translated 2 units right and then rotated 180° about the origin. Where does A end up?

A″ = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-F1-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.A.5 • Set 9 • FORWARD 1 • Q3/5

True or false?

Reflecting a figure across the x-axis and then across the y-axis gives the same image as reflecting it across the line $y = x$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-F1-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.A.5 • Set 9 • FORWARD 1 • Q4/5

Point P(1, 3) is translated 2 units left and then reflected across the line y = x. Where does it end up?

A. (3, 1)

B. (−1, 3)

C. (1, −3)

D. (3, −1)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-F1-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.A.5 • Set 9 • FORWARD 1 • Q5/5

SET 9 • HSG-CO.A.5

FORWARD 2

Draw rotations and reflections off the grid with compass, protractor, and straightedge • High School — Geometry • Geometry: Congruence • HSG-CO.A.5

Use a compass and protractor. Rotate point P 60° counterclockwise about point O. Mark and label the image P′.

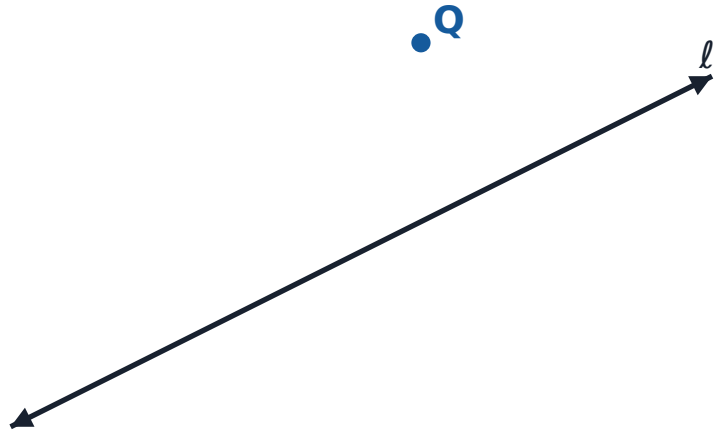


REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-F2-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.A.5 • Set 9 • FORWARD 2 • Q1/5

Use a compass and straightedge. Reflect point Q across line ℓ. Mark and label the image Q′.

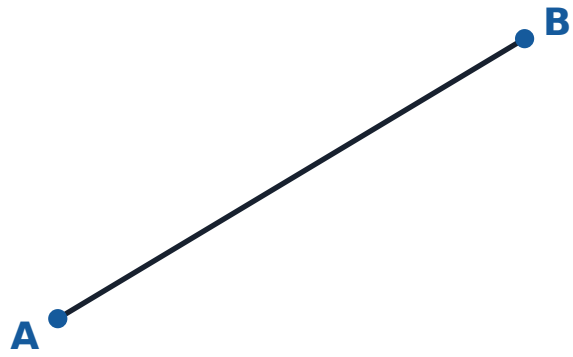


REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-F2-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.A.5 • Set 9 • FORWARD 2 • Q2/5

Use a compass and protractor. Rotate segment AB 90° clockwise about point A. Draw and label the image AB′.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-F2-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.A.5 • Set 9 • FORWARD 2 • Q3/5

Use a compass and straightedge. Rotate point P 180° about point O. Mark and label the image P′.

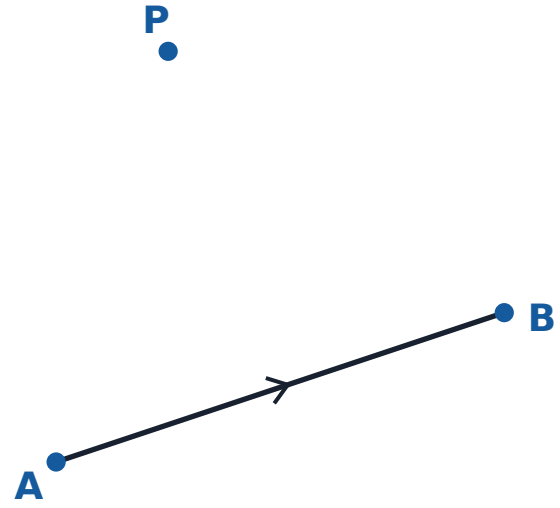


REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-F2-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.A.5 • Set 9 • FORWARD 2 • Q4/5

Use a compass and straightedge. The translation maps A to B. Draw the image P′ of point P (P is not on line AB).



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-F2-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.A.5 • Set 9 • FORWARD 2 • Q5/5

Specify a sequence of transformations that carries one figure onto another

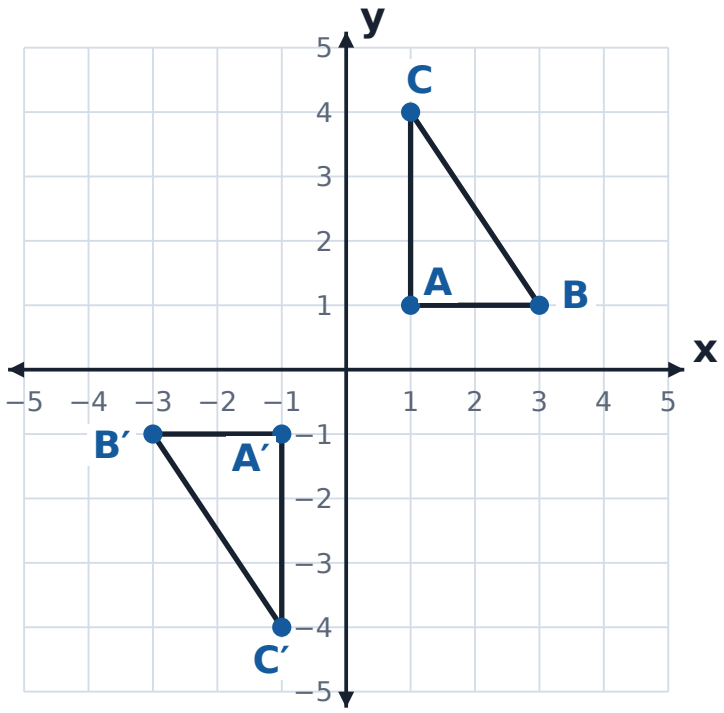
MAIN	Geometry main question	**HSG-CO.A.5**
BACKWARD 1	Understand congruence as the result of one rigid motion	**8.G.A.2**
BACKWARD 2	Find images with coordinate rules for sequences	**8.G.A.3**
BACKWARD 3	Read the orientation (clockwise or counterclockwise order) of a tr…	**8.G.A.1**
FORWARD 1	Use the fewest transformations and justify	**HSG-CO.A.5**
FORWARD 2	Specify a similarity sequence: a dilation and rigid motions	**HSG-SRT.A.2**

SET 10 • HSG-CO.A.5

MAIN

Geometry main question • High School — Geometry • Geometry: Congruence • HSG-CO.A.5

Describe a sequence of rigid transformations that carries triangle ABC onto triangle A′B′C′ (A to A′, B to B′, C to C′).



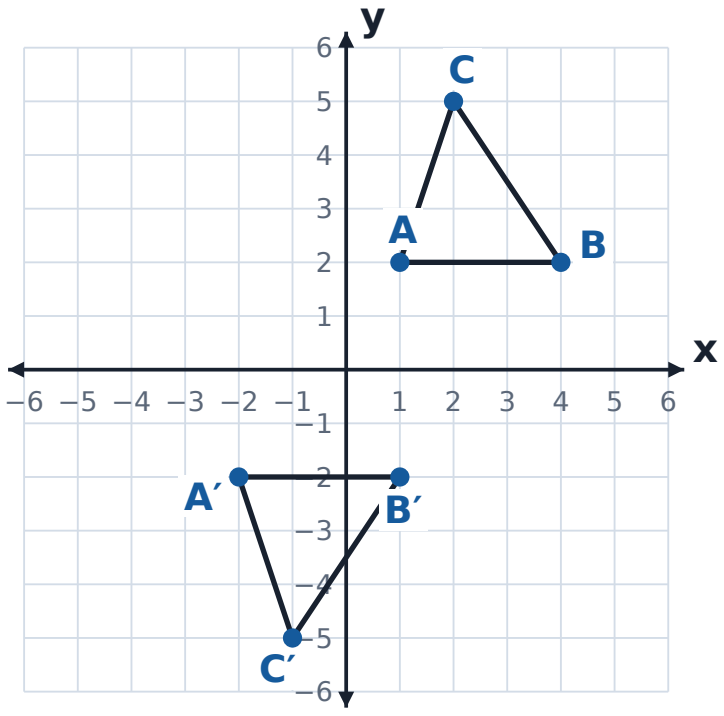
Sequence: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-M-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.A.5 • Set 10 • MAIN • Q1/5

Describe a sequence of rigid transformations that carries triangle ABC onto triangle A′B′C′.



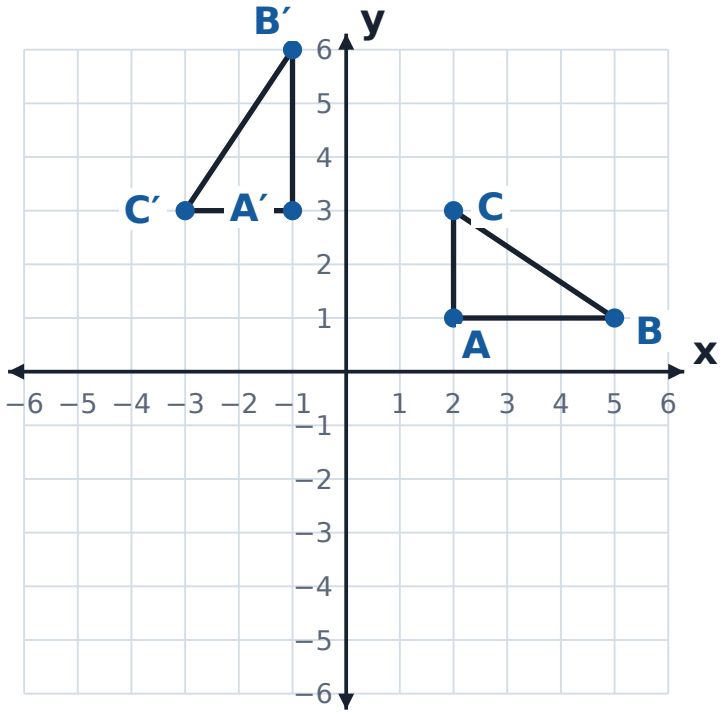
Sequence: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-M-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.A.5 • Set 10 • MAIN • Q2/5

On the grid, triangle A′B′C′ is congruent to triangle ABC. Describe a sequence of rigid transformations that carries ABC onto A′B′C′, with A → A′, B → B′, C → C′.



Sequence: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-M-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.A.5 • Set 10 • MAIN • Q3/5

Right triangle ABC has A(0, 0), B(2, 0), C(0, 3). Triangle A′B′C′ has A′(4, 0), B′(4, 2), C′(1, 0). Which sequence carries ABC onto A′B′C′?

A. Reflect across the y-axis, then translate 4 units right

B. Rotate 90° counterclockwise about the origin, then translate 4 units right

C. Rotate 90° clockwise about the origin, then translate 4 units right

D. Translate 4 units right, then rotate 90° counterclockwise about the origin

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-M-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.A.5 • Set 10 • MAIN • Q4/5

Triangle PQR has P(1, 1), Q(4, 1), R(1, 3). Triangle P′Q′R′ has P′(1, −1), Q′(4, −1), R′(1, −3). Describe a sequence of rigid transformations that carries PQR onto P′Q′R′, and explain why it must include a reflection.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-M-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.A.5 • Set 10 • MAIN • Q5/5

SET 10 • HSG-CO.A.5

BACKWARD 1

Understand congruence as the result of one rigid motion • Grade 8 • Geometry • 8.G.A.2

Triangle 2 is the image of triangle 1 after one rigid motion. Every point moved 3 units right. Which motion was it?

A. A reflection

B. A translation

C. A rotation

D. A dilation

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-B1-Q1

Grade 8 • Geometry • 8.G.A.2 • Set 10 • BACKWARD 1 • Q1/5

A triangle and its image face opposite ways, as in a mirror. Which kind of rigid motion was most likely used?

Motion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-B1-Q2

Grade 8 • Geometry • 8.G.A.2 • Set 10 • BACKWARD 1 • Q2/5

True or false?

If one figure is the image of another under a sequence of rigid motions, the figures are congruent.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-B1-Q3

Grade 8 • Geometry • 8.G.A.2 • Set 10 • BACKWARD 1 • Q3/5

True or false?

A figure and its image under a rigid motion can have different areas.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-B1-Q4

Grade 8 • Geometry • 8.G.A.2 • Set 10 • BACKWARD 1 • Q4/5

Segment AB goes from (0, 0) to (2, 0). Its image goes from (0, 0) to (0, 2). Which single rigid motion could do this?

A. A translation 2 units up

B. A reflection across the x-axis

C. A 90° counterclockwise rotation about the origin

D. A 180° rotation about the origin

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-B1-Q5

Grade 8 • Geometry • 8.G.A.2 • Set 10 • BACKWARD 1 • Q5/5

SET 10 • HSG-CO.A.5

BACKWARD 2

Find images with coordinate rules for sequences • Grade 8 • Geometry • 8.G.A.3

Point (2, −3) is reflected across the y-axis. Find its image.

Image: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-B2-Q1

Grade 8 • Geometry • 8.G.A.3 • Set 10 • BACKWARD 2 • Q1/5

Point $(-1, 4)$ is rotated 90° counterclockwise about the origin. Find its image.

Image: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-B2-Q2

Grade 8 • Geometry • 8.G.A.3 • Set 10 • BACKWARD 2 • Q2/5

Point (5, 1) is translated 2 units left and 3 units down. Find its image.

A. (3, −2)

B. (7, 4)

C. (7, −2)

D. (3, 4)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-B2-Q3

Grade 8 • Geometry • 8.G.A.3 • Set 10 • BACKWARD 2 • Q3/5

True or false?

Point (0, 3) rotated 180° about the origin is (0, −3).

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-B2-Q4

Grade 8 • Geometry • 8.G.A.3 • Set 10 • BACKWARD 2 • Q4/5

True or false?

Point (4, 1) reflected across the line $y = x$ is $(4, -1)$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-B2-Q5

Grade 8 • Geometry • 8.G.A.3 • Set 10 • BACKWARD 2 • Q5/5

SET 10 • HSG-CO.A.5

BACKWARD 3

Read the orientation (clockwise or counterclockwise order) of a triangle • Grade 8 • Geometry • 8.G.A.1

Triangle ABC has A(0, 0), B(4, 0), C(0, 3). Going A → B → C, is the order clockwise or counterclockwise?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-B3-Q1

Grade 8 • Geometry • 8.G.A.1 • Set 10 • BACKWARD 3 • Q1/5

True or false?

A reflection reverses the orientation of a triangle.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-B3-Q2

Grade 8 • Geometry • 8.G.A.1 • Set 10 • BACKWARD 3 • Q2/5

True or false?

A rotation reverses the orientation of a triangle.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-B3-Q3

Grade 8 • Geometry • 8.G.A.1 • Set 10 • BACKWARD 3 • Q3/5

Triangle PQR is clockwise. After two reflections, its image is

A. counterclockwise

B. either, depending on the lines

C. clockwise

D. not a triangle

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-B3-Q4

Grade 8 • Geometry • 8.G.A.1 • Set 10 • BACKWARD 3 • Q4/5

Triangle ABC is counterclockwise. It is translated and then reflected. Is the final image clockwise or counterclockwise?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-B3-Q5

Grade 8 • Geometry • 8.G.A.1 • Set 10 • BACKWARD 3 • Q5/5

SET 10 • HSG-CO.A.5

FORWARD 1

Use the fewest transformations and justify • High School — Geometry • Geometry: Congruence • HSG-CO.A.5

Two reflections are done one after the other, across parallel lines that are 3 units apart. Which single transformation has the same result?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-F1-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.A.5 • Set 10 • FORWARD 1 • Q1/5

Two reflections are done one after the other, across lines that meet at point O and form a 40° angle. Which single transformation has the same result?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-F1-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.A.5 • Set 10 • FORWARD 1 • Q2/5

Triangle ABC is clockwise and triangle A′B′C′ is congruent to it and counterclockwise. Explain why no single rotation or translation carries ABC onto A′B′C′.

Explanation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-F1-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.A.5 • Set 10 • FORWARD 1 • Q3/5

True or false?

Every rigid motion of the plane can be written as a sequence of at most three reflections.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-F1-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.A.5 • Set 10 • FORWARD 1 • Q4/5

Triangle A′B′C′ is the image of triangle ABC under some rigid motion, and both are counterclockwise. What is the fewest number of reflections that can always carry ABC onto A′B′C′?

A. 2

B. 4

C. 3

D. 1

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-F1-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.A.5 • Set 10 • FORWARD 1 • Q5/5

SET 10 • HSG-CO.A.5

FORWARD 2

Specify a similarity sequence: a dilation and rigid motions • High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2

Triangle ABC has A(0, 0), B(2, 0), C(0, 1). Triangle A′B′C′ has A′(0, 0), B′(0, 6), C′(−3, 0). Describe a sequence (a dilation and a rigid motion) that carries ABC onto A′B′C′.

Sequence: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-F2-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 10 • FORWARD 2 • Q1/5

Square S has vertices (0, 0), (1, 0), (1, 1), (0, 1). Square T has vertices (2, 2), (4, 2), (4, 4), (2, 4). Describe a similarity sequence carrying S onto T.

Sequence: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-F2-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 10 • FORWARD 2 • Q2/5

True or false?

Any two figures that a sequence of rigid motions and a dilation carries one onto the other are congruent.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-F2-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 10 • FORWARD 2 • Q3/5

Segment AB is from (0, 0) to (3, 0). Segment CD is from (1, 1) to (1, 7). Which sequence carries AB onto CD (A to C, B to D)?

A. Rotate 90° counterclockwise about the origin, then translate 1 right and 1 up

B. Dilate by 2 about the origin, then translate 1 right and 1 up

C. Dilate by 3 about the origin, rotate 90° counterclockwise, then translate 1 right and 1 up

D. Dilate by 2 about the origin, rotate 90° counterclockwise about the origin, then translate 1 right and 1 up

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-F2-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 10 • FORWARD 2 • Q4/5

Triangle PQR has sides 3, 4, 5. Triangle XYZ has sides 6, 8, 10. What scale factor must the dilation in a similarity sequence from PQR to XYZ have?

Scale factor: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-F2-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 10 • FORWARD 2 • Q5/5

Predict the effect of a rigid motion on a figure

MAIN	Geometry main question	**HSG-CO.B.6**
BACKWARD 1	Know what rigid motions keep: lines, lengths, angles, and parallel l…	**8.G.A.1**
BACKWARD 2	Match the vertices and sides of a figure with its image	**8.G.A.2**
FORWARD 1	Predict the effect of a sequence of rigid motions	**HSG-CO.B.6**
FORWARD 2	Use a rigid motion to conclude that parts of a figure are congruent	**HSG-CO.B.7**

SET 11 • HSG-CO.B.6

MAIN

Geometry main question • High School — Geometry • Geometry: Congruence • HSG-CO.B.6

Triangle ABC has AB = 7, BC = 5, and m∠B = 48°. It is reflected across a line ℓ. What are A′B′ and m∠B′?

A. 14 and 48°

B. 7 and 132°

C. 7 and 48°

D. They depend on where ℓ is

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-M-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.B.6 • Set 11 • MAIN • Q1/5

Quadrilateral PQRS has PQ ∥ SR. It is rotated 90° about a point O. What is true about P′Q′ and S′R′? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-M-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.B.6 • Set 11 • MAIN • Q2/5

Point A is 4 cm from line m. Triangle ABC is reflected across m. How far is A′ from m, and on which side of m is it?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-M-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.B.6 • Set 11 • MAIN • Q3/5

Triangle ABC is labeled counterclockwise (A → B → C). It is translated 5 units right and then reflected across a line m. Is the final image A″B″C″ labeled clockwise or counterclockwise? Explain each step.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-M-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.B.6 • Set 11 • MAIN • Q4/5

Segment AB lies on line ℓ. A 180° rotation about a point O on ℓ maps AB to A′B′. Where does A′B′ lie? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-M-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.B.6 • Set 11 • MAIN • Q5/5

SET 11 • HSG-CO.B.6

BACKWARD 1

Know what rigid motions keep: lines, lengths, angles, and parallel lines • Grade 8 • Geometry • 8.G.A.1

True or false?

Under a rotation, the image of a line is a line.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-B1-Q1

Grade 8 • Geometry • 8.G.A.1 • Set 11 • BACKWARD 1 • Q1/5

True or false?

A reflection can turn two parallel lines into two lines that meet.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-B1-Q2

Grade 8 • Geometry • 8.G.A.1 • Set 11 • BACKWARD 1 • Q2/5

A 52° angle is translated 6 units to the right. What is the measure of its image?

Measure: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-B1-Q3

Grade 8 • Geometry • 8.G.A.1 • Set 11 • BACKWARD 1 • Q3/5

A 9 cm segment is rotated 30° about one of its endpoints. How long is its image?

A. 30 cm

B. 4.5 cm

C. 39 cm

D. 9 cm

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-B1-Q4

Grade 8 • Geometry • 8.G.A.1 • Set 11 • BACKWARD 1 • Q4/5

Name two properties of a figure that a reflection does NOT change.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-B1-Q5

Grade 8 • Geometry • 8.G.A.1 • Set 11 • BACKWARD 1 • Q5/5

SET 11 • HSG-CO.B.6

BACKWARD 2

Match the vertices and sides of a figure with its image • Grade 8 • Geometry • 8.G.A.2

Triangle ABC is rotated to triangle A′B′C′. Which side of the image corresponds to side BC?

Side: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-B2-Q1

Grade 8 • Geometry • 8.G.A.2 • Set 11 • BACKWARD 2 • Q1/5

Quadrilateral PQRS is reflected to P′Q′R′S′. Which angle corresponds to ∠Q?

A. ∠R′

B. ∠P′

C. ∠Q′

D. ∠S′

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-B2-Q2

Grade 8 • Geometry • 8.G.A.2 • Set 11 • BACKWARD 2 • Q2/5

True or false?

If triangle ABC is translated to triangle A′B′C′, then AB = A′B′.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-B2-Q3

Grade 8 • Geometry • 8.G.A.2 • Set 11 • BACKWARD 2 • Q3/5

Triangle DEF is reflected to D′E′F′. DE = 6 and EF = 9. Find D′E′ and E′F′.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-B2-Q4

Grade 8 • Geometry • 8.G.A.2 • Set 11 • BACKWARD 2 • Q4/5

True or false?

After a rotation, side A′B′ of the image always lies on top of side AB.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-B2-Q5

Grade 8 • Geometry • 8.G.A.2 • Set 11 • BACKWARD 2 • Q5/5

SET 11 • HSG-CO.B.6

FORWARD 1

Predict the effect of a sequence of rigid motions • High School — Geometry • Geometry: Congruence • HSG-CO.B.6

Triangle ABC is reflected across line ℓ and then across line m, where $\ell \parallel$ m. Is the final image clockwise or counterclockwise if ABC is counterclockwise? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-F1-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.B.6 • Set 11 • FORWARD 1 • Q1/5

A figure is rotated 70° counterclockwise about point O and then rotated 50° clockwise about the same point O. Which single rigid motion has the same effect?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-F1-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.B.6 • Set 11 • FORWARD 1 • Q2/5

True or false?

A translation that is not the identity has exactly one fixed point.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-F1-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.B.6 • Set 11 • FORWARD 1 • Q3/5

Which rigid motion keeps exactly one point fixed?

A. A translation 3 units right

B. A reflection across line ℓ

C. A rotation of 90° about point O

D. The identity

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-F1-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.B.6 • Set 11 • FORWARD 1 • Q4/5

Line ℓ is reflected across line m, and ℓ is perpendicular to m. What is the image of ℓ? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-F1-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.B.6 • Set 11 • FORWARD 1 • Q5/5

SET 11 • HSG-CO.B.6

FORWARD 2

Use a rigid motion to conclude that parts of a figure are congruent • High School — Geometry • Geometry: Congruence • HSG-CO.B.7

A rotation about O maps triangle OAB onto triangle OCD, with A → C and B → D. OA = 5 and AB = 7. Find OC and CD.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-F2-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.B.7 • Set 11 • FORWARD 2 • Q1/5

A reflection across line ℓ maps triangle PQR onto triangle PSR (Q → S, and P and R are on ℓ). $m\angle PQR = 64°$. Find $m\angle PSR$.

Measure: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-F2-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.B.7 • Set 11 • FORWARD 2 • Q2/5

True or false?

If a rigid motion maps segment AB onto segment CD, then AB could be longer than CD.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-F2-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.B.7 • Set 11 • FORWARD 2 • Q3/5

A translation maps triangle KLM onto triangle K′L′M′. Which conclusion is NOT justified?

A. KL ∥ K′L′ or KL is on line K′L′

B. ∠L ≅ ∠L′

C. KL is on the same line as K′L′

D. KL = K′L′

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-F2-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.B.7 • Set 11 • FORWARD 2 • Q4/5

Explain why the image of an isosceles triangle under any rigid motion is also isosceles.

Explanation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-F2-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.B.7 • Set 11 • FORWARD 2 • Q5/5

Decide whether two figures are congruent using rigid motions

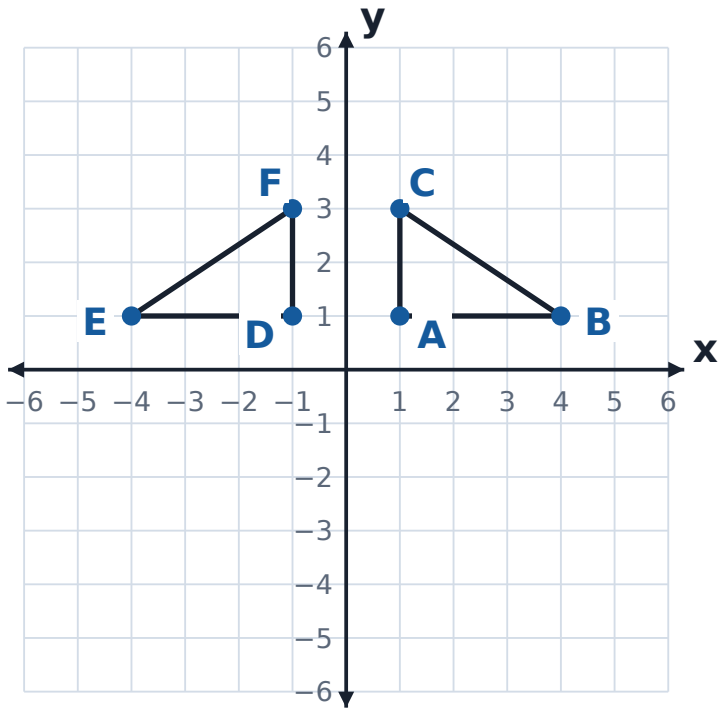
MAIN	Geometry main question	**HSG-CO.B.6**
BACKWARD 1	Recognize congruent figures as images under rigid motions	**8.G.A.2**
BACKWARD 2	Find lengths on a grid with the Pythagorean Theorem	**8.G.B.8**
FORWARD 1	Decide congruence for figures given by measurements, including…	**HSG-CO.B.6**
FORWARD 2	Use an established congruence to find unknown parts	**HSG-CO.B.7**

SET 12 • HSG-CO.B.6

MAIN

Geometry main question • High School — Geometry • Geometry: Congruence • HSG-CO.B.6

Is triangle ABC congruent to triangle DEF? If so, give a rigid motion that carries ABC onto DEF. If not, explain why.



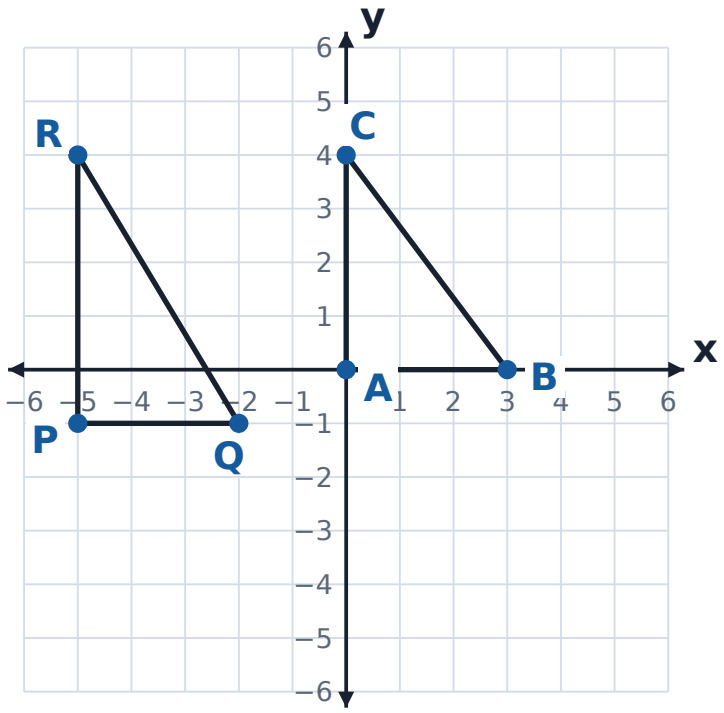
Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-M-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.B.6 • Set 12 • MAIN • Q1/5

Is triangle ABC congruent to triangle PQR? If so, give a rigid motion that carries one onto the other. If not, explain why.



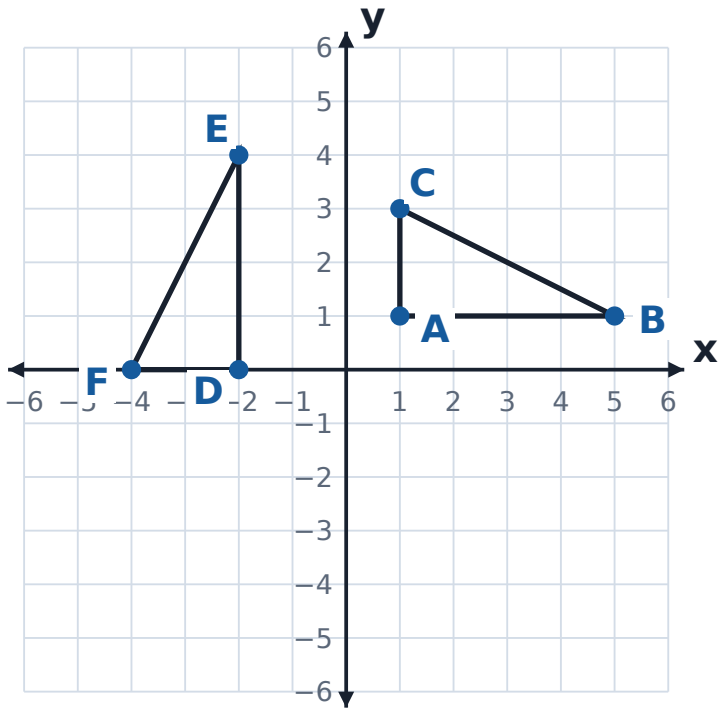
Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-M-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.B.6 • Set 12 • MAIN • Q2/5

Is triangle ABC congruent to triangle DEF? If so, give a rigid motion (or sequence) that carries ABC onto DEF. If not, explain why.



Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-M-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.B.6 • Set 12 • MAIN • Q3/5

Quadrilateral Q has sides 3, 4, 3, 4 and four right angles. Quadrilateral R has sides 3, 4, 3, 4 and two 70° angles. Is Q congruent to R? Explain using rigid motions.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-M-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.B.6 • Set 12 • MAIN • Q4/5

Rectangle R is 4 cm wide and 6 cm tall. Rectangle S is 6 cm wide and 4 cm tall. Are they congruent? Give a rigid motion or explain why not.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-M-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.B.6 • Set 12 • MAIN • Q5/5

SET 12 • HSG-CO.B.6

BACKWARD 1

Recognize congruent figures as images under rigid motions • Grade 8 • Geometry • 8.G.A.2

True or false?

If one figure is a translation of another, the figures are congruent.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-B1-Q1

Grade 8 • Geometry • 8.G.A.2 • Set 12 • BACKWARD 1 • Q1/5

True or false?

Two figures with the same shape but different sizes are congruent.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-B1-Q2

Grade 8 • Geometry • 8.G.A.2 • Set 12 • BACKWARD 1 • Q2/5

Which pair of figures must be congruent?

A. A square and a larger square

B. A triangle and its reflection

C. A circle of radius 2 and a circle of radius 3

D. A triangle and a dilation of it by 2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-B1-Q3

Grade 8 • Geometry • 8.G.A.2 • Set 12 • BACKWARD 1 • Q3/5

A triangle has sides 5, 6, and 7. It is rotated 90°. What are the side lengths of its image?

Lengths: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-B1-Q4

Grade 8 • Geometry • 8.G.A.2 • Set 12 • BACKWARD 1 • Q4/5

Is a figure congruent to its image after a reflection followed by a translation?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-B1-Q5

Grade 8 • Geometry • 8.G.A.2 • Set 12 • BACKWARD 1 • Q5/5

SET 12 • HSG-CO.B.6

BACKWARD 2

Find lengths on a grid with the Pythagorean Theorem • Grade 8 • Geometry • 8.G.B.8

Find the length of the segment from (1, 2) to (5, 2).

Length: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-B2-Q1

Grade 8 • Geometry • 8.G.B.8 • Set 12 • BACKWARD 2 • Q1/5

Find the length of the segment from (0, 0) to (3, 4).

Length: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-B2-Q2

Grade 8 • Geometry • 8.G.B.8 • Set 12 • BACKWARD 2 • Q2/5

Find the length of the segment from $(-2, 1)$ to $(1, 5)$.

Length: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-B2-Q3

Grade 8 • Geometry • 8.G.B.8 • Set 12 • BACKWARD 2 • Q3/5

True or false?

The segments from (0, 0) to (2, 1) and from (0, 0) to (1, 2) have the same length.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-B2-Q4

Grade 8 • Geometry • 8.G.B.8 • Set 12 • BACKWARD 2 • Q4/5

True or false?

The segment from (0, 0) to (3, 3) has length 3.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-B2-Q5

Grade 8 • Geometry • 8.G.B.8 • Set 12 • BACKWARD 2 • Q5/5

SET 12 • HSG-CO.B.6

FORWARD 1

Decide congruence for figures given by measurements, including curved figures • High School — Geometry • Geometry: Congruence • HSG-CO.B.6

Circle P has radius 3 and circle Q has radius 3. Explain, using a rigid motion, why they are congruent.

Explanation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-F1-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.B.6 • Set 12 • FORWARD 1 • Q1/5

Two quadrilaterals each have sides 5, 5, 5, 5. One is a square and the other has a 60° angle. Are they congruent? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-F1-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.B.6 • Set 12 • FORWARD 1 • Q2/5

True or false?

Two triangles with the same perimeter must be congruent.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-F1-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.B.6 • Set 12 • FORWARD 1 • Q3/5

Sector P has radius 4 and central angle 50°. Sector Q has radius 4 and central angle 50°, drawn in another place and turned. Are they congruent? Explain with a rigid motion.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-F1-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.B.6 • Set 12 • FORWARD 1 • Q4/5

Angle 1 measures 35° and angle 2 measures 35°. Describe a rigid motion that carries angle 1 onto angle 2.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-F1-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.B.6 • Set 12 • FORWARD 1 • Q5/5

SET 12 • HSG-CO.B.6

FORWARD 2

Use an established congruence to find unknown parts • High School — Geometry • Geometry: Congruence • HSG-CO.B.7

A rigid motion carries quadrilateral ABCD onto quadrilateral EFGH with A → E, B → F, C → G, D → H. AB = 6, ∠C = 110°, and GH = 4. Find EF, ∠G, and CD.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-F2-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.B.7 • Set 12 • FORWARD 2 • Q1/5

Triangle RST is congruent to triangle XYZ by a rigid motion with R → X, S → Y, T → Z. RS = 2x + 3 and XY = 15. Find x.

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-F2-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.B.7 • Set 12 • FORWARD 2 • Q2/5

True or false?

If a rigid motion carries triangle ABC onto triangle DEF with A → D, B → E, C → F, then ∠B ≅ ∠F.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-F2-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.B.7 • Set 12 • FORWARD 2 • Q3/5

A reflection carries pentagon PQRST onto pentagon P′Q′R′S′T′. Which pair must be congruent?

A. ∠S and ∠S′

B. PQ and RS

C. QR and S′T′

D. ∠S and ∠T′

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-F2-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.B.7 • Set 12 • FORWARD 2 • Q4/5

Triangle ABC is congruent to triangle CDA by a rigid motion with A → C, B → D, C → A. m∠BAC = 3y° and m∠DCA = 42°. Find y.

y = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-F2-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.B.7 • Set 12 • FORWARD 2 • Q5/5

Congruent triangles: a rigid motion exactly when all corresponding parts are congruent

MAIN	Geometry main question	**HSG-CO.B.7**
BACKWARD 1	Read a congruence statement to match corresponding parts	**8.G.A.2**
BACKWARD 2	Rigid motions keep lengths and angle measures	**8.G.A.1**
BACKWARD 3	Move a point onto another and a ray onto another with rigid motio…	**8.G.A.3**
FORWARD 1	Find which three corresponding parts are enough	**HSG-CO.B.8**
FORWARD 2	Use congruent triangles to prove a relationship in a figure	**HSG-SRT.B.5**

SET 13 • HSG-CO.B.7

MAIN

Geometry main question • High School — Geometry • Geometry: Congruence • HSG-CO.B.7

A rigid motion carries $\triangle ABC$ onto $\triangle DEF$ with $A \to D$, $B \to E$, $C \to F$. Explain why $AB = DE$, $BC = EF$, $CA = FD$, $\angle A \cong \angle D$, $\angle B \cong \angle E$, and $\angle C \cong \angle F$.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-M-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.B.7 • Set 13 • MAIN • Q1/5

In $\triangle ABC$ and $\triangle DEF$, $AB = DE$, $BC = EF$, $CA = FD$, $\angle A \cong \angle D$, $\angle B \cong \angle E$, and $\angle C \cong \angle F$. Explain how to build a rigid motion that carries $\triangle ABC$ onto $\triangle DEF$.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-M-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.B.7 • Set 13 • MAIN • Q2/5

The rotation of 90° counterclockwise about the origin carries $\triangle ABC$, with A(1, 0), B(4, 0), C(1, 2), onto $\triangle A'B'C'$. Find A′, B′, C′, check that all six pairs of corresponding parts are congruent, and explain why the definition of congruence guarantees this for every rigid motion.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-M-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.B.7 • Set 13 • MAIN • Q3/5

△ABC has A(0, 0), B(3, 0), C(0, 4). △DEF has D(5, 1), E(5, 4), F(1, 1). Show that all six pairs of corresponding parts (A ↔ D, B ↔ E, C ↔ F) are congruent, and then give a rigid motion that carries △ABC onto △DEF.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-M-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.B.7 • Set 13 • MAIN • Q4/5

Explain why the converse direction needs the sides as well as the angles: give two triangles with all three pairs of angles congruent for which no rigid motion carries one onto the other, and say which part of the definition fails.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-M-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.B.7 • Set 13 • MAIN • Q5/5

SET 13 • HSG-CO.B.7

BACKWARD 1

Read a congruence statement to match corresponding parts • Grade 8 • Geometry • 8.G.A.2

△ABC ≅ △XYZ. Which angle corresponds to ∠B?

Angle: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-B1-Q1

Grade 8 • Geometry • 8.G.A.2 • Set 13 • BACKWARD 1 • Q1/5

△ABC ≅ △XYZ. Which side corresponds to AC?

Side: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-B1-Q2

Grade 8 • Geometry • 8.G.A.2 • Set 13 • BACKWARD 1 • Q2/5

△PQR ≅ △LMN. Which pair corresponds?

A. QR and MN

B. QR and LN

C. PQ and MN

D. PR and LM

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-B1-Q3

Grade 8 • Geometry • 8.G.A.2 • Set 13 • BACKWARD 1 • Q3/5

True or false?

$\triangle DEF \cong \triangle GHI$ means that $\angle F \cong \angle I$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-B1-Q4

Grade 8 • Geometry • 8.G.A.2 • Set 13 • BACKWARD 1 • Q4/5

True or false?

$\triangle DEF \cong \triangle GHI$ means that $DE = HI$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-B1-Q5

Grade 8 • Geometry • 8.G.A.2 • Set 13 • BACKWARD 1 • Q5/5

SET 13 • HSG-CO.B.7

BACKWARD 2

Rigid motions keep lengths and angle measures • Grade 8 • Geometry • 8.G.A.1

True or false?

A rotation keeps the measure of every angle.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-B2-Q1

Grade 8 • Geometry • 8.G.A.1 • Set 13 • BACKWARD 2 • Q1/5

Segment PQ is 12 units long. It is reflected and then translated. How long is the image?

Length: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-B2-Q2

Grade 8 • Geometry • 8.G.A.1 • Set 13 • BACKWARD 2 • Q2/5

A 71° angle is reflected across a line. Its image measures

A. 289°

B. 109°

C. 71°

D. 19°

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-B2-Q3

Grade 8 • Geometry • 8.G.A.1 • Set 13 • BACKWARD 2 • Q3/5

True or false?

A translation can change the length of a segment if it moves the segment far enough.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-B2-Q4

Grade 8 • Geometry • 8.G.A.1 • Set 13 • BACKWARD 2 • Q4/5

Which three basic rigid motions keep lengths and angle measures?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-B2-Q5

Grade 8 • Geometry • 8.G.A.1 • Set 13 • BACKWARD 2 • Q5/5

SET 13 • HSG-CO.B.7

BACKWARD 3

Move a point onto another and a ray onto another with rigid motions • Grade 8 • Geometry • 8.G.A.3

Which translation moves (2, 3) onto (5, −1)?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-B3-Q1

Grade 8 • Geometry • 8.G.A.3 • Set 13 • BACKWARD 3 • Q1/5

A ray starts at the origin and points along the positive x-axis. Which rotation about the origin makes it point along the positive y-axis?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-B3-Q2

Grade 8 • Geometry • 8.G.A.3 • Set 13 • BACKWARD 3 • Q2/5

After a rotation, point C is below line DE but should be above it, and D and E are on the line. Which motion moves C across without moving D or E?

A. A rotation about D

B. A translation along DE

C. A rotation about the midpoint of DE

D. A reflection across line DE

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-B3-Q3

Grade 8 • Geometry • 8.G.A.3 • Set 13 • BACKWARD 3 • Q3/5

True or false?

Reflecting across a line keeps every point of that line fixed.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-B3-Q4

Grade 8 • Geometry • 8.G.A.3 • Set 13 • BACKWARD 3 • Q4/5

True or false?

Translating a triangle can keep exactly one of its vertices fixed.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-B3-Q5

Grade 8 • Geometry • 8.G.A.3 • Set 13 • BACKWARD 3 • Q5/5

SET 13 • HSG-CO.B.7

FORWARD 1

Find which three corresponding parts are enough • High School — Geometry • Geometry: Congruence • HSG-CO.B.8

Must two triangles be congruent if all three pairs of corresponding sides are congruent? Use the result of the full six-part test.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-F1-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 13 • FORWARD 1 • Q1/5

Give an example of two triangles with two pairs of congruent sides and a pair of congruent angles that are NOT congruent.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-F1-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 13 • FORWARD 1 • Q2/5

True or false?

Knowing that all three pairs of corresponding angles are congruent is enough to prove two triangles congruent.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-F1-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 13 • FORWARD 1 • Q3/5

Which set of corresponding parts is NOT always enough for congruence?

A. Three sides

B. Two sides and an angle not between them

C. Two sides and the angle between them

D. Two angles and the side between them

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-F1-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 13 • FORWARD 1 • Q4/5

Two triangles have AB = DE, ∠A ≅ ∠D, and ∠B ≅ ∠E. Must ∠C ≅ ∠F? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-F1-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 13 • FORWARD 1 • Q5/5

SET 13 • HSG-CO.B.7

FORWARD 2

Use congruent triangles to prove a relationship in a figure • High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5

In quadrilateral ABCD, AB = CB and AD = CD. Prove that diagonal BD bisects ∠ABC.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-F2-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 13 • FORWARD 2 • Q1/5

Segments AC and BD bisect each other at E. Prove that AB = CD.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-F2-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 13 • FORWARD 2 • Q2/5

In $\triangle PQR$ and $\triangle STU$, $PQ = ST$, $QR = TU$, and $PR = SU$. The triangles are congruent. What must be true of $\angle R$ and $\angle U$? Why?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-F2-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 13 • FORWARD 2 • Q3/5

True or false?

Once two triangles are proved congruent, only the parts used in the proof are known to be congruent.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-F2-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 13 • FORWARD 2 • Q4/5

To prove that point M is the midpoint of segment XY using congruent triangles, which pair of parts must you show is congruent?

A. ∠M and ∠X

B. ∠X and ∠Y

C. XM and YM

D. XY and XM

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-F2-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 13 • FORWARD 2 • Q5/5

Explain SAS congruence with rigid motions

MAIN	Geometry main question	**HSG-CO.B.8**
BACKWARD 1	Identify the included angle and included side	**7.G.A.2**
BACKWARD 2	Rigid motions map rays, lengths, and angles to equal ones	**8.G.A.1**
BACKWARD 3	Choose a translation, rotation, or reflection to move a point or ray…	**8.G.A.2**
FORWARD 1	Prove triangles congruent by SAS inside a larger figure	**HSG-SRT.B.5**
FORWARD 2	Decide whether given information is enough for SAS, and name a…	**HSG-SRT.B.5**

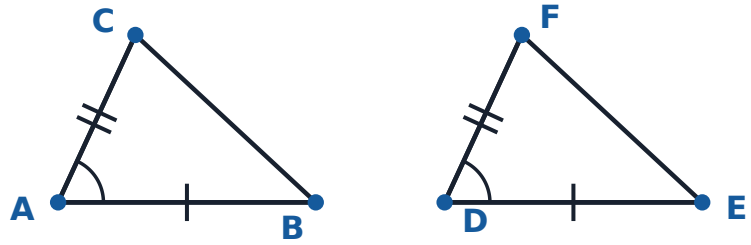
SET 14 • HSG-CO.B.8

MAIN

Geometry main question • High School — Geometry • Geometry: Congruence • HSG-CO.B.8

In $\triangle ABC$ and $\triangle DEF$, $AB = DE$, $AC = DF$, and $\angle A \cong \angle D$. Explain, using rigid motions, why $\triangle ABC \cong \triangle DEF$.

Work space



Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-M-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 14 • MAIN • Q1/5

In $\triangle JKL$ and $\triangle MNO$, $JK = MN$, $JL = MO$, and $\angle J \cong \angle M$. $\triangle MNO$ is a mirror image of $\triangle JKL$ (its vertices run the opposite way around). Explain, using rigid motions, why $\triangle JKL \cong \triangle MNO$, and say at which step a reflection is needed.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-M-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 14 • MAIN • Q2/5

△ABC has A(0, 0), B(4, 0), C(1, 3). △DEF has D(2, −1), E(2, 3), F(−1, 0). AB = DE = 4, AC = DF = √10, and ∠A ≅ ∠D. Name a translation and a rotation that carry A to D and ray AB onto ray DE, and explain, using only the SAS facts, why C must then land on F.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-M-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 14 • MAIN • Q3/5

In quadrilateral ABCD, AB = AD and diagonal AC bisects ∠BAD. Explain, using one reflection, why △ABC ≅ △ADC.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-M-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 14 • MAIN • Q4/5

Segments AC and BD bisect each other at E (A, E, C on one line and B, E, D on another). Explain, using a 180° rotation about E, why $\triangle AEB \cong \triangle CED$.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-M-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 14 • MAIN • Q5/5

SET 14 • HSG-CO.B.8

BACKWARD 1

Identify the included angle and included side • Grade 7 • Geometry • 7.G.A.2

In △ABC, which angle is included between sides AB and AC?

Angle: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-B1-Q1

Grade 7 • Geometry • 7.G.A.2 • Set 14 • BACKWARD 1 • Q1/5

In △PQR, which side is included between ∠P and ∠Q?

Side: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-B1-Q2

Grade 7 • Geometry • 7.G.A.2 • Set 14 • BACKWARD 1 • Q2/5

In △XYZ, which angle is NOT included between sides XY and YZ?

A. Both ∠X and ∠Y

B. ∠X

C. ∠Y

D. None of the angles is excluded

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-B1-Q3

Grade 7 • Geometry • 7.G.A.2 • Set 14 • BACKWARD 1 • Q3/5

True or false?

Two sides of 5 cm and 7 cm with a 40° angle between them determine one triangle.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-B1-Q4

Grade 7 • Geometry • 7.G.A.2 • Set 14 • BACKWARD 1 • Q4/5

True or false?

Two sides of 5 cm and 7 cm with a 40° angle that is not between them always determine one triangle.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-B1-Q5

Grade 7 • Geometry • 7.G.A.2 • Set 14 • BACKWARD 1 • Q5/5

SET 14 • HSG-CO.B.8

BACKWARD 2

Rigid motions map rays, lengths, and angles to equal ones • Grade 8 • Geometry • 8.G.A.1

True or false?

Under a rigid motion, the image of a ray is a ray.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-B2-Q1

Grade 8 • Geometry • 8.G.A.1 • Set 14 • BACKWARD 2 • Q1/5

A 30° angle is rotated about its vertex. What is the measure of the image angle?

Measure: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-B2-Q2

Grade 8 • Geometry • 8.G.A.1 • Set 14 • BACKWARD 2 • Q2/5

A point is 5 units from D along a ray. After a rotation about D, the image point is

A. closer to D

B. not on the image ray

C. 5 units from D on the image ray

D. farther from D

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-B2-Q3

Grade 8 • Geometry • 8.G.A.1 • Set 14 • BACKWARD 2 • Q3/5

True or false?

A reflection across line DE can move point D.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-B2-Q4

Grade 8 • Geometry • 8.G.A.1 • Set 14 • BACKWARD 2 • Q4/5

Segment AB is translated so that A lands on D. AB = 9. How far is the image of B from D?

Distance: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-B2-Q5

Grade 8 • Geometry • 8.G.A.1 • Set 14 • BACKWARD 2 • Q5/5

SET 14 • HSG-CO.B.8

BACKWARD 3

Choose a translation, rotation, or reflection to move a point or ray onto a target • Grade 8 • Geometry • 8.G.A.2

Which kind of rigid motion moves point A onto point D without turning anything?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-B3-Q1

Grade 8 • Geometry • 8.G.A.2 • Set 14 • BACKWARD 3 • Q1/5

A and D are already together. Which kind of rigid motion turns ray AB onto ray DE without moving D?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-B3-Q2

Grade 8 • Geometry • 8.G.A.2 • Set 14 • BACKWARD 3 • Q2/5

D and E are fixed, and point C is on the wrong side of line DE. Which motion fixes D and E and moves C across the line?

A. A translation along DE

B. A rotation of 90° about D

C. A rotation of 180° about E

D. A reflection across line DE

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-B3-Q3

Grade 8 • Geometry • 8.G.A.2 • Set 14 • BACKWARD 3 • Q3/5

True or false?

A rotation about point D keeps D fixed.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-B3-Q4

Grade 8 • Geometry • 8.G.A.2 • Set 14 • BACKWARD 3 • Q4/5

True or false?

A translation that moves A onto D also keeps A fixed.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-B3-Q5

Grade 8 • Geometry • 8.G.A.2 • Set 14 • BACKWARD 3 • Q5/5

SET 14 • HSG-CO.B.8

FORWARD 1

Prove triangles congruent by SAS inside a larger figure • High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5

In quadrilateral ABCD, AB ∥ DC and AB = DC. Diagonal AC is drawn. Prove that △ABC ≅ △CDA.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-F1-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 14 • FORWARD 1 • Q1/5

Point M is the midpoint of segment AB, and line ℓ is perpendicular to AB at M. Point P is on ℓ, and P ≠ M. Prove that △PMA ≅ △PMB.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-F1-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 14 • FORWARD 1 • Q2/5

Segments AC and BD cross at E, with AE = CE and BE = DE. Name the congruent triangles and the criterion.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-F1-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 14 • FORWARD 1 • Q3/5

True or false?

In an SAS proof, a side that the two triangles share can be one of the pairs of congruent sides.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-F1-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 14 • FORWARD 1 • Q4/5

Segments PQ and RS bisect each other at T. Which reason gives the included angles for SAS?

A. The angles are both right angles

B. Vertical angles are congruent

C. Base angles are congruent

D. Corresponding angles are congruent

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S14-F1-Q5**

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 14 • FORWARD 1 • Q5/5

SET 14 • HSG-CO.B.8

FORWARD 2

Decide whether given information is enough for SAS, and name a missing condition • High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5

In △ABC and △DEF, AB = DE and BC = EF. Name ONE more pair of congruent parts that would prove congruence by SAS.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-F2-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 14 • FORWARD 2 • Q1/5

In $\triangle PQR$ and $\triangle XYZ$, $PQ = XY$, $PR = XZ$, and $\angle Q \cong \angle Y$. Is this enough to prove the triangles congruent? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-F2-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 14 • FORWARD 2 • Q2/5

True or false?

In $\triangle ABC$ and $\triangle ADC$ (sharing AC), $AB = AD$ and $\angle BCA \cong \angle DCA$. These facts prove $\triangle ABC \cong \triangle ADC$ by SAS.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-F2-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 14 • FORWARD 2 • Q3/5

Kites ABCD and EFGH have AB = EF, BC = FG, and ∠B ≅ ∠F. Which triangles are congruent by SAS?

A. △BCD and △FGH

B. △ABC and △EFG

C. △ABD and △EFH

D. △ACD and △EGH

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-F2-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 14 • FORWARD 2 • Q4/5

Two triangles have sides 7 and 9 with a 50° angle between them. A student says they need not be congruent because the third sides were not measured. Is the student right? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-F2-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 14 • FORWARD 2 • Q5/5

Explain ASA congruence with rigid motions

MAIN	Geometry main question	**HSG-CO.B.8**
BACKWARD 1	Two angles and the included side determine a triangle	**7.G.A.2**
BACKWARD 2	Two different lines meet in at most one point	**8.EE.C.8.a**
BACKWARD 3	Rigid motions keep rays and angle measures	**8.G.A.1**
FORWARD 1	Explain AAS from ASA and the triangle angle sum	**HSG-CO.C.10**
FORWARD 2	Prove congruence with ASA or AAS in overlapping triangles	**HSG-SRT.B.5**

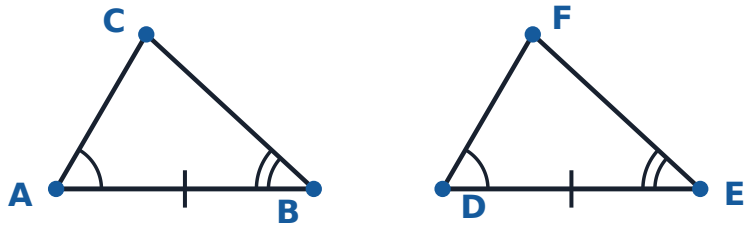
SET 15 • HSG-CO.B.8

MAIN

Geometry main question • High School — Geometry • Geometry: Congruence • HSG-CO.B.8

In $\triangle ABC$ and $\triangle DEF$, $\angle A \cong \angle D$, $AB = DE$, and $\angle B \cong \angle E$. Explain, using rigid motions, why $\triangle ABC \cong \triangle DEF$.

Work space



Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-M-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 15 • MAIN • Q1/5

In $\triangle PQR$ and $\triangle STU$, $\angle P \cong \angle S$, $PQ = ST$, and $\angle Q \cong \angle T$. $\triangle STU$ is a mirror image of $\triangle PQR$ (its vertices run the opposite way around). Explain, using rigid motions, why $\triangle PQR \cong \triangle STU$, including where the reflection is used and why R lands on U.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-M-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 15 • MAIN • Q2/5

△ABC has A(0, 0), B(4, 0), C(1, 3). △DEF has D(5, 5), E(1, 5), F(4, 2). ∠A ≅ ∠D, AB = DE, and ∠B ≅ ∠E. The 180° rotation about (2.5, 2.5) sends A to D and B to E. Explain, using only the ASA facts (not the coordinates of C), why it must send C to F.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-M-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 15 • MAIN • Q3/5

In $\triangle ABC$, $\angle B \cong \angle C$. Let ℓ be the perpendicular bisector of BC. Explain, using the reflection across ℓ and the ASA argument, why the reflection carries $\triangle ABC$ onto $\triangle ACB$, and conclude that AB = AC.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-M-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 15 • MAIN • Q4/5

Segments AD and BC cross at E, which is the midpoint of AD, and AB ∥ CD (B and C on opposite sides of AD). Explain, using a 180° rotation about E, why △ABE ≅ △DCE.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-M-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 15 • MAIN • Q5/5

SET 15 • HSG-CO.B.8

BACKWARD 1

Two angles and the included side determine a triangle • Grade 7 • Geometry • 7.G.A.2

True or false?

Two angles of 40° and 60° with a 5 cm side between them determine exactly one triangle (up to position).

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-B1-Q1

Grade 7 • Geometry • 7.G.A.2 • Set 15 • BACKWARD 1 • Q1/5

True or false?

Any two angles with a side between them make a triangle.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-B1-Q2

Grade 7 • Geometry • 7.G.A.2 • Set 15 • BACKWARD 1 • Q2/5

A triangle has angles of 50° and 70° at the ends of a 6 cm side. What is the third angle?

Angle: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-B1-Q3

Grade 7 • Geometry • 7.G.A.2 • Set 15 • BACKWARD 1 • Q3/5

Which set of conditions determines exactly one triangle?

A. Angles 30° and 80° with a 4 cm side between them

B. One angle of 45° and nothing else

C. Angles 30°, 60°, and 90°

D. Angles 100° and 90° with a 3 cm side between them

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-B1-Q4

Grade 7 • Geometry • 7.G.A.2 • Set 15 • BACKWARD 1 • Q4/5

Two angles of a triangle are 95° and 90°. Can the triangle exist? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-B1-Q5

Grade 7 • Geometry • 7.G.A.2 • Set 15 • BACKWARD 1 • Q5/5

SET 15 • HSG-CO.B.8

BACKWARD 2

Two different lines meet in at most one point • Grade 8 • Expressions & Equations • 8.EE.C.8.a

How many points do the lines $y = x + 1$ and $y = -x + 3$ have in common?

Number: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-B2-Q1

Grade 8 • Expressions & Equations • 8.EE.C.8.a • Set 15 • BACKWARD 2 • Q1/5

True or false?

Two different lines can meet in exactly two points.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-B2-Q2

Grade 8 • Expressions & Equations • 8.EE.C.8.a • Set 15 • BACKWARD 2 • Q2/5

Lines $y = 2x + 1$ and $y = 2x - 4$

A. never meet

B. are the same line

C. meet at one point

D. meet at two points

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-B2-Q3

Grade 8 • Expressions & Equations • 8.EE.C.8.a • Set 15 • BACKWARD 2 • Q3/5

True or false?

If two lines share two points, they are the same line.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-B2-Q4

Grade 8 • Expressions & Equations • 8.EE.C.8.a • Set 15 • BACKWARD 2 • Q4/5

Where do the lines $x = 2$ and $y = -3$ meet?

Point: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-B2-Q5

Grade 8 • Expressions & Equations • 8.EE.C.8.a • Set 15 • BACKWARD 2 • Q5/5

SET 15 • HSG-CO.B.8

BACKWARD 3

Rigid motions keep rays and angle measures • Grade 8 • Geometry • 8.G.A.1

A 64° angle is translated and then reflected. What is the measure of its image?

Measure: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-B3-Q1

Grade 8 • Geometry • 8.G.A.1 • Set 15 • BACKWARD 3 • Q1/5

True or false?

A rigid motion maps an angle to an angle with the same measure.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-B3-Q2

Grade 8 • Geometry • 8.G.A.1 • Set 15 • BACKWARD 3 • Q2/5

A ray from E makes a 35° angle with ray ED above line DE. How many such rays are there?

A. Exactly one

B. Two

C. Infinitely many

D. None

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-B3-Q3

Grade 8 • Geometry • 8.G.A.1 • Set 15 • BACKWARD 3 • Q3/5

True or false?

After a rotation, an angle can become larger if its sides are longer.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-B3-Q4

Grade 8 • Geometry • 8.G.A.1 • Set 15 • BACKWARD 3 • Q4/5

Segment AB = 4 cm is rotated so that A lands on D. Where is B, relative to D?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-B3-Q5

Grade 8 • Geometry • 8.G.A.1 • Set 15 • BACKWARD 3 • Q5/5

SET 15 • HSG-CO.B.8

FORWARD 1

Explain AAS from ASA and the triangle angle sum • High School — Geometry • Geometry: Congruence • HSG-CO.C.10

In $\triangle ABC$ and $\triangle DEF$, $\angle A \cong \angle D$, $\angle B \cong \angle E$, and $BC = EF$ (the side is NOT between the angles). Prove $\triangle ABC \cong \triangle DEF$.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-F1-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 15 • FORWARD 1 • Q1/5

Two triangles have angles 40° and 75° and a 9 cm side opposite the 75° angle. Must they be congruent? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-F1-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 15 • FORWARD 1 • Q2/5

In the argument above, how does AAS follow from ASA? Name the theorem used to get the third pair of angles.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-F1-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 15 • FORWARD 1 • Q3/5

In $\triangle PQR$ and $\triangle XYZ$, $\angle P \cong \angle X$, $\angle Q \cong \angle Y$, and $PR = XZ$. Which criterion proves congruence?

A. ASA

B. AAS

C. SAS

D. SSS

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-F1-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 15 • FORWARD 1 • Q4/5

Which corresponding side would make AAS apply if $\angle A \cong \angle D$ and $\angle C \cong \angle F$?

Side: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-F1-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 15 • FORWARD 1 • Q5/5

SET 15 • HSG-CO.B.8

FORWARD 2

Prove congruence with ASA or AAS in overlapping triangles • High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5

Segments AD and BC cross at E. AB ∥ CD and E is the midpoint of AD. Prove that △ABE ≅ △DCE.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-F2-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 15 • FORWARD 2 • Q1/5

In $\triangle ABC$, the bisector of $\angle A$ is perpendicular to BC at D. Prove that AB = AC.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-F2-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 15 • FORWARD 2 • Q2/5

In overlapping triangles ABE and ACD, ∠A is shared, AB = AC, and ∠B ≅ ∠C. Which criterion proves △ABE ≅ △ACD?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-F2-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 15 • FORWARD 2 • Q3/5

True or false?

An angle shared by two overlapping triangles cannot be used in a congruence proof, because it is the same angle.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-F2-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 15 • FORWARD 2 • Q4/5

In $\triangle JKL$ and $\triangle JML$, $\angle KJL \cong \angle MJL$ and $\angle JLK \cong \angle JLM$. Which side completes ASA?

A. $JL = JL$

B. $JK = JM$

C. $KL = ML$

D. $JK = KL$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S15-F2-Q5**

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 15 • FORWARD 2 • Q5/5

Explain SSS congruence with rigid motions

MAIN	Geometry main question	**HSG-CO.B.8**
BACKWARD 1	Three side lengths determine a triangle (triangle inequality)	**7.G.A.2**
BACKWARD 2	Find the points at given distances from two points (two circles)	**7.G.B.4**
BACKWARD 3	Rigid motions keep every length	**8.G.A.1**
FORWARD 1	Explain the Hypotenuse-Leg criterion for right triangles	**HSG-SRT.B.5**
FORWARD 2	Decide which criteria guarantee congruence, with counterexampl…	**HSG-SRT.B.5**

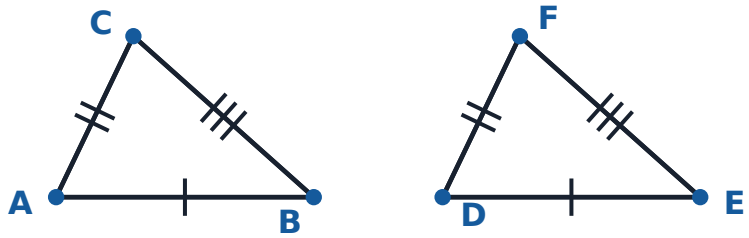
SET 16 • HSG-CO.B.8

MAIN

Geometry main question • High School — Geometry • Geometry: Congruence • HSG-CO.B.8

In $\triangle ABC$ and $\triangle DEF$, $AB = DE$, $BC = EF$, and $CA = FD$. Explain, using rigid motions, why $\triangle ABC \cong \triangle DEF$.

Work space



Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-M-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 16 • MAIN • Q1/5

In $\triangle ABC$ and $\triangle DEF$, $AB = DE$, $BC = EF$, and $CA = FD$. A translation and a rotation have moved A onto D and B onto E, and the image C' of C is on the opposite side of line DE from F, with $DC' = DF$ and $EC' = EF$. Explain why the reflection across line DE carries C' onto F. Justify each step without using SSS (it is what is being explained).

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-M-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 16 • MAIN • Q2/5

△ABC has A(0, 0), B(5, 0), C(1, 3). △DEF has D(1, 1), E(1, 6), F(−2, 2). Show that the three pairs of corresponding sides are congruent, and explain, using rigid motions, why the triangles are congruent.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-M-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 16 • MAIN • Q3/5

Quadrilateral ABCD has AB = AD and CB = CD (a kite). Explain, using the reflection across line AC, why $\triangle ABC \cong \triangle ADC$.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-M-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 16 • MAIN • Q4/5

A triangle has sides 4, 5, and 6. Its 6-unit side is placed on segment DE (DE = 6). Explain why the third vertex can be in exactly two places, one on each side of line DE, and why the reflection across DE carries one place onto the other.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-M-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 16 • MAIN • Q5/5

SET 16 • HSG-CO.B.8

BACKWARD 1

Three side lengths determine a triangle (triangle inequality) • Grade 7 • Geometry • 7.G.A.2

True or false?

Side lengths 3, 4, and 8 can form a triangle.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-B1-Q1

Grade 7 • Geometry • 7.G.A.2 • Set 16 • BACKWARD 1 • Q1/5

True or false?

Side lengths 5, 6, and 10 form exactly one triangle (up to position).

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-B1-Q2

Grade 7 • Geometry • 7.G.A.2 • Set 16 • BACKWARD 1 • Q2/5

Which three lengths can form a triangle?

A. 5, 5, 11

B. 4, 5, 7

C. 2, 3, 6

D. 1, 1, 3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-B1-Q3

Grade 7 • Geometry • 7.G.A.2 • Set 16 • BACKWARD 1 • Q3/5

Two sides of a triangle are 6 and 10. Give the range of possible lengths for the third side.

Range: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-B1-Q4

Grade 7 • Geometry • 7.G.A.2 • Set 16 • BACKWARD 1 • Q4/5

Can two different-shaped triangles both have sides 7, 8, and 9? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-B1-Q5

Grade 7 • Geometry • 7.G.A.2 • Set 16 • BACKWARD 1 • Q5/5

SET 16 • HSG-CO.B.8

BACKWARD 2

Find the points at given distances from two points (two circles) • Grade 7 • Geometry • 7.G.B.4

Point P is 5 units from A and 5 units from B, and A and B are 6 units apart. How many such points P are there in the plane?

Number: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-B2-Q1

Grade 7 • Geometry • 7.G.B.4 • Set 16 • BACKWARD 2 • Q1/5

True or false?

In a given plane containing A, all points 3 cm from A form a circle with center A and radius 3 cm.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-B2-Q2

Grade 7 • Geometry • 7.G.B.4 • Set 16 • BACKWARD 2 • Q2/5

Circles of radius 4 centered at A and B meet at two points. AB = 6. Where are the two points?

A. Both on line AB

B. One on each side of line AB

C. At A and B

D. Both on the same side of AB

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-B2-Q3

Grade 7 • Geometry • 7.G.B.4 • Set 16 • BACKWARD 2 • Q3/5

True or false?

Two circles always meet in exactly two points.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-B2-Q4

Grade 7 • Geometry • 7.G.B.4 • Set 16 • BACKWARD 2 • Q4/5

Points A and B are 10 apart. Is there a point 3 units from A and 4 units from B? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-B2-Q5

Grade 7 • Geometry • 7.G.B.4 • Set 16 • BACKWARD 2 • Q5/5

SET 16 • HSG-CO.B.8

BACKWARD 3

Rigid motions keep every length • Grade 8 • Geometry • 8.G.A.1

Triangle ABC has sides 3, 5, and 7. After a rotation and a reflection, what are the side lengths of the image?

Lengths: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-B3-Q1

Grade 8 • Geometry • 8.G.A.1 • Set 16 • BACKWARD 3 • Q1/5

True or false?

Under a rigid motion, the distance from the image of A to the image of C equals AC.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-B3-Q2

Grade 8 • Geometry • 8.G.A.1 • Set 16 • BACKWARD 3 • Q2/5

Which can a rigid motion change?

A. The perimeter

B. The length of a side

C. The area

D. The position of a triangle

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-B3-Q3

Grade 8 • Geometry • 8.G.A.1 • Set 16 • BACKWARD 3 • Q3/5

True or false?

A reflection doubles the distance from a point to the line of reflection.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-B3-Q4

Grade 8 • Geometry • 8.G.A.1 • Set 16 • BACKWARD 3 • Q4/5

A point C is 6 units from D. After a reflection across a line through D, how far is the image of C from D?

Distance: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-B3-Q5

Grade 8 • Geometry • 8.G.A.1 • Set 16 • BACKWARD 3 • Q5/5

SET 16 • HSG-CO.B.8

FORWARD 1

Explain the Hypotenuse-Leg criterion for right triangles • High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5

Right triangles ABC and DEF have right angles at C and F, hypotenuses AB = DE, and legs AC = DF. Prove that $\triangle ABC \cong \triangle DEF$.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-F1-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 16 • FORWARD 1 • Q1/5

Two right triangles each have hypotenuse 13 and one leg 5. Are they congruent? What is the other leg?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-F1-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 16 • FORWARD 1 • Q2/5

True or false?

Hypotenuse-Leg is a special case of SSA that does guarantee congruence.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-F1-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 16 • FORWARD 1 • Q3/5

Which information proves two right triangles congruent?

A. One pair of acute angles is equal

B. Their right angles are equal

C. Their hypotenuses are equal and one pair of legs is equal

D. Their hypotenuses are equal

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-F1-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 16 • FORWARD 1 • Q4/5

In right triangles PQR and XYZ (right angles at R and Z), PQ = XY = 10 and QR = YZ = 6. Find PR and XZ.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-F1-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 16 • FORWARD 1 • Q5/5

SET 16 • HSG-CO.B.8

FORWARD 2

Decide which criteria guarantee congruence, with counterexamples • High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5

Explain with a counterexample why AAA is not a congruence criterion.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-F2-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 16 • FORWARD 2 • Q1/5

Explain with a counterexample why SSA is not a congruence criterion.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-F2-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 16 • FORWARD 2 • Q2/5

Which list contains only valid triangle congruence criteria?

A. SSS, SAS, ASA, AAS

B. SSS, SSA, ASA, AAS

C. SSA, AAA, SAS, HL

D. SAS, AAA, ASA, SSS

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-F2-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 16 • FORWARD 2 • Q3/5

True or false?

If two triangles have the same three side lengths, they have the same three angle measures.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-F2-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 16 • FORWARD 2 • Q4/5

True or false?

Two triangles with two pairs of congruent sides and equal areas must be congruent.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-F2-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 16 • FORWARD 2 • Q5/5

SET 17 • HSG-CO.C.9 • CONGRUENCE: PROVING THEOREMS

Prove that vertical angles are congruent

MAIN	Geometry main question	**HSG-CO.C.9**
BACKWARD 1	Angles in a linear pair add to 180°	**7.G.B.5**
BACKWARD 2	Use properties of equality as reasons	**HSA-REI.A.1**
BACKWARD 3	Identify vertical angles and linear pairs in a figure	**7.G.B.5**
FORWARD 1	Use the vertical angle theorem in multistep proofs	**HSG-CO.C.9**
FORWARD 2	Prove statements about the bisectors of angles formed by two lin…	**HSG-CO.C.9**

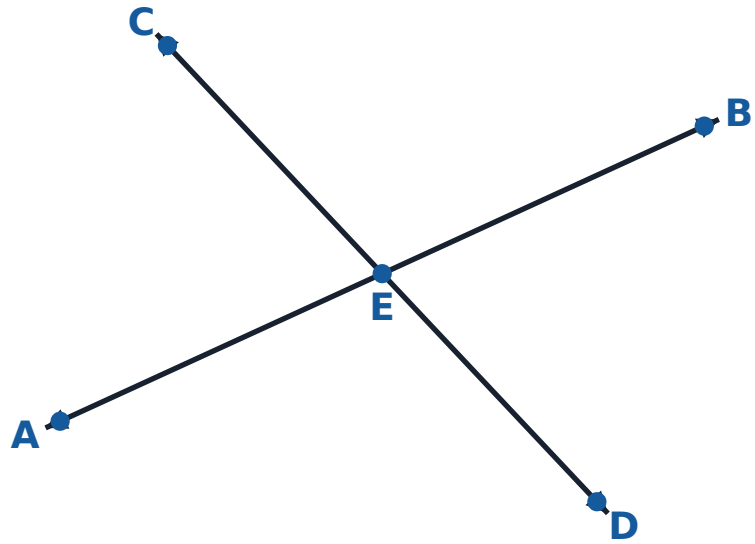
SET 17 • HSG-CO.C.9

MAIN

Geometry main question • High School — Geometry • Geometry: Congruence • HSG-CO.C.9

Lines AB and CD intersect at E. Prove that ∠AEC ≅ ∠BED.

Work space



Conclusion: _____

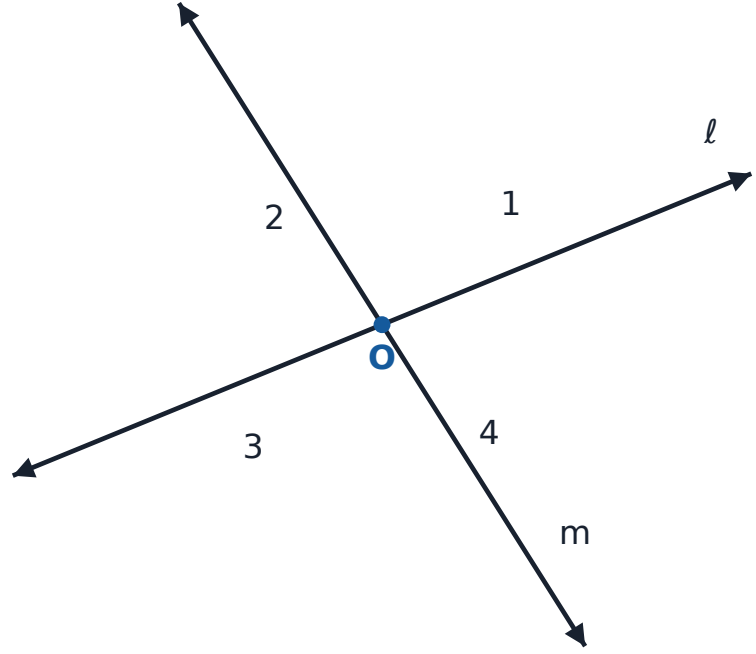
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-M-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 17 • MAIN • Q1/5

Lines ℓ and m meet at O, forming $\angle 1$, $\angle 2$, $\angle 3$, and $\angle 4$ in order around O. Prove that $\angle 2 \cong \angle 4$.

Work space



Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-M-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 17 • MAIN • Q2/5

Lines PQ and RS intersect at T (P, T, Q on one line and R, T, S on the other). Prove that ∠PTR ≅ ∠QTS.

Work space

Conclusion: _____

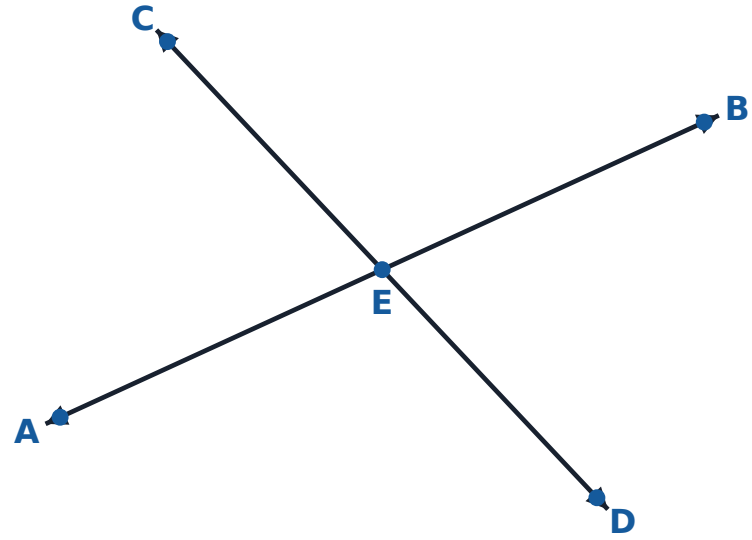
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-M-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 17 • MAIN • Q3/5

Lines AB and CD intersect at E. Explain, using a rigid motion, why $\angle AEC \cong \angle BED$.

Work space



Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-M-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 17 • MAIN • Q4/5

Two lines intersect, forming vertical angles $\angle 1$ and $\angle 3$, and $\angle 2$ is adjacent to both. Suppose $m\angle 1 = x°$. Prove that $m\angle 3 = x°$ for every value of x, giving a reason for each step.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-M-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 17 • MAIN • Q5/5

SET 17 • HSG-CO.C.9

BACKWARD 1

Angles in a linear pair add to 180° • Grade 7 • Geometry • 7.G.B.5

Two angles form a linear pair. One measures 128°. Find the other.

Measure: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-B1-Q1

Grade 7 • Geometry • 7.G.B.5 • Set 17 • BACKWARD 1 • Q1/5

∠PQR and ∠RQS form a linear pair, and m∠PQR = 3 · m∠RQS. Find m∠RQS.

Measure: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-B1-Q2

Grade 7 • Geometry • 7.G.B.5 • Set 17 • BACKWARD 1 • Q2/5

True or false?

The two angles of a linear pair are always supplementary.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-B1-Q3

Grade 7 • Geometry • 7.G.B.5 • Set 17 • BACKWARD 1 • Q3/5

Points A, B, C are on a line with B between A and C, and ray BD is drawn. Which angles form a linear pair?

A. ∠DBC and ∠ABC

B. ∠BAD and ∠BCD

C. ∠ABD and ∠DBC

D. ∠ABD and ∠ABC

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-B1-Q4

Grade 7 • Geometry • 7.G.B.5 • Set 17 • BACKWARD 1 • Q4/5

True or false?

Two angles that add to 180° must form a linear pair.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-B1-Q5

Grade 7 • Geometry • 7.G.B.5 • Set 17 • BACKWARD 1 • Q5/5

SET 17 • HSG-CO.C.9

BACKWARD 2

Use properties of equality as reasons • High School — Algebra I (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.1

From $m\angle 1 + 40° = m\angle 2 + 40°$, which property gives $m\angle 1 = m\angle 2$?

Property: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-B2-Q1

High School — Algebra I (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.1 • Set 17 • BACKWARD 2 • Q1/5

If $x + y = 180$ and $y + z = 180$, which property lets you write $x + y = y + z$?

Property: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-B2-Q2

High School — Algebra I (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.1 • Set 17 • BACKWARD 2 • Q2/5

Which property justifies: if a = b, then b = a?

A. Reflexive property

B. Symmetric property

C. Distributive property

D. Transitive property

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-B2-Q3

High School — Algebra I (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.1 • Set 17 • BACKWARD 2 • Q3/5

True or false?

The reflexive property says that any quantity equals itself, such as $m\angle A = m\angle A$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-B2-Q4

High School — Algebra I (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.1 • Set 17 • BACKWARD 2 • Q4/5

True or false?

From $2x = 2y$ you may conclude $x = y$ by the subtraction property of equality.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-B2-Q5

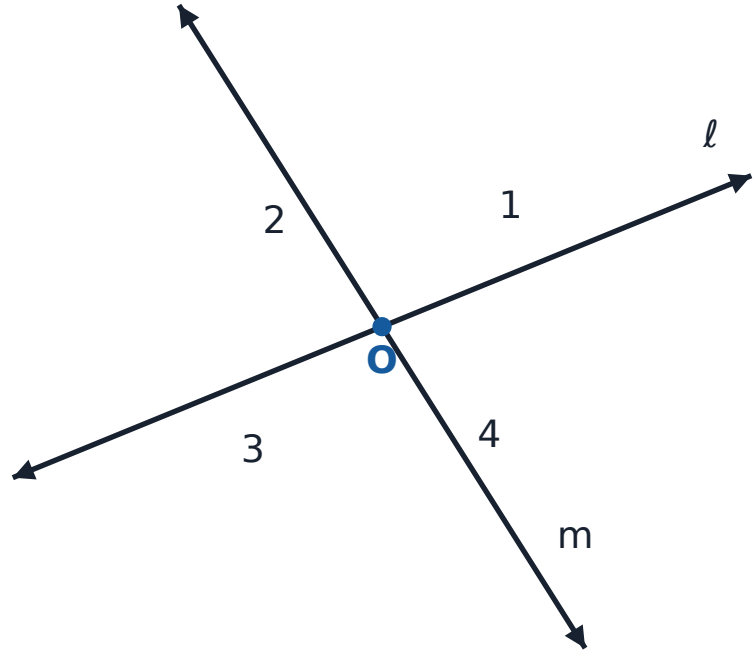
High School — Algebra I (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.1 • Set 17 • BACKWARD 2 • Q5/5

SET 17 • HSG-CO.C.9

BACKWARD 3

Identify vertical angles and linear pairs in a figure • Grade 7 • Geometry • 7.G.B.5

Lines ℓ and m meet at O, forming $\angle 1$, $\angle 2$, $\angle 3$, $\angle 4$ in order. Which angle is vertical to $\angle 1$?



A. $\angle 3$

B. $\angle 1$

C. $\angle 4$

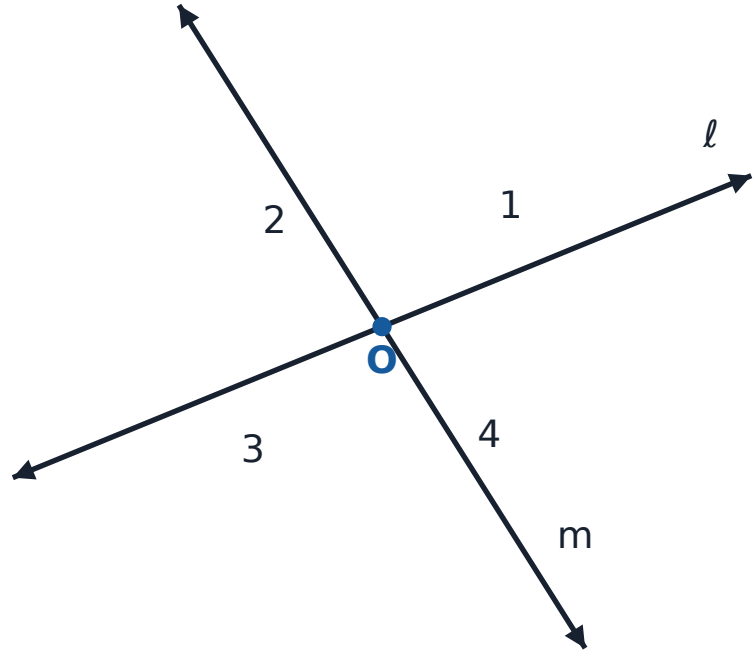
D. $\angle 2$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-B3-Q1

Grade 7 • Geometry • 7.G.B.5 • Set 17 • BACKWARD 3 • Q1/5

Lines ℓ and m meet at O, forming $\angle 1$, $\angle 2$, $\angle 3$, $\angle 4$ in order. Name the two angles that form linear pairs with $\angle 2$.



Angles: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-B3-Q2

Grade 7 • Geometry • 7.G.B.5 • Set 17 • BACKWARD 3 • Q2/5

True or false?

Vertical angles share a side.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-B3-Q3

Grade 7 • Geometry • 7.G.B.5 • Set 17 • BACKWARD 3 • Q3/5

True or false?

When two lines cross, they form two pairs of vertical angles.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-B3-Q4

Grade 7 • Geometry • 7.G.B.5 • Set 17 • BACKWARD 3 • Q4/5

Two lines cross. One angle is 63°. Find its vertical angle and the angles next to it.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-B3-Q5

Grade 7 • Geometry • 7.G.B.5 • Set 17 • BACKWARD 3 • Q5/5

SET 17 • HSG-CO.C.9

FORWARD 1

Use the vertical angle theorem in multistep proofs • High School — Geometry • Geometry: Congruence • HSG-CO.C.9

Lines AB and CD intersect at E, and ray EF bisects ∠AEC. Ray EG is the ray opposite ray EF. Prove that ray EG bisects ∠BED.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-F1-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 17 • FORWARD 1 • Q1/5

Three lines meet at point O, forming six angles. Two of the angles that are next to each other are 50° and 70°. Prove that the six angles are 50°, 70°, 60°, 50°, 70°, 60° in order.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-F1-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 17 • FORWARD 1 • Q2/5

Two lines meet. One angle is (4x + 10)° and its vertical angle is (6x − 30)°. Find x and all four angles.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-F1-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 17 • FORWARD 1 • Q3/5

True or false?

If one angle formed by two crossing lines is a right angle, all four angles are right angles.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-F1-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 17 • FORWARD 1 • Q4/5

Lines AB and CD meet at E with m∠AEC = 2x and m∠CEB = 3x. What is m∠BED?

A. 72°

B. 144°

C. 36°

D. 108°

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-F1-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 17 • FORWARD 1 • Q5/5

SET 17 • HSG-CO.C.9

FORWARD 2

Prove statements about the bisectors of angles formed by two lines • High School — Geometry • Geometry: Congruence • HSG-CO.C.9

∠ABD and ∠DBC form a linear pair. Ray BE bisects ∠ABD and ray BF bisects ∠DBC. Prove that ∠EBF is a right angle.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-F2-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 17 • FORWARD 2 • Q1/5

Is the converse of the vertical angle theorem true: "If two angles are congruent, they are vertical angles"? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-F2-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 17 • FORWARD 2 • Q2/5

Two lines meet at O. Prove that the bisectors of a pair of vertical angles lie on one line.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-F2-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 17 • FORWARD 2 • Q3/5

True or false?

The bisectors of a pair of vertical angles are perpendicular.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-F2-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 17 • FORWARD 2 • Q4/5

Two lines meet, and the bisectors of two adjacent angles are drawn. What angle do the bisectors form?

A. 45°

B. It depends on the lines

C. 180°

D. 90°

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-F2-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 17 • FORWARD 2 • Q5/5

Prove the angle theorems for parallel lines cut by a transversal

MAIN	Geometry main question	**HSG-CO.C.9**
BACKWARD 1	Name and use angles formed by parallel lines and a transversal	**8.G.A.5**
BACKWARD 2	Vertical angles and linear pairs as reasons	**HSG-CO.C.9**
BACKWARD 3	The parallel postulate and the definition of parallel lines	**HSG-CO.A.1**
FORWARD 1	Prove the converse: congruent alternate interior angles make line...	**HSG-CO.C.9**
FORWARD 2	Use parallel-line theorems to prove the triangle angle sum	**HSG-CO.C.10**

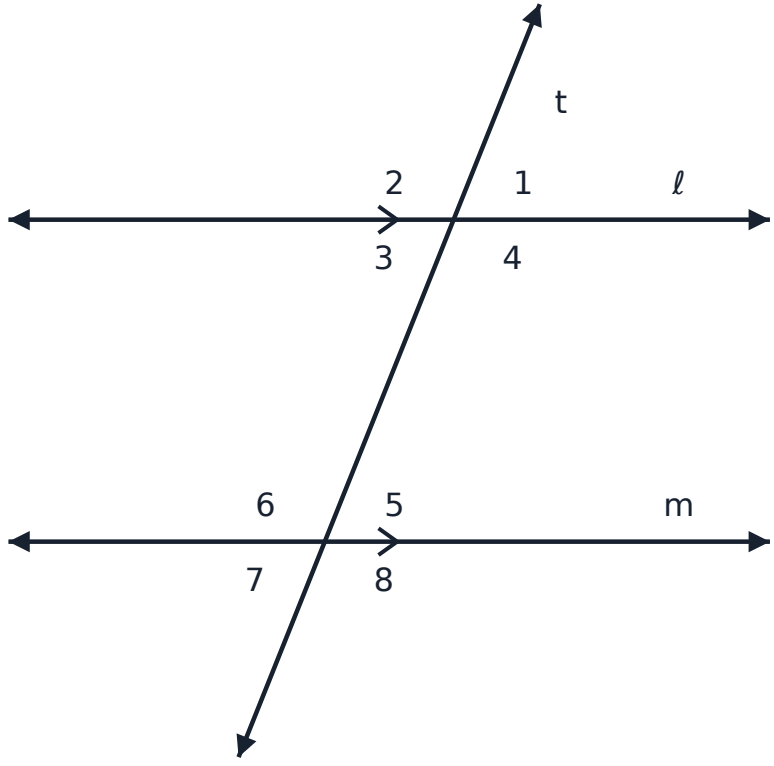
SET 18 • HSG-CO.C.9

MAIN

Geometry main question • High School — Geometry • Geometry: Congruence • HSG-CO.C.9

Lines ℓ and m are parallel and are cut by transversal t, crossing ℓ at P and m at Q. Explain, using a translation, why corresponding angles $\angle 1$ and $\angle 5$ are congruent. (This argument uses only the parallel postulate and rigid motions, so the later items may use its result.)

Work space



Conclusion: _____

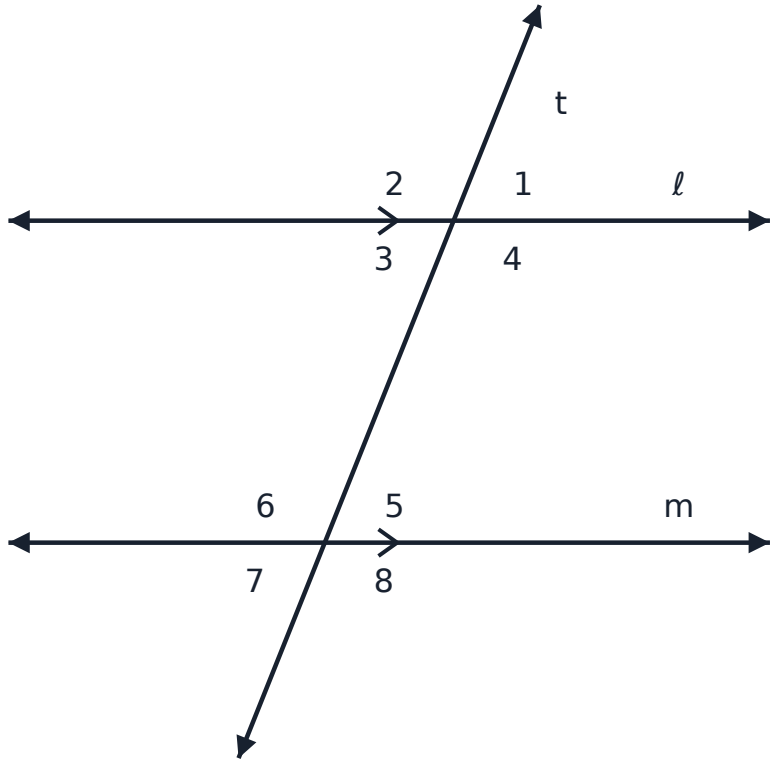
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-M-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 18 • MAIN • Q1/5

Lines ℓ and m are parallel and are cut by transversal t. Use the corresponding angles theorem (proved with a translation) to prove that alternate interior angles $\angle 3$ and $\angle 5$ are congruent.

Work space



Conclusion: _____

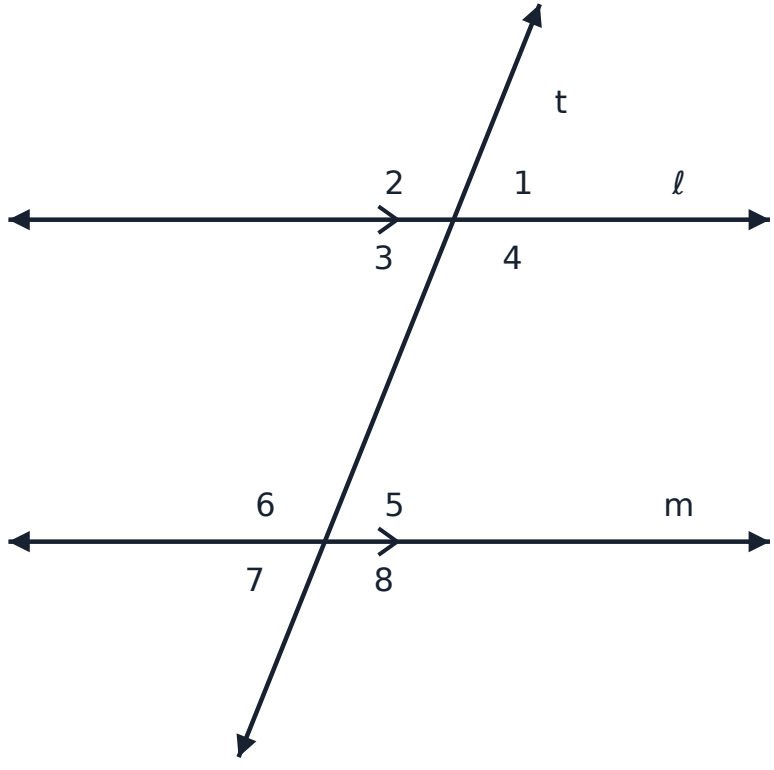
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-M-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 18 • MAIN • Q2/5

Lines ℓ and m are parallel and are cut by transversal t. Use the corresponding angles theorem to prove that alternate exterior angles $\angle 2$ and $\angle 8$ are congruent.

Work space



Conclusion: _____

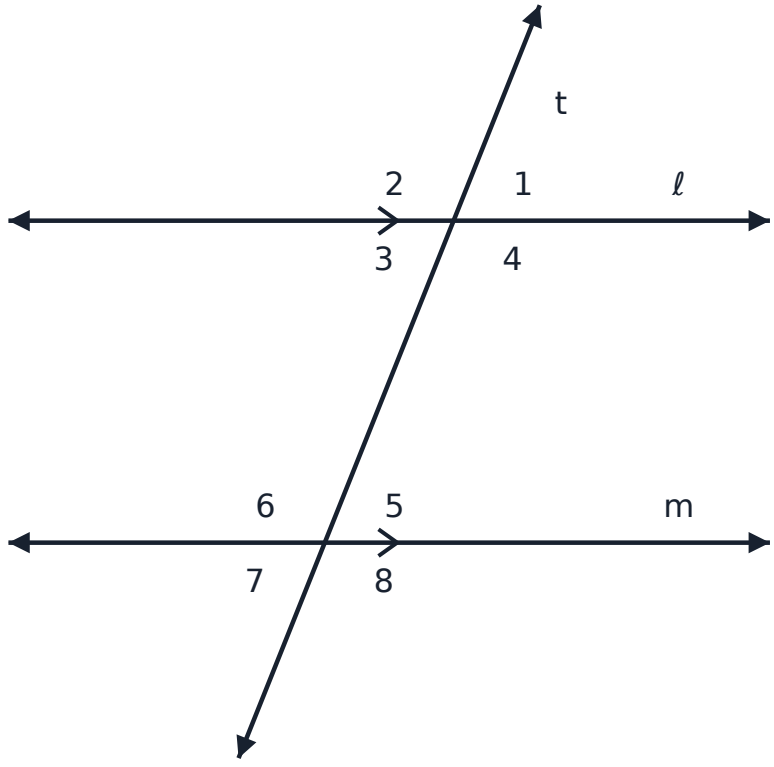
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-M-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 18 • MAIN • Q3/5

Lines ℓ and m are parallel and are cut by transversal t. Prove that same-side interior angles $\angle 4$ and $\angle 5$ are supplementary.

Work space



Conclusion: _____

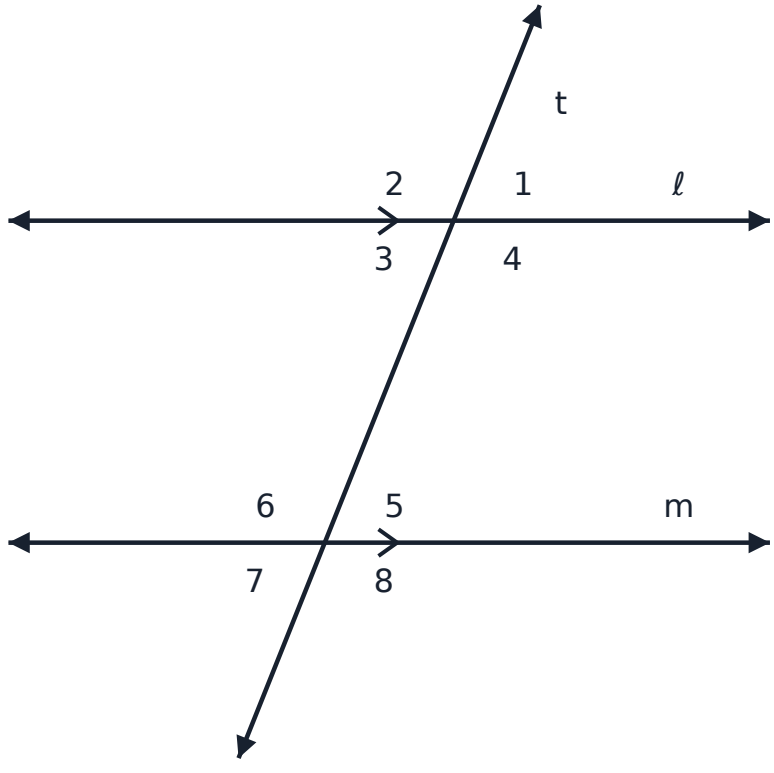
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-M-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 18 • MAIN • Q4/5

Lines ℓ and m are parallel and are cut by transversal t at P (on ℓ) and Q (on m). Let M be the midpoint of PQ. Explain, using the 180° rotation about M, why alternate interior angles $\angle 3$ and $\angle 5$ are congruent.

Work space



Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-M-Q5

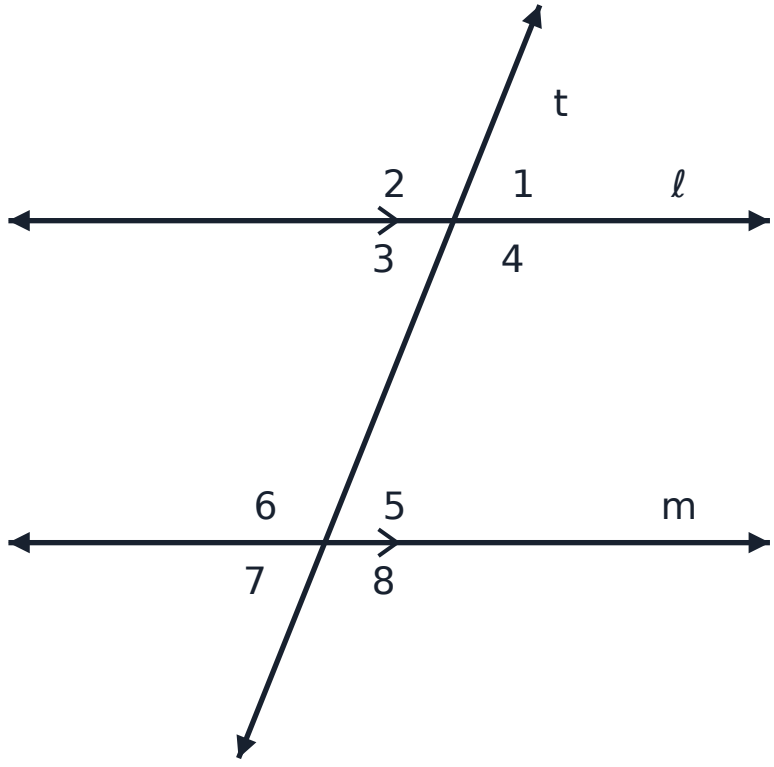
High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 18 • MAIN • Q5/5

SET 18 • HSG-CO.C.9

BACKWARD 1

Name and use angles formed by parallel lines and a transversal • Grade 8 • Geometry • 8.G.A.5

Lines $\ell \parallel m$ are cut by transversal t. $\angle 4$ measures $70°$. Find the measure of $\angle 6$.



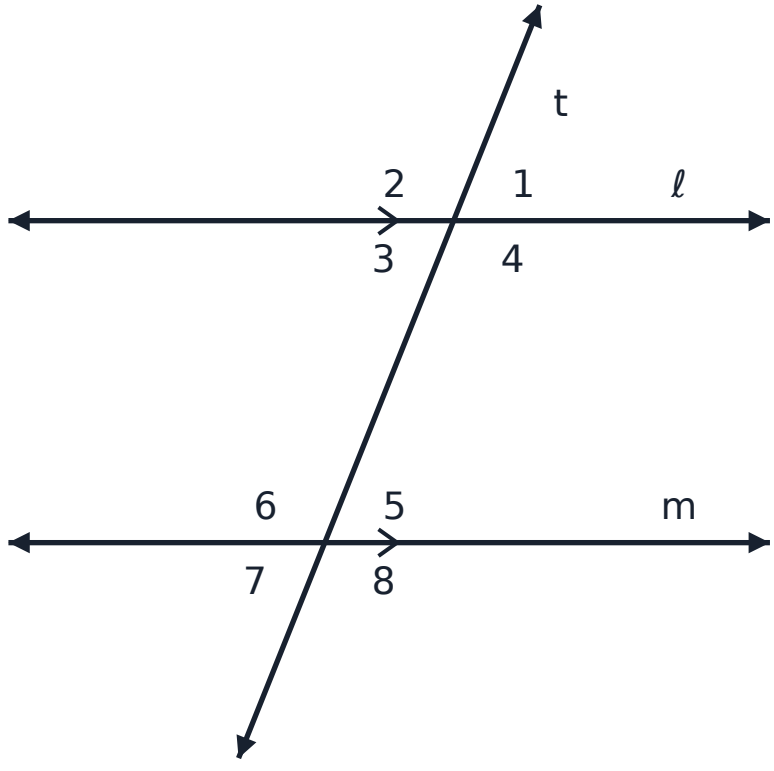
Measure: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-B1-Q1

Grade 8 • Geometry • 8.G.A.5 • Set 18 • BACKWARD 1 • Q1/5

Lines $\ell \parallel m$ are cut by transversal t. Which angle corresponds to $\angle 2$?



A. $\angle 7$

B. $\angle 8$

C. $\angle 6$

D. $\angle 5$

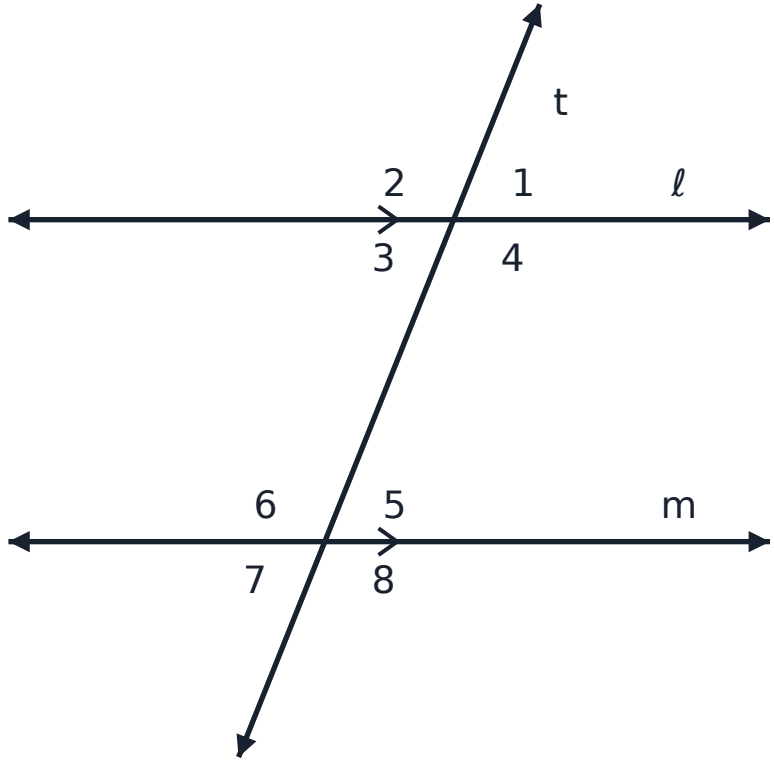
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-B1-Q2

Grade 8 • Geometry • 8.G.A.5 • Set 18 • BACKWARD 1 • Q2/5

True or false?

Lines $\ell \parallel m$ are cut by transversal t. $\angle 4$ and $\angle 5$ are same-side interior angles, so they add to 180°.



A. True **B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-B1-Q3

Grade 8 • Geometry • 8.G.A.5 • Set 18 • BACKWARD 1 • Q3/5

True or false?

When two lines are cut by a transversal, corresponding angles are congruent even if the lines are not parallel.

A. True

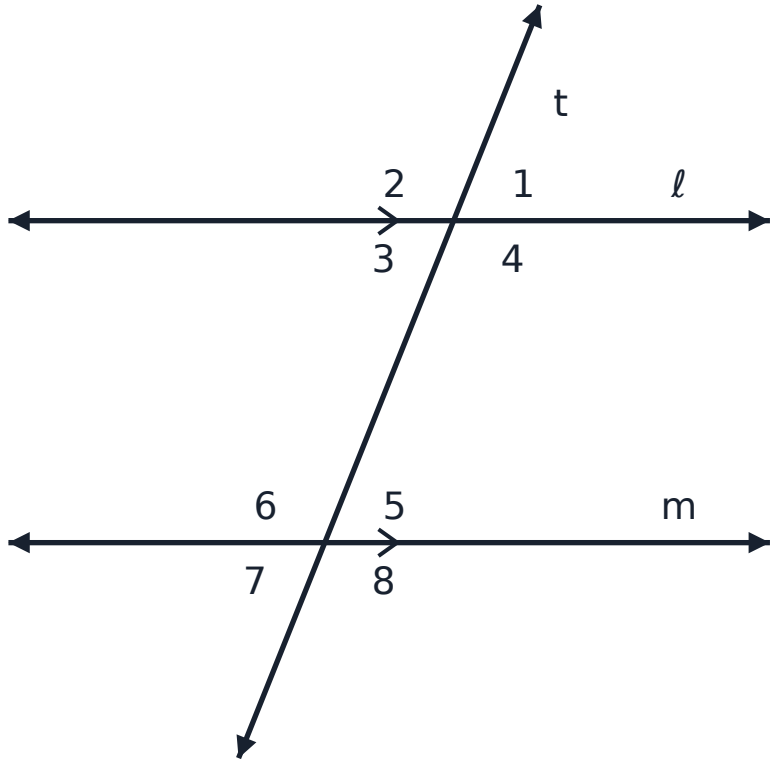
B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-B1-Q4

Grade 8 • Geometry • 8.G.A.5 • Set 18 • BACKWARD 1 • Q4/5

Lines $\ell \parallel m$ are cut by transversal t. $\angle 8$ measures 112°. Find the measure of $\angle 2$.



Measure: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-B1-Q5

Grade 8 • Geometry • 8.G.A.5 • Set 18 • BACKWARD 1 • Q5/5

SET 18 • HSG-CO.C.9

BACKWARD 2

Vertical angles and linear pairs as reasons • High School — Geometry • Geometry: Congruence • HSG-CO.C.9

Two lines cross. Name the reason two opposite angles are congruent.

Reason: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-B2-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 18 • BACKWARD 2 • Q1/5

Name the reason two adjacent angles on a line add to 180°.

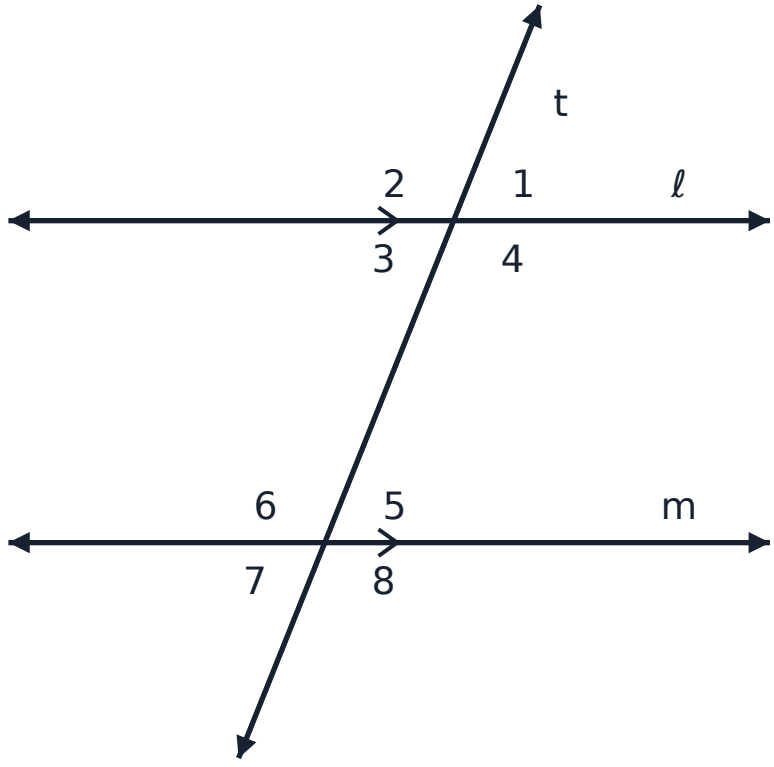
Reason: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-B2-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 18 • BACKWARD 2 • Q2/5

At the top crossing, $\angle 1 = 110°$. What is $\angle 2$ and why?



A. 110°, linear pair

B. 70°, linear pair

C. 110°, vertical angles

D. 70°, vertical angles

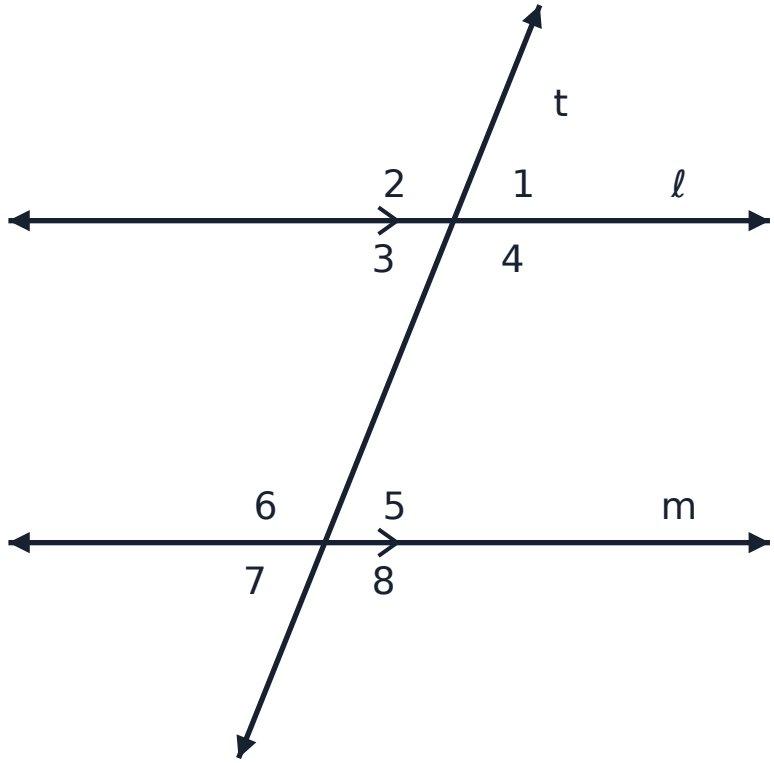
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-B2-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 18 • BACKWARD 2 • Q3/5

True or false?

Angles 1 and 3 are vertical angles in the figure.



A. True

B. False

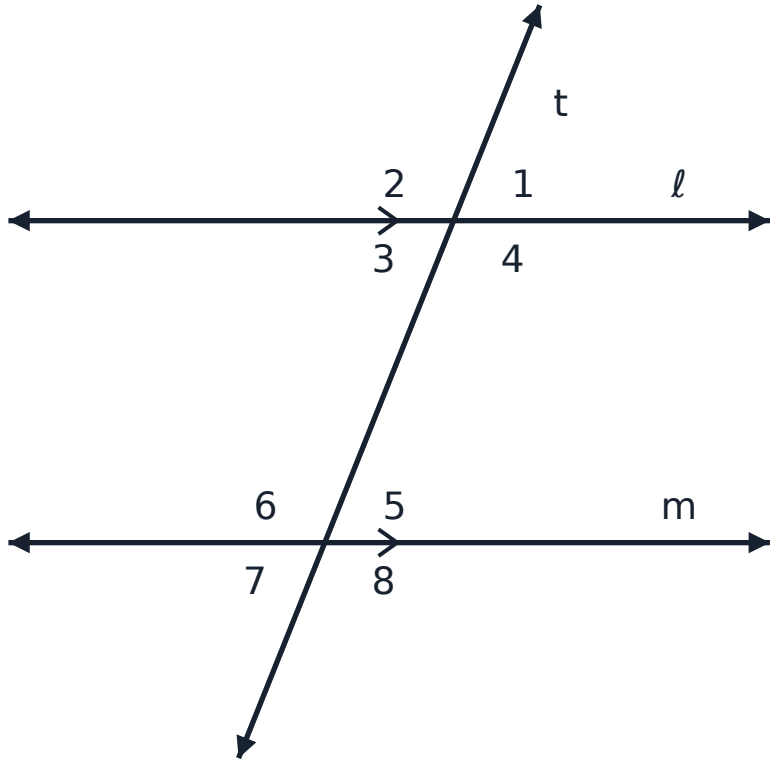
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-B2-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 18 • BACKWARD 2 • Q4/5

True or false?

Angles 1 and 4 are vertical angles in the figure.



A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-B2-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 18 • BACKWARD 2 • Q5/5

SET 18 • HSG-CO.C.9

BACKWARD 3

The parallel postulate and the definition of parallel lines • High School — Geometry • Geometry: Congruence • HSG-CO.A.1

True or false?

Through a point not on a line, there is exactly one line parallel to it (in a plane).

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-B3-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 18 • BACKWARD 3 • Q1/5

Line ℓ is parallel to line m. Line n passes through a point of ℓ and is also parallel to m. What can you conclude about n?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-B3-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 18 • BACKWARD 3 • Q2/5

Which pair of lines in a plane are parallel?

A. $x = 2$ and $y = 2$

B. $y = x$ and $y = -x$

C. $y = 2x + 1$ and $y = -2x + 1$

D. $y = 2x + 1$ and $y = 2x - 5$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-B3-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 18 • BACKWARD 3 • Q3/5

True or false?

Two lines in a plane that never meet are parallel.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-B3-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 18 • BACKWARD 3 • Q4/5

Line $\ell \parallel$ line m. A translation maps m to a line through a point of ℓ. Must that image line be parallel to m or the same as m?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-B3-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 18 • BACKWARD 3 • Q5/5

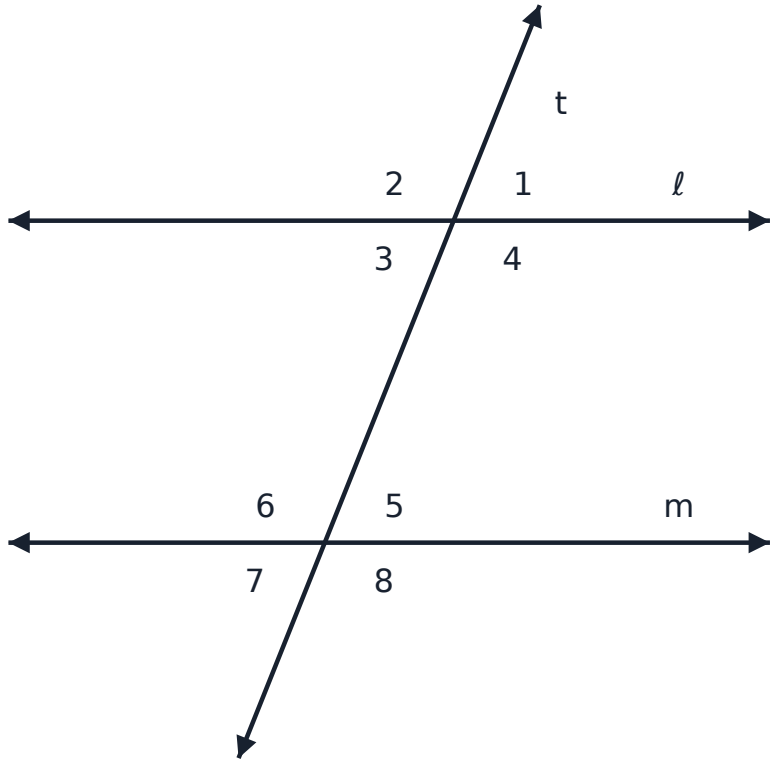
SET 18 • HSG-CO.C.9

FORWARD 1

Prove the converse: congruent alternate interior angles make lines parallel • High School — Geometry • Geometry: Congruence • HSG-CO.C.9

Lines ℓ and m are cut by transversal t, and alternate interior angles $\angle 3$ and $\angle 5$ are congruent. Prove that $\ell \parallel$ m.

Work space



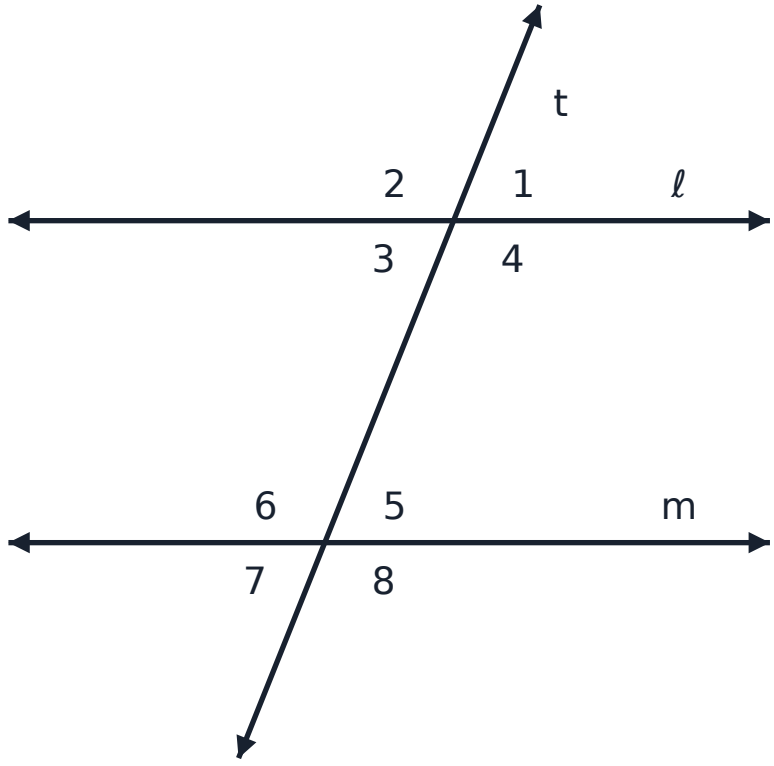
Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-F1-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 18 • FORWARD 1 • Q1/5

Lines ℓ and m are cut by transversal t with $m\angle 4 = (5x - 20)°$ and $m\angle 6 = (3x + 30)°$. For what value of x is $\ell \parallel m$? Explain.



Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-F1-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 18 • FORWARD 1 • Q2/5

True or false?

If same-side interior angles are supplementary, the lines are parallel.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-F1-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 18 • FORWARD 1 • Q3/5

Which fact proves that lines ℓ and m are parallel?

A. A pair of corresponding angles is congruent

B. A pair of vertical angles is congruent

C. A linear pair adds to 180°

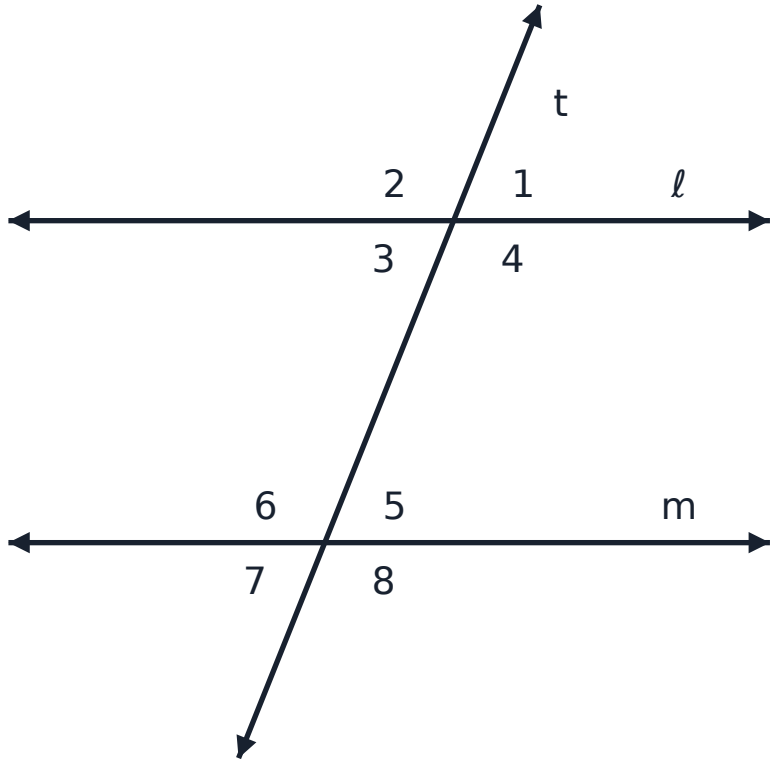
D. Two angles at the same crossing are equal

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-F1-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 18 • FORWARD 1 • Q4/5

Lines ℓ and m are cut by a transversal. $m\angle 1 = 72°$ and $m\angle 7 = 108°$. Are ℓ and m parallel? Explain.



Answer: _____

SET 18 • HSG-CO.C.9

FORWARD 2

Use parallel-line theorems to prove the triangle angle sum • High School — Geometry • Geometry: Congruence • HSG-CO.C.10

Prove that the angles of triangle ABC add to 180°. Begin by drawing the line through B parallel to AC.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-F2-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 18 • FORWARD 2 • Q1/5

In triangle ABC, a line through B parallel to AC makes a 50° angle with BA (on A's side) and a 65° angle with BC (on C's side). Find all three angles of the triangle.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-F2-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 18 • FORWARD 2 • Q2/5

True or false?

The proof of the triangle angle sum works without any fact about parallel lines.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-F2-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 18 • FORWARD 2 • Q3/5

Which theorem is used to show the new angles at B equal ∠A and ∠C?

A. Alternate interior angles theorem

B. Isosceles triangle theorem

C. Triangle inequality

D. Vertical angles theorem

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-F2-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 18 • FORWARD 2 • Q4/5

Explain why the exterior angle at C of triangle ABC equals m∠A + m∠B.

Explanation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-F2-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 18 • FORWARD 2 • Q5/5

SET 19 • HSG-CO.C.9 • CONGRUENCE: PROVING THEOREMS

Prove the perpendicular bisector theorem and its converse

MAIN	Geometry main question	**HSG-CO.C.9**
BACKWARD 1	Use the definitions of midpoint and perpendicular bisector	**HSG-CO.A.1**
BACKWARD 2	Use SAS and SSS to prove triangles congruent	**HSG-CO.B.8**
BACKWARD 3	Corresponding parts of congruent triangles and right angles from…	**HSG-CO.B.7**
FORWARD 1	Use the perpendicular bisector to find points equidistant from two…	**HSG-GPE.B.5**
FORWARD 2	Prove that the perpendicular bisectors of a triangle meet at one p…	**HSG-C.A.3**

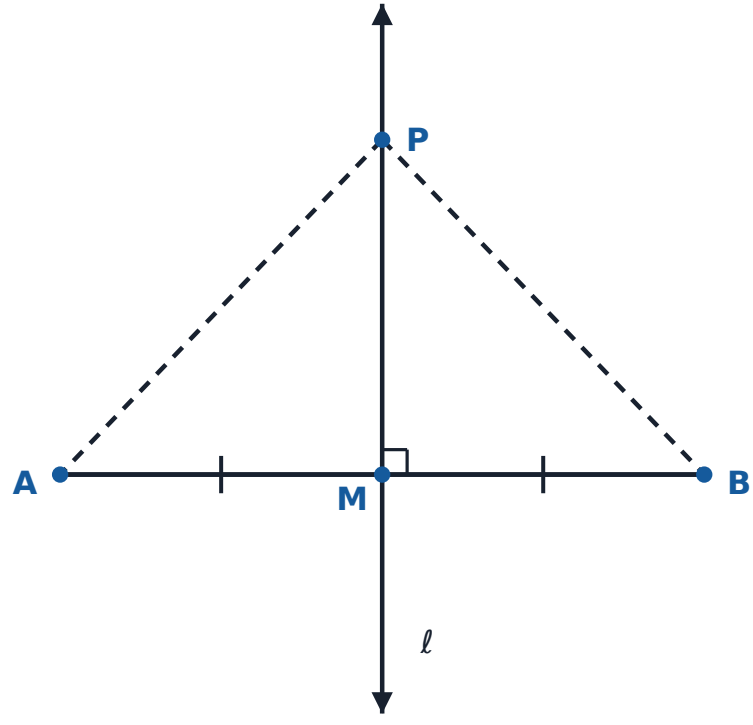
SET 19 • HSG-CO.C.9

MAIN

Geometry main question • High School — Geometry • Geometry: Congruence • HSG-CO.C.9

Line ℓ is the perpendicular bisector of segment AB, meeting AB at M. Point P is on ℓ. Prove that PA = PB.

Work space



Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-M-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 19 • MAIN • Q1/5

Point P is not on line AB, and PA = PB. Let M be the midpoint of AB. Prove that line PM is perpendicular to AB, so P is on the perpendicular bisector of AB.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-M-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 19 • MAIN • Q2/5

Point P is not on line AB, and PA = PB. The bisector of ∠APB meets AB at M. Prove that line PM is the perpendicular bisector of AB (a second proof of the converse, using SAS).

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-M-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 19 • MAIN • Q3/5

In a coordinate plane, A(−a, 0) and B(a, 0) with a > 0. Prove that every point P(0, y) on the y-axis is equidistant from A and B, and that every point P(x, y) with PA = PB has x = 0.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-M-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 19 • MAIN • Q4/5

A(1, 1) and B(5, 3). Prove that every point on the line $y = -2x + 8$ is equidistant from A and B, and that every point equidistant from A and B lies on $y = -2x + 8$.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-M-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 19 • MAIN • Q5/5

SET 19 • HSG-CO.C.9

BACKWARD 1

Use the definitions of midpoint and perpendicular bisector • High School — Geometry • Geometry: Congruence • HSG-CO.A.1

M is the midpoint of AB and AB = 14. Find AM.

AM = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-B1-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 19 • BACKWARD 1 • Q1/5

True or false?

The perpendicular bisector of AB passes through the midpoint of AB and is perpendicular to AB.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-B1-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 19 • BACKWARD 1 • Q2/5

Line ℓ meets AB at M with AM = MB, but ℓ is not perpendicular to AB. What is ℓ?

A. Not related to AB

B. Parallel to AB

C. The perpendicular bisector of AB

D. A bisector of AB that is not the perpendicular bisector

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-B1-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 19 • BACKWARD 1 • Q3/5

Line ℓ is perpendicular to AB at M, and AM = 3, MB = 5. Is ℓ the perpendicular bisector of AB? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-B1-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 19 • BACKWARD 1 • Q4/5

True or false?

A segment has exactly one perpendicular bisector in a plane.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-B1-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 19 • BACKWARD 1 • Q5/5

SET 19 • HSG-CO.C.9

BACKWARD 2

Use SAS and SSS to prove triangles congruent • High School — Geometry • Geometry: Congruence • HSG-CO.B.8

In $\triangle PMA$ and $\triangle PMB$, PM is shared, AM = BM, and $\angle PMA \cong \angle PMB$. Which criterion proves they are congruent?

Criterion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-B2-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 19 • BACKWARD 2 • Q1/5

In $\triangle PMA$ and $\triangle PMB$, $PA = PB$, $AM = BM$, and PM is shared. Which criterion proves they are congruent?

Criterion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-B2-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 19 • BACKWARD 2 • Q2/5

Two triangles share a side and have two other pairs of congruent sides. Which criterion applies?

A. SAS

B. ASA

C. SSS

D. AAS

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-B2-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 19 • BACKWARD 2 • Q3/5

True or false?

A side shared by two triangles counts as a pair of congruent sides.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-B2-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 19 • BACKWARD 2 • Q4/5

True or false?

SSA is a valid congruence criterion.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-B2-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 19 • BACKWARD 2 • Q5/5

SET 19 • HSG-CO.C.9

BACKWARD 3

Corresponding parts of congruent triangles and right angles from a linear pair • High School — Geometry • Geometry: Congruence • HSG-CO.B.7

△PMA ≅ △PMB. Which angle corresponds to ∠PMA?

Angle: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-B3-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.B.7 • Set 19 • BACKWARD 3 • Q1/5

Two congruent angles form a linear pair. What is each angle's measure?

Measure: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-B3-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.B.7 • Set 19 • BACKWARD 3 • Q2/5

True or false?

If $\triangle PMA \cong \triangle PMB$, then $PA = PB$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-B3-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.B.7 • Set 19 • BACKWARD 3 • Q3/5

$\triangle XYZ \cong \triangle XWZ$. Which pair of segments must be congruent?

A. YZ and XW

B. YZ and WZ

C. XY and XZ

D. XZ and YW

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-B3-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.B.7 • Set 19 • BACKWARD 3 • Q4/5

True or false?

Two angles in a linear pair are always congruent.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-B3-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.B.7 • Set 19 • BACKWARD 3 • Q5/5

SET 19 • HSG-CO.C.9

FORWARD 1

Use the perpendicular bisector to find points equidistant from two points • High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5